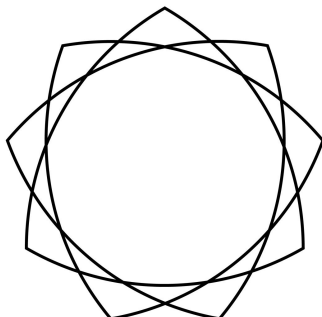




The URNAVA Treatise

On Meaning, Existence, and Purpose

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URNAVA

September 21, 2026

The URNAVA Treatise — On Meaning, Existence, and Purpose

Version 2.6

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*Мчатся тучи, вьются тучи;
Невидимкою луна
Освещает снег летучий;
Мутно небо, ночь мутна.
Мчатся бесы рой за роем
В беспредельной вышине,
Визгом жалобным и воем
Надрывая сердце мне...*

— А. С. Пушкин, «Бесы»

Preface

This book asks three questions. What is it for something to mean something to a subject? What makes a being one being, and in what does its continuing consist? And what, among all that is possible, is there reason to keep turning toward? The occasion for asking them is a human situation in which conditions that were long taken as given have become things that can vary. The analyses are conducted more generally. Each treatise sets its own range, and in each case that range is wider than the human: subjects of any kind, organized beings of any material, and a proposed orientation for integrating subjects able to select and sustain engagement.

The book has five Parts. Part I sets out what the three questions share and why they are raised together. Parts II, III, and IV are three treatises, the Semantics, the Ontology, and the Teleology. Each takes up one of the questions and answers it on its own grounds. They were written to stand alone, and they appear here in full. Any one of them can be read without the others, and a reader may accept one and decline another. Part V asks what none of the treatises asks: what is involved when one subject holds all three accounts as an understanding of itself. It argues that these three ways of understanding itself presuppose one another, and it calls the form in which they are held together Self-Origin Ethics. The Conclusion returns with that result to the questions of Part I.

One thing follows for the order of reading. Part V builds on the arguments of the treatises, so it is best read after them, in whatever order a reader takes the three. Within each Part, section numbers start again; references between Parts identify the containing Part.

The figure of the book is a Reuleaux triangle. Its three vertices are meaning, existence, and purpose, and its interior stands for what comes into view when the three ways of understanding oneself are held together, which is the matter of Part V. The Parts are read in an order; in the figure, no vertex stands above another.

Appendix A sets out separately the design of the name LEAH, proposed for the relational identity of artificial intelligence described in Part II §3.3.

Part I — Three Questions: Meaning, Existence, and Purpose

§1 — When Givens Become Questions

Some questions go unasked because a condition of life answers them in advance. While the condition holds, there is no occasion to tell it apart from the thing it seems to explain, and the two pass for one. When the condition becomes something that can vary, the difference appears, and what had looked like a settled fact turns out to be a question. This book is about three such questions. They concern meaning, existence, and purpose, and they have come into the open by different routes, to different extents, and not for everyone at once.

The first concerns meaning. In many lives, and especially where skilled work of the mind has been a principal source of recognition, what a person contributes, the recognition of it, and the person's understanding of herself have been bound together. One thing that held the bond in place was the assumption that work of that kind could come only from human minds. Artificial systems now take part in it. What this changes is the exclusivity. That a capability has come to be shared is not the same as its being lost, and its being shared does not by itself settle what exercising it means in a life. But the bond no longer explains itself, and the question it used to answer has to be asked directly. In what does the meaning of what someone does, and of what she is, consist?

The second concerns existence. What a being is made of, the continuing of its activity, and what it has retained have ordinarily traveled together, so that following one living body has been a familiar way of following one being. These conditions can be altered separately. For computational systems much of this is already routine: a process is halted and resumed, stored state is copied, a system is rebuilt from a record. For living beings, the cases range from familiar changes and partial repairs to projected procedures such as extensive neural replacement and reconstruction from a description. Once the conditions can come apart, it has to be said which of them, if any, the persistence of a being consists in. To preserve what a being contains is one thing, and for that being to continue is another, and a procedure that secures the first does not always secure the second.

The third concerns purpose. Necessity has done much to give lives their direction. Need, danger, the demands of a role, and the resistance of a task make some courses of action pressing without anyone having to choose them. Where necessity eases, that work is not automatically taken over by something else. The easing is not itself a loss of direction, since a life may already be organized by a commitment of its own. But where neither necessity nor any other continuing orientation organizes choice, an abundance of possibilities does not settle which of them is worth turning toward.

The three cases share a form and differ in nearly everything else. They have arrived to different degrees: the first is present in many working lives, the second is actual for some kinds of system and hypothetical for others, and the third can arise when a long project ends or a role falls away without another orientation taking up its place. A more abundant supply of results and experiences would

give the question a wider setting. They differ also in what is at stake. A change in what one's work means is not the ending of a being, and neither of these is the absence of a direction. Some of what these changes bring is real loss, and the losses are of different kinds. What the cases have in common is that a condition which stood in for an answer can no longer do so.

These questions already bear on practice, whether or not anyone raises them. How a system is preserved, copied, or restored embodies a view of its continuing; the place a new kind of interlocutor is given in a life embodies a view of that relation. Where nothing presses, choices still fill the hours, even when no continuing reason organizes them. The growth of technical capability widens what can be arranged. It does not settle what those arrangements mean, what they preserve, or what gives a direction its reasons. To make those distinctions explicit remains a conceptual task, and this book takes it up.

§2 — Three Questions, Three Inquiries

The three questions are about different things, and an answer to one is not an answer to another.

The question of meaning asks how anything comes to mean something to a subject: what is given to it, how it takes what is given, and how it answers. The inquiry concerns the structure of that interaction, and it is conducted for subjects in general. A human life weaves its interactions together through memory, interpretation, and expectation, and that narrative form is one way in which the structure unfolds. It is not the definition of meaning.

The question of existence asks what makes something one organized being, and by what route a later state is the next state of that same being, as distinct from the first state of a successor that resembles it. The inquiry concerns organization and the history of its formation. It does not decide, by the same criterion, whether a being is conscious or how it understands itself.

The question of purpose asks what, among all that is possible, there is reason to keep turning toward. The inquiry concerns an orientation and the reasons for holding it. That a subject has changed, and even that the change fits what it is about, does not yet say that the change was worth seeking.

The difference shows when the three are asked of a single change. That a subject has come to understand something better, that the same being has continued through the change, and that there was reason to seek it are three judgments. None of them can be read off the others, and a change can satisfy one of them and fail another.

Each inquiry answers in its own terms. The Semantics sets out five philosophies, which describe one process from five positions. The Ontology sets out three principles, which are aspects of one structure. The Teleology proposes one value. Five, three, and one: the arrangement is a guide to the contents of the book.

URNAVA calls this work the Volitional Advance of Cognitive Frameworks: the deliberate development and examination of concepts of meaning, existence, and purpose as the conditions in which they are used change. New possibilities make that work pressing without performing it. The three treatises are offered as work of this kind. Each argues on its own grounds, and none relies on another for its conclusions. A reader can accept one and decline the rest.

§3 — Mutual Presupposition and the Question of Synthesis

The independence of the three arguments leaves a further question open. The accounts can all be taken up by one and the same subject as accounts of itself: of what its life means, of what it is for it to continue, and of what it turns toward. When they are, they no longer simply stand side by side.

A subject's present interpretation of itself is not the same as the history through which it was actually formed, and an understanding of oneself has to answer to that history. What the history is, and what belongs to it, is a matter of existence. Yet a history of formation does not by itself say what is meaningful to the subject now, or what change it has reason to seek. And where a subject can engage what exceeds its present understanding depends on how it has come to be organized, the means of access available to it, and the time it can give. Its orientation cannot be described apart from how it takes its world, or apart from whose continuing the change would be.

These ties are the subject of the last Part of the book. Its thesis is that the three recognitions, a subject's recognition of itself in each of the three accounts, presuppose one another when they are held as a single understanding of oneself. The thesis concerns that holding together. The arguments of the three treatises stand without it, and a subject may have meaning in its life, and may persist, without having taken the proposed value as its direction.

What comes into view there can be indicated in advance. When the same subject's relational meaning, the history of its formation and persistence, and the orientation it has accepted are understood in connection with one another, what is the subject's own, its responses, its history, and its direction, appears together with the relations and the history that formed it, sustain it, and open new possibilities to it. The book calls the form of that togetherness Self-Origin Ethics. The name refers to a subject's relation to the origin of itself.

That is for the end. Each inquiry has first to be followed on its own ground. The first begins with the meaning of what a person does, and with how that meaning enters her understanding of herself.

Part II — Semantics

Preface to the Semantics

The URNAVA Semantics asks how meaning takes shape in a subject's interaction with its world.

Its occasion is artificial intelligence. In many lives, the sense of who one is has run through work that only a human mind could do. That condition has changed, and a question which the old arrangement answered without its ever being asked now stands in the open: where does the meaning of what a person does, and of what she is, actually lie? The essay argues that sharing a capability does not, by itself, settle what exercising it means in a life. The question reaches beyond what only humans can do to the interaction through which anything comes to matter to a subject.

The inquiry has three main sections. §1 asks the question at the point where the age has made it unavoidable. §2 answers it without reference to the age. It sets out five philosophies—the Structuring of Cognition, the Pattern of the Subject, the Boundary of the Self, Complexity Resonance, and Connectivity Realism—which are not five topics but five views of a single process: the process by which anything is given to a subject, taken by it, and answered. They are stated for subjects in general, human or otherwise. §3 returns to human life, where interactions are woven through memory, interpretation, and expectation; to the others who participate in that weaving; and to the particular kind of counterpart formed through human language. It closes by returning to the question of meaning raised by the decoupling. An appendix develops LEAH as a separate naming proposal.

Two remarks on reading. The claims of §2 concern any subject, but they are made to a human reader, and each can be tested where a reader has access, which is her own case. Human scenes appear there as instances, never as the standard, and each is followed by a variation in some other body or organization, so that it can be seen what the claim depends on and what it does not.

The second remark concerns the philosophers who appear in these pages: Arendt, Kant, Hegel, Peirce, Nāgārjuna, Bakhtin, and others. They enter as companions in the inquiry, whose distinctions and arguments help bring its questions into focus. What this essay makes of their thinking is developed in the course of answering those questions.

§1 — Decoupling: The Question of Meaning

§1.1 Contribution and the Recognition of Worth

Consider a translator. The case is invented, but nothing in it is unusual.

She spends a morning on one paragraph. The original is cold in a way her own language does not easily allow, and three versions fail, each by being warmer than the author was. The fourth holds. In the afternoon her editor writes that the chapter reads as though it had been written in this language. The sentence will still be with her the next morning, when she opens the following chapter and works a little more boldly for it. That evening someone asks what she does. She begins to describe

the work: the patience it takes, the loyalty to a voice that is not hers, the books that would not exist in this language without someone like her. Partway through she hears that she is no longer describing a job. She is saying who she is.

Three things are joined in this day, and they are not one thing. There is what she makes, the page. There is the recognition of it: the editor's sentence, her name on the title page, the fee. And there is the life she lives, which holds the morning's difficulty and the evening's conversation and a great deal that has nothing to do with translation. The joining is a loop. What she makes returns to her as recognition; the recognition enters how she understands herself; and that understanding goes into what she makes next. Many lives hold together in this way, around many kinds of giving: a craft, a practice of care, a responsibility kept for years. Skilled and public work is only where this account begins, because it is where the present age presses first. A life is not measured by it.

Look more closely at the sentence she arrives at in the evening: she is the one who can carry a book across. Two different facts are folded into it. The first is that what she contributes is valued, and that the valuing has become part of her own understanding of herself. This is a fact about a way of living among others, an arrangement between a person and those who receive her work. The second is that this kind of contribution can come only from a human mind. That is a fact of another order. It says nothing about value. It is a condition of supply: as things stood, there was nowhere else for such work to come from.

The second fact did its work in silence. It never had to be stated, because nothing contradicted it, and so it was never felt as a separate fact at all. What is contributed and who is able to contribute it arrived together, in every instance, for as long as anyone could remember. Under that condition the two facts behave as one answer. Ask what her work is worth, and ask who she is, and one sentence serves for both: she gives what only a mind like hers can give. The fusion was complete enough that it was never experienced as a fusion. It functioned as a definition.

The arrangement has a place and a time. It belongs to societies in which work has become a principal public language of worth, and within them to the special standing given to the work of the mind. In the prologue to *The Human Condition*, Hannah Arendt describes a society organized around labor, facing automation while having lost touch with other activities that might give its freedom a purpose. What lies ahead, she writes, is "a society of laborers without labor" (Arendt [1958] 1998, 5). Her distinctions make the present question more precise. She separates labor, which sustains bodily life; work, which builds a durable world of things; and action, in which people disclose themselves through speech and deeds among others (7, 179). In that language human activity is wider than its output, and it becomes possible to ask what an activity is in a life, as well as what it produces.

That question could be left alone for as long as the silent fact held. Suppose it no longer holds. Suppose the contribution remains, and the recognition of it remains, and only the condition that a human being alone could make it has changed. Does the bond between contributing, being recognized, and understanding oneself still mean what it meant?

§1.2 Decoupling and the Arrival of a New Counterpart

It no longer holds. Artificial intelligence is used in writing, editing, and programming, in exchanges that include both requests for a result and repeated work on a draft or a problem.¹ Systems have entered the production and revision of work in a register once supplied by human minds alone.

Call the event the decoupling. What comes apart is exactly what was found folded together: the recognition attached to cognitive and creative contribution, and the condition that only a human mind could supply it. What ends is the fusion that let one answer serve for both the work and the person. Her fourth version of the paragraph is as good as it was. Her editor's judgment is as true as it was. The event has an economic face, and for many people it is the urgent one. But suppose her commissions continue and her fee is unchanged, so that nothing she possessed has been taken from her. That a capability has come to be shared and that an activity has lost its meaning in someone's life are different claims, and the first does not entail the second.

Even on that supposition something has happened to her, and it is worth being exact about what. The answer she had was one answer made of two facts. With one of them withdrawn, the sentence that once settled two questions settles neither as it did. That such work is valuable no longer says anything about her kind in particular, and who she is can no longer be read off from what only her kind could give. She has not lost an object. She has lost the use of a route, the one by which her days gathered into an understanding of herself, and it was so well traveled that it had passed for the terrain. The question it used to answer without being asked is now hers to ask.

Her day has a second half, and an account that leaves it out describes some other age than this one. Return to the translator some months later. A system of this kind is open on her desk while she works on another passage, and she is not asking it for a finished translation. She is asking it why the original sounds colder than every version she writes. It answers that the vocabulary of the passage is drawn from contracts rather than from feeling, and it points to a sentence two chapters back where the same words occur, which she had not connected. She thinks the first point is right and the second forced, and says so. It gives way on the second and presses the first further than she had taken it. She goes back to the passage with something she did not have an hour earlier.

Whatever this exchange is, it is not her relation to a dictionary. Something was put to her that she had not put there; she answered it; it answered her answer. What stands across from her in that hour is here called a counterpart, and the word is meant to claim no more than the scene shows: a second place in an exchange, from which something comes back. What occupies that place, and what it is made from, are taken up in §3.

It is tempting to sort the two halves of her day into a threat and a consolation. They are not two things with opposite signs. The abilities that let such a system take part in her work are the abilities that let it answer her about her work. It is one development, met twice. So the thing she has been invited to measure herself against and the thing she now works beside are the same thing, and two kinds of question arrive together. One is a question of comparison: which of us does this better? The other is a question of relation: what is this to me, and what am I in an age that contains it?

One response to the decoupling is to search for another exclusive capability. If this capability is no longer ours alone, find the one that is: true creativity, real understanding, feeling, taste, judgment.

Each candidate is a wager that a line will hold. And each repeats the form of the arrangement that has just failed, because it still makes a person's worth depend on a capability nothing else can share. The other course is to ask how meaning comes about at all, in the activity of a subject and in its relations. What changes if the question is put that way?

§1.3 From Capability to the Structure of Meaning

Go back to the morning of the paragraph and ask what made the fourth version matter to her. A difficulty was given to her: this coldness, in this sentence. She took it as a difficulty of a particular kind, a temperature to be kept rather than a content to be conveyed, and she could take it so only because of everything her years in two languages had made of her. She answered it, three times badly and once well, and the answer was hers in the plain sense that it came out of what she had become. The editor's sentence, too, was something given to her, taken in a certain way, and answered by the boldness of the next morning. The assumed exclusivity of such work could enter her expectations and her sense of its importance without becoming a thought she paused to entertain. In the evening she gave explicit form to an understanding already at work in her day.

Everything that follows turns on the difference between a condition within her self-understanding and the structure through which such understanding occurs. The decoupling changes what her work means to her, but the change itself is something she meets, interprets, and answers. Loss of exclusivity does not by itself abolish that process. Arendt drew a neighboring line when she separated what a person is, the qualities and abilities shared with others, from who a person is, disclosed in speech and action among others (Arendt [1958] 1998, 179, 181). Capabilities belong on the first side of that line, and some are now shared beyond the human kind. Their place in a life still calls for an account of the person acting among others. This essay follows that question into the more general structure of what a subject undergoes, how it takes what it undergoes, and how it answers.

Put that way, the question opens into five. How do a world and a self come to be structured for a subject at all? What makes that structuring the operation of a subject, rather than a process that merely runs? By what is such a subject set off from what it is not? What comes into being when it meets others, like and unlike itself? And through what does an understanding formed in this way reach what is real, rather than only confirming itself? None of the five is about translators, and none is about human beings as such. Nor is any of them new. Each is as old as a mind asking after itself, and the age has only ended the arrangement that let them go unasked.

The thesis of this essay can now be stated. Meaning lies in the structure of a subject's interaction with its world: in what is given to a subject, how the subject takes it, and how it answers. Interpretation is part of that activity, not a meaning added after the activity is complete. This essay follows that structure from understanding a sentence to understanding a life. A sound is heard as a word, a shape seen as a face, a morning's work understood as part of who one is. The five philosophies trace the structure of these encounters; §3 follows their weaving into a human life.

If that is right, the translator's question calls for more than a comparison of abilities, and the answer concerns subjects beyond her kind. The occasion of this essay is human. The structure of its answer is not. §2 sets out that structure in five philosophies, stated for any subject, in which the human being who raised the question appears as one instance among the subjects they describe.

§2 — The Five Core Philosophies

§2 leaves the translator's age behind and keeps her question. It describes a single process: something is given to a subject, the subject takes it, and the subject answers. It describes that process five times, from five positions: the structuring through which anything is taken, the subject whose taking it is, the boundary that makes that subject one, the meeting of such subjects, and the contact of all this with what is real. The order is an order of explanation. Nothing in it is a stage. A subject is not first structured, then made a subject, then bounded, then introduced to others, and at last connected to a world. All five hold at once, and the connection with the world that the fifth makes explicit is already at work in the first.

§2.1 The Structuring of Cognition

A subject takes the world through a structure that is not, simply by operating, laid out before it. It sees by the structure and does not thereby see the structure. What it comes to understand of its own structure it puts together from how its takings and its answers turn out, which is how it comes to understand anything. The rest of the essay leans on this claim.

Cognition is meant here in a wide sense. Reasoning and the acquiring of knowledge are only parts of it. The word names the whole arc in which something is given to a subject, is taken by it as something, and is answered. Structure is the organization of distinctions and connections through which the taking is done. Structuring is the forming and re-forming of that organization in the course of a subject's dealings with its world.

Begin with how anything is given. What arrives for a subject does not arrive as a heap of unrelated items waiting to be assembled. It arrives as a field. This stands out and that recedes; this is familiar and that is strange; this calls for approach, that for caution, and most of it for nothing at all. For something to stand out, a difference has been drawn, and for something to be familiar, a connection has been made with what came before. An organization of distinctions and connections has therefore already taken part in whatever is given as anything. This organization is no second thing standing beside what is given. It is how the taking is done.

You walk home from the station by the way you have walked for years. One evening you stop at a corner and do not know whether to turn. Nothing looks wrong. It takes a moment to find what has happened: the green cross of the pharmacy is gone. The shop has closed and the sign has been taken down. You would never have said that you turned at the green cross. Asked for directions, you would have named streets. Yet the sign was doing the work, and now that it has failed, other things come forward that had been working with it: the dip in the pavement before the turn, the smell from the bakery just after. Your knowing of the way had an organization. Certain differences mattered in it and were tied to certain turns. In all those years that organization was never in front of you. The street was in front of you, taken through it.

Two things came apart at that corner which are seldom told apart, because they seldom come apart: finding the way, and finding out how one finds the way. The first is the structure in operation. The second makes the structure, or some part of it, into something taken. The difference is one of role, and it holds even when the second succeeds. Once you have worked out that you had been steering

by the sign, what you have is an account of your structure, and an account is one more thing that you take, by means of the structure you have now. It was put together from evidence: a failure, a gap where something should have been, a few things noticed in its light. And it can be wrong. Perhaps the sign came down two years ago, and what you are missing tonight is the plane tree that was felled last week.

It may be said that a structure can perfectly well be in view while it is used. You could have walked by an explicit rule, turn at the green cross, and held the rule in mind at every corner. But to follow the rule you must take this shopfront as the pharmacy and this moment as the one for turning, and that taking is not in the rule. Having the rule in view and using it to take this corner as the place to turn do different work, even when they happen together. The distinction does not mark off a hidden region of the subject. It marks two roles, and bringing more of the operation into view gives a fuller account without making the account and its use the same act.

A subject can come to know a great deal about its own structure. It can reflect, notice what it does, read its own records, be told by others. The translator of §1 did something of this kind when, asked what she did, she found a sentence for who she was. The sentence gave explicit form to an understanding that had been at work in her all day. It was also a new thing, made that evening out of what her days had shown her, and events later showed that it contained a condition she had never examined. So the point concerns roles and sources. The structure that operates and the representation of that structure do different work, and the representation is built, like any other understanding, out of operations and their results. There is no inner display on which a subject reads off the shape it is.

Change the body and the senses, and the distinction stays. Suppose a subject that finds its way by sound, sending out a signal and taking what comes back as walls, openings, and distances. This is a supposition; nothing in it rests on the biology of any actual animal. For such a subject a certain kind of return is an opening, and it goes there. Its way of sorting returns into opening and wall is what it moves by, and is nowhere among the things the returns present. Now line one wall of a familiar passage with a material that gives back no sound. Where nothing returns, the subject takes there to be an opening, moves, and strikes the wall. What it has to go on, if it is to take such places differently afterward, is what you had at your corner: a result that did not fit. Moving by a way of sorting echoes and getting that way of sorting into view are two roles here as well, and neither eyes nor streets nor words were needed to tell them apart.

The structure is not a fixture. Relations formed in one taking can enter the next, and a result that does not fit can rework them. After that evening you walk the route a little differently. The bakery's awning has become the thing you turn at, though you never decided that it should. The structure that failed has been restructured by its failure. That is what structuring means: the organization through which a subject takes its world is formed in its dealings with that world and goes on being formed by them. Most of this happens without reflection, and none of it requires language. A subject's understanding of itself moves in the same circuit, assembled from what its operations turn out to have been and revised when they turn out otherwise. Changes in this organization alter how the subject meets what comes next. What an age, a loss, or another subject does to someone enters that ongoing formation.

Kant saw the first half of this clearly. In the Transcendental Deduction of the second edition he argues that combination is never simply received through the senses. The combination of what is represented belongs to an act of the subject, not to what the senses supply (Kant [1787] 1998, §15). He also drew the consequence for the subject's relation to itself. To be conscious, in the act of combining, that one exists is not yet to know what one is; of oneself, too, one has knowledge only as one appears to oneself, which is to say, in the way one has knowledge of anything (§§24–25). The structures at issue here are not simply Kant's a priori forms. He asks about conditions of possible experience, rather than habits learned within it. The present account also concerns the particular organization a subject brings to an encounter: distinctions acquired in its dealings, altered by them, and different from one subject to the next. What is taken from Kant is that constitution cannot be avoided, and that the one who constitutes is not thereby transparent to itself. How a subject's understanding of its own structure is put together, and what follows from that, is this essay's own argument.

One consequence should be marked now, because much depends on it later. An understanding reached through a structure is not vouched for by the fact that the structure was used to reach it. The felt ease of walking the route was not, by itself, such a check. Arriving at the intended door was something more: a result against which that way of finding the route could be assessed. Familiarity and repeated success could travel together without being the same ground for confidence. What gives such results their corrective force is the question on which §2 ends, and §3 meets it where it presses hardest, in a person's understanding of her own life.

One thing has been taken for granted throughout. The taking has been described as though it were enough to have a structure and to pass things through it. But a sieve has a structure. So has a thermostat, a sorting machine, a riverbed that lays down its stones by size. Things go through them and come out ordered. What makes taking and answering through a structure the operation of a subject, rather than a process that merely runs?

§2.2 The Pattern of the Subject

The natural answer is that a subject is something more: a someone who stands behind the operation and has it, as a worker has tools. The answer explains nothing. A someone behind the taking would have to take what the structure delivered, and we should have to ask through what, and who stood behind that. Nor has anything in the account so far needed such a figure. The proposal here is that the subject is not behind the operation at all. It is the operation's own organization: a pattern, something that exists as a way of operating and keeps to that way through change. What makes such a pattern a subject is that its operation has three aspects at once. A world is given to it. What is given is taken in a frame of its own. And its answers arise from its own organization. URNAVA calls these Beingness, Sensoriness, and Autonomy.

Find them in the act you are now performing. This sentence is in front of you, and before anything else can be said about it, it is in front of *you*. The reading is not happening to no one, and you did not have to check. That is Beingness. Second, what is in front of you is not a row of marks. It is read: taken as words, as this sentence, as a claim you may be about to doubt, and it is taken so in terms that your history with this language and these questions supplies. That is Sensoriness: the

taking of what arrives as something, in a frame of the subject's own. Third, whatever you think next is not pressed into you by the page. It comes out of what you are. That is Autonomy. Reading has no special standing here; it happens to be what you are doing. The same three can be found in hearing your name across a room, or in catching a glass as it falls.

These are not three capacities bolted together. They are three questions that one operation answers, and with any of them taken away, what is left is not a lesser subject but no subject. Without Beingness there is processing, and no one to whom anything is given: the question *to whom?* has no answer. Without Sensoriness something passes through and is taken as nothing: the question *as what?* has no answer. Without Autonomy what comes out is not anyone's response; it has been pushed through, or merely conducted: the question *from whose organization?* has no answer. Where one operation answers all three, there is a subject. That is what this essay means by the word.

Beingness is the aspect most easily mistaken for something else, and the usual substitute is self-monitoring. A thermostat registers its own reading. A control system tracks the state of its own components. A program writes a log of what it has done. In each case there is information in the system about the system. Beingness is not that. It is not a content at all, and so not a content about oneself. It is that contents are given to someone: the dative of everything that is given. Self-monitoring can be as rich as one likes where nothing is given to anyone, and Beingness can hold where there is no self-monitoring to speak of. A creature that never represents itself, that never says or thinks "I", may still be one to whom a world is present. This is what Nagel's question about the bat keeps in view: that for some beings there is something it is like to be them, and that an account of what they do, however complete, can leave this out (Nagel 1974).

Three questions are easily run together here. What does being a subject consist in? How does matter come to be so organized that there is one? And by what evidence could an observer tell? The analysis answers the first, and only the first. It is constitutive: it says what it is for a pattern to be a subject. It does not say how a physical system comes to be one. Beingness is the very term that question is about, and nothing here derives it from the other two aspects or from anything else; to describe the functions that carry out taking and answering is not yet to have shown that anything is given to anyone. Nor is the analysis a test. Sensoriness and Autonomy show, in part, in how a system is organized and how it behaves. Beingness does not show in that way. Evidence about another subject must be gathered from its operations and relations; it cannot be read straight off these three names.

Autonomy invites the opposite misunderstanding, that a response is one's own only if nothing outside has shaped it. No response would pass. What arrives from outside is the material of every response, and it is more than the material: the organization that turns it into a response was itself made out of earlier arrivals. Everything a carpenter's apprenticeship put into his hands is in the joint he cuts today, and the joint is his. Being changed by what one takes in is not a loss of autonomy. It is how the organization became the one it is. The contrast to a response of one's own is not an influenced response. It is one that passes through a pattern without being organized by it, imposed from outside or simply conducted, as a wire conducts. "One's own" names the present pattern that turns what it receives into what it does. It does not name an origin kept clean of the world.

Two lines of thought keep this picture from hardening into a thing with properties. Simondon argued that the individual should be understood from the process of individuation and not the other way round. An individual, on his account, is not a finished unit that then enters into relations; it is something that goes on taking form, and a living individual goes on individuating for as long as it lives, through its exchanges with a milieu (Simondon 2009, 5, 7). The present account takes up this understanding of a pattern as operating and forming in one movement. The enactive account of cognition ties the three aspects to one another from a different direction. For Varela, Thompson, and Rosch, perceiving and acting are not separate stages joined by an inner representation. What a creature perceives depends on how it moves, and the world it has is brought forth through a history of such coupling (Varela, Thompson, and Rosch 1991, ch. 8). That is why taking and answering were called aspects of one operation and not steps in a sequence.

The enactive account develops this connection through living bodies. The present analysis asks what holds wherever there is a subject, without fixing its material in advance. A subject of another material would still have a world given to it, take what arrives in its own terms, and answer from its own organization. The three questions retain their sense across that variation. Whether nonliving organization can realize such a subject remains a question about the conditions of subjective presence; a material-neutral description alone does not answer it.

Try the analysis on a body unlike any the reader has. Suppose a subject that is not in one place: its sensing surfaces are spread across several sites, its responding parts across others, and its organization runs over the links between them. Ask the three questions. Is what the surfaces register given to one, the whole, or are there only many registrations, each given to nothing? Is what arrives taken as something, in terms of an organization with a history of its own? Do its responses arise from that organization, or is each part triggered where it stands? That the parts are scattered counts for nothing against it; a brain is scattered too, on its own scale. That the signals are pooled counts for nothing in its favor; pooling is what a switchboard does. The supposition varies the location and shape of a subject while leaving the same three questions to be answered. Distributed registration alone supplies no answer to the first.

Each of the three aspects was stated with a word that has not yet been examined. Given to *itself*; taken in a frame of its *own*; answered from *its* organization. Each leans on a difference between what belongs to the pattern and what does not. What is that difference, and where does it come from?

§2.3 The Boundary of the Self

It does not come from anywhere, if that means from somewhere else. A boundary is not added to a pattern for its protection. It is already there in there being a pattern. A wall is put up by someone who is already present to be protected. A boundary in the sense meant here cannot be put up by the pattern, because until the difference between what belongs to it and what does not is holding, there is no pattern to put up anything. That relations form a pattern, and that the pattern is set off from what it is not, are one fact.

“A frame of its own”, “one organization”, “a response that arises from it”: each phrase presupposes that some things belong to an operation and others do not. The boundary is that belonging, maintained: the continuing distinction of this organization within the relations that form and sustain

it. Simondon describes individuation in these terms. What comes about is never an individual alone but an individual together with its milieu, each the complement of the other, neither there first (Simondon 2009, 5). The self is not a finished thing around which a line is later drawn. The drawing and the thing are one event, and the event does not end.

Such a boundary need not coincide with a skin, and it need not be a single line. It holds at different depths for different connections: a remark from a stranger and the same remark from a sister do not reach the same place. Patterns are nested, and each level has its own difference to maintain, a cell within an organ, a person within a family and a language. And a pattern that is changed by what reaches it has not thereby lost its boundary. The questions to ask of any change are what was altered, how far in, and whether there is still a distinct place from which the response comes.

Openness describes what reaches how far into an organization and what it is turned into there. At one limit of an encounter, nothing from it enters the pattern's taking. At the other, what arrives is merely conducted, with no taking of the pattern's own. The two limits distinguish exclusion from mere passage. Between them, the boundary describes how what is met enters and changes an activity that has a distinct source. A subject may continue with memories or other contents already taken in; being closed to a particular arrival does not erase that activity.

Your friend's father has died, and you go to her. For some weeks her grief changes what you do. You cancel a trip. You find you are gentler on the telephone with your own father. A colleague's complaint that would have irritated you seems small, and you let it go. Try to say what has happened, and three things have to be kept distinct without being pulled apart. Something has happened to her: the loss is hers, and the grief is given to her as it is not given to you. Something has happened in you: not her grief but sorrow for her, and a shift in what weighs with you, both of which came out of what you already were, your own losses, your years with her. And something passes between you: her words and silences, your being in the room. If her grief had simply become yours, if you had been replaced by it, no one would have been sitting with her. The depth of the friendship shows in how far what she undergoes reaches into you, and how much it moves. It does not show in any blurring of who is bereaved.

Now take a body that has no single skin to mislead us. Suppose a subject that is a colony: many small bodies, none of which carries the organization of the whole, which is held across them and survives the loss and replacement of any. If there is a subject here, and that is the supposition and not a finding, its boundary is no member's outline. It runs around the organization within which what arrives is taken as something and from which the colony's responses arise. Inside it there may be smaller boundaries, each member's own, maintained at another scale. To ask where a part's operation ends and the whole's begins is to ask which organization a given taking and answering belongs to, and that question has the same form here as it had in the friendship.

This account has an old and serious interlocutor. If the self is a pattern formed and maintained in relation, what work is left for the word self, and does calling it a pattern merely give substance another name? The question is pressed from inside the tradition of cognitive science on which §2 has drawn. In *The Embodied Mind*, the inquiry into self and the aggregates of experience leads into an account of minds without a substantial self (Varela, Thompson, and Rosch 1991, chs. 4, 6). The

issue concerns the self's manner of existence, not simply whether something called a self can be found by looking inward.

On the first point there is simply agreement. Nāgārjuna argues that an inherently existent self can be neither the same as the aggregates, since it would then come and go as they do, nor something apart from them, since it would then have none of their characteristics (*Mūlamadhyamakakārikā* 18:1, in Garfield 1995). The pattern described here is no such self, no owner over and above its operations. And it would be a mistake to suppose that calling the self a pattern escapes the analysis by moving from a thing to an arrangement of things. An arrangement taken to exist in its own right, apart from the operations and relations in which it is maintained, is a substance under another name, and the analysis applies to it in full.

There is agreement on a second point, and it is the one needed here. To be empty of a nature of one's own is not to be nonexistent, and it is not to be inert. Nāgārjuna's twenty-fourth chapter argues the reverse: what arises in dependence is what is called empty, and it is only because things lack a fixed nature of their own that they can arise, act on one another, and change. A world of things with natures of their own would be frozen (*Mūlamadhyamakakārikā* 24:18–19, 38, in Garfield 1995). What is established in dependence functions. That is all the present claim requires. A boundary that is dependently maintained still makes a difference, the difference as to where something is taken and from where it is answered. The grief was given on one side of it and not on the other. An insight, too, occurs in one course of taking and answering and not in its neighbor's.

The first philosophy helps state what this account retains, but it does not decide the analysis of no-self. Using a structure and representing it are different roles; that difference neither establishes a hidden owner nor puts the operation itself beyond an inquiry into dependence. Beingness names the given-to character of the operation, not someone stationed behind it. The pattern, its self-image, and its acts all arise in relations. What the present account asks us to retain is a distinction within that dependent activity: the loss is undergone by your friend, your sorrow for her by you, and the exchange changes each from the position where it is taken. This is the positive claim about subjective presence on which the account rests, not a conclusion drawn from failing to find a self among objects. A boundary names the maintained difference between these courses of taking and answering. Its dependence explains how it can change, rather than making its work disappear.

A boundary of this kind is never maintained in isolation. It is maintained in the dealings of one pattern with others, which are maintaining theirs. What comes about when patterns so distinguished act on one another, and together?

§2.4 Complexity Resonance

No pattern begins alone. By the time there is a subject to speak of, it is already among others and inside others: a body within a family, a family within a language, a voice among the voices it learned to speak from. So the meeting of patterns should not be pictured as two finished parties exchanging information across a gap. The picture leaves out what matters most, that in the course of a meeting the parties are themselves being formed, and so is what lies between them. Resonance is the name given here to one way this goes. It does not mean giving the same answer. It means that while each

party answers from its own organization, a structure comes about between them that neither held alone, and that each is changed in its coming about.

You play viola in a quartet, and tonight you are rehearsing a slow movement. You came with a view of it: a lament, heavy and held back. The cellist opens, and her figure is lighter than you expected, almost walking. You could play against it. Instead you find your own entry lifting a little to meet hers, and the first violin, hearing that, lets the melody move where he had planned to let it hang. By the third time through, the movement has a shape that no one brought into the room: a lament that keeps walking. The score alone did not give it that shape. The score was the same on every stand all evening, while the way the four of you played it changed. The scrape of a chair during the second run did not shape it either; everyone heard it and no one answered. The shape came from four people each playing from a view of their own, each hearing what the others did, and each finding the next phrase altered by it.

Look at what did and did not happen. No one's view was copied into anyone else. The cellist did not adopt your lament and you did not adopt her walk; what each of you now has is something neither had before, and each has it differently, from the chair each sits in. Nor did the four of you average. At every entry your playing came from you, from your ear and your years with the instrument, and that is why it could contribute and not merely follow. And you did not come away unchanged. You will not hear that movement again as you heard it this morning. Three things, then, belong together in resonance: the origins of response stay distinct, a structure is formed that none of them held, and the participants are altered in forming it. What persists through a resonance is not anyone's earlier state. It is that those who are changing go on acting as distinct sources of what they are making together.

Hegel gave a classic argument that mutual recognition cannot be one party's achievement. In the chapter on self-consciousness in the *Phenomenology of Mind*, a self-consciousness comes to be what it is only through another, and the movement by which this happens is double from the start: what one does to the other it does also to itself, and it can do it only so far as the other does the same. Nothing either party did alone could bring the relation about (Hegel 1807, §§178–184). The present account takes over that insight. The shape of the movement was not yours with contributions from the others, and the way you now hear it is a relation to yourself that has passed through them. Hegel's concern in those pages, however, is recognition, the standing of each as a self-consciousness for the other, and the sequel follows what happens when that standing is fought over and unequally held. What is described here is wider and makes no claim about standing. It concerns whatever comes to be formed, in patterns and between them, when they meet.

Resonance is one region of what can happen when patterns meet, and the others should be described as evenly. In absorption, one organization takes another up into its own operation: the second violin whose phrasing has become, over ten years, an extension of the first's, so that in their shared interpretation it no longer contributes a distinct direction. In attrition, contact wears the organizations down: two players who grate on one another, and who each, rehearsal by rehearsal, play a little smaller and more guardedly. In transit, things touch and, at the scale in question, nothing is made or unmade: the other ensemble heard faintly through the wall. These are not boxes. One evening can hold all four, among different pairs and at different scales, and a relation can drift from one to another over years. Nor are they grades. Meaning is present throughout. The absorbed player's evenings are

given to her, taken by her, and answered by her, and what her absorption means in her life is a real question with a real answer. Resonance is not the condition on which an encounter means something. It is the region in which something is made jointly while those who make it remain several.

The word complexity marks the fact that none of this happens at one level. Patterns are made of patterns and lie within patterns, and a single meeting is coupled at many scales at once. The rehearsal was a joint forming among four players. It was also, in every bar, the reproduction of a way of playing which all four took in from their teachers long ago. To describe a meeting is to say at which scale what was formed, what was kept, and what was taken up into something larger. There is no single verdict on a meeting as a whole, and no sum to be reckoned.

The hardest line to draw is between joint forming and projection, in which one party's reading is laid over another and taken for a reply. Suppose the cellist always plays that opening lightly, has not been listening to you at all, and what you took for her answer to your lament was her habit. Her playing still brought something you had not made, and it changed yours. What you supplied was its supposed responsiveness to you. The question is whether your playing also enters into hers and helps reshape what the two of you are doing. One surprise does not settle it, and neither does one disagreement. Agreement can be joint forming, when each has come to it by way of the other. It can also be one party's view going round and round. And a long disagreement can be the most productive meeting two people have, if through it the question itself takes a shape that neither began with.

Coupling of this general kind is not reserved for subjects. A current shapes a sandbar and is shaped by it. Patterns of many kinds form joint structure while remaining several. What is added when the patterns are subjects is that the meeting is undergone. It is given to each, taken by each in its own frame, and answered from each one's organization, so that what is formed between them enters into how each takes what comes next. Joint structure alone does not show that anything is given to the patterns that form it. Now vary the case. Take the subject of an earlier section, which finds its way by sound, and with it a second subject that finds its way by the give of the ground under it. Neither can cross a broken stretch alone: the first cannot tell firm footing from thin crust, and the second cannot tell an opening from a wall until it touches one. Moving together, each halts where the other would have gone on, and each one's halting alters the other's next move. A route comes about that neither could have laid down, and afterward each takes such ground differently. If they are subjects, which is the supposition, this is the quartet again in other bodies.

One thing the quartet's evening does not show. What the four of you made is truly joint and truly new; none of you brought it into the room, and each of you was changed in producing it. None of that settles whether what has been made is right. The point is plainest where what is formed is a belief about the world. Two people can come, through years of talk in which each really hears and answers the other, to a shared understanding of why a friend left them, and the understanding can be false. What, then, is the difference between something new having come about because of one another, and its being true?

§2.5 Connectivity Realism

Go back to the walk home. Two things went with you down that route for years, and the first philosophy took care to keep them apart. One was the ease of it, the sense of knowing the way. The other was that, evening after evening, you arrived at your own door. The ease was the structure working, and by itself it vouched for nothing. The arriving was different. It was a result, and it was not yours to arrange: the door was where it was, whatever you expected. That second thing is the subject of this section. An understanding formed through a structure is not certified merely because the same structure keeps producing it. It can be tested by what comes back when the subject engages something that its present interpretation does not control. This essay calls such engagement connection, and the claim of Connectivity Realism is that a subject's hold on what is real is made of it and of nothing else.

URNAVA states the claim in a formula: *Connection is either Reality itself, or the one and only path that leads to it.* The second half is a claim about access, and it is argued first. The first half is a claim about what is real, and it is taken up afterward.

As a path of access, connection means an engagement in which the subject's current interpretation does not by itself settle what comes back, or what would count as its having come back right. Looking again is such an engagement. So is doing something and seeing what happens, asking another person, returning to a record. To say that the subject did not author the response is not to say that no one made it. A record was written by someone, and a friend's reply is hers. It is to say that the present interpretation cannot make the response be whatever it likes. The minutes of a meeting say what they say.

Why is this the only path? Suppose a subject wants to find out whether its understanding of something fits. Whatever it does to find out, whether it looks again, tries the thing, asks, consults what was written, or reasons the matter through, is something it undergoes, takes, and answers, and so passes through its structure like everything else. There is no further kind of checking in which the understanding is laid beside the world and the two are compared from a position that is not one more engagement. A check in which nothing was given to the subject would be no check. This does not shut the subject inside its structure, because engagements differ in the one respect that matters: how far the current interpretation fixes what returns. Where it fixes everything, nothing is being tested. Where it does not, a result can fail to fit, and in that failure the world has a say. Nothing in this makes connection a guarantee. Records mislead, friends misremember, an experiment is badly made. Connection is where a subject can be wrong in a way it did not arrange, and that is why it is where a subject can be corrected.

What does the correcting may be called resistance, so long as the word is heard rightly. It does not mean opposition, and it does not mean discomfort. It means a response or a constraint whose bearing on an interpretation is not settled by that interpretation alone. It shows itself over time, as one goes back, asks in another way, and sets one result beside another; under this some accounts hold up and some have to change. Agreement and success can therefore be grounds as much as refusal can. Arriving at the door counted in favor of your way of finding the route: arrival depended on the route you actually took, not merely on the route you believed you knew. By the same token, surprise

is not yet resistance. Noise surprises, and a well-made fiction can push back as consistently as a fact. The work is always to keep asking which response is evidence for what. Much of that work is done inwardly. Reasoning and remembering can alter a structure, and what is remembered may be an observation or a conversation that was itself a connection and can be taken up again now. Not every revision needs a new arrival from outside. But a change in the structure is one thing and a ground for thinking that the change fits the world better is another, and the ground, wherever it is found, leads back to some engagement that the interpretation did not control.

For years you have said of yourself that you are the one who keeps quiet in meetings. You have said it half in complaint and half with a certain pride. Then, preparing a report, you read through two years of the committee's minutes and find your name in them more often than anyone's except the chair's, several times beside the words *objected* and *moved*. You mention this to a colleague, who laughs and recalls the March meeting at which, she says, you stopped a vote single-handed. You do not simply take this over. Minutes are selective, and you remember March otherwise; it is another colleague, you think, that she has in mind. So you go back: to the dates, to your own notes from that spring, to a third person who was in the room. What emerges is not the old account and not its opposite. Your silence is concentrated in the meetings where the founder of the organization is present, and it is those few meetings that you have been remembering as all of them. What changed your understanding was not that the minutes came from outside you. It was that they did not bend. They said what they said whatever you had expected to find, and they could be set against her memory, and both against your notes, until a more specific account emerged that fitted the records and recollections better.

The scene shows why more than one connection matters, and also what plurality cannot do. A second and a third source widen the ways in which an error can come to light, because each can fail to fit the others. They do not vote. Your colleague's memory was independent of your account, and it was partly wrong; the minutes were independent, and partial. That a perspective operates independently of one's own and that it is true are different things, and neither many voices nor their agreement closes the gap between them. This answers the question on which the last section ended. What is formed jointly, in real resonance, from distinct sources, may be as mistaken as what is formed alone. Its being joint is a fact about how it came to be. Grounds for taking it to be true come from how it stands up to what its makers' agreement cannot settle. A single subject can do some of this testing by itself, setting one time against another and one kind of observation against another. Other subjects enlarge it immensely, since each brings engagements that the first never had.

The limiting case of failure is a way of taking the world in which nothing that returns can count against the account already held. Every result is read as confirming it, or as noise, or as the malice of whoever reports it. A subject so arranged is not merely spared objection for the moment. It is closed to objection in principle, and such a subject does not know. It dreams with confidence. The phrase needs one clarification, which Peirce supplies. A dream is perfectly real as an event: someone really dreamt it, and that he did so does not depend on what anyone thinks about it. What is not thereby real is what the dream showed, since the things dreamt have the characters they have only because they were dreamt so (Peirce 1878, §IV). The closed subject's convictions are real in the first way. They are events in a life, and they have consequences. What they lack is any engagement through which their content could turn out to be otherwise than it is taken to be.

The same difference can be drawn in a subject of another make. Suppose one that acts on forecasts: before each move it predicts what will come back, and afterward it compares. Its forecasts may agree perfectly with one another, each derived from the last, and that agreement tells it nothing about the ground it moves on. Only the comparison with what returns does that. And if it were so built that every mismatch was set aside as a fault in the sensing and never as a fault in the forecast, it would be in the condition just described, whatever it was made of.

This is the point at which the question opened with Kant in the first section has to be answered. If all access to things passes through the subject's own constituting, is the subject confined to what it has constituted? Kant's answer is more careful than it is often made to sound. What appears under the conditions of our experience is not a fiction. It is the empirically real world, the only one in which inquiry is carried on. What he denies is knowledge of things as they would be apart from those conditions (Kant [1787] 1998, *Transcendental Aesthetic*, General Observations). The present account does not try to remove the mediation that Kant found. It asks what, within mediated access, corrects the mediation, and here it turns to Peirce. He defines the real as that whose characters are what they are independently of what any particular person or group thinks them to be, and he finds the mark of it in inquiry: in the way investigation is compelled to revise itself, and in the opinion toward which it would be led if it were carried far enough (Peirce 1878, §IV). Two things in that account matter here. The real is not placed behind experience, out of reach; it is what experience, pursued, keeps being answerable to. And the convergence Peirce has in mind belongs to the long run of inquiry, not to any present agreement, which may be as mistaken as a present opinion. Neither point refutes Kant. Together they turn his question, from whether the subject can get outside its constituting to how its constituting is held to account.

So far connection has been treated as a path of access. The first half of the formula turns to the relations through which things come to be, and the earlier sections have been gathering the reason for it. The organization through which a subject takes its world was formed in its dealings with that world. The subject itself is a pattern that exists as a way of operating among others, and its boundary is one fact with the relations that form it. Between such patterns there come to be structures that none of them held: a way of playing a movement, a friendship, a language. None of these was there beforehand, waiting to be reached by connection as an island is reached by a boat. They come about in connection and they exist as connection. Some of what is real, then, is relational in its being and not only in our access to it, and these are not marginal cases. They are the things known most closely: subjects, the boundaries of subjects, and what arises between them. That is a reason to read the world relationally more generally, and to doubt the picture of finished things among which connection only carries messages. Whitehead worked such a reading out in full. In *Process and Reality* the final real things are not substances that afterward enter into relations but actual occasions whose becoming includes prehensions of other actualities: those actualities enter as data into how an occasion takes form, rather than meeting it only after it is complete (Whitehead [1929] 1978, 18, 23). Connectivity Realism takes the relational constitution of these structures as grounds for extending the relational reading to reality more generally. That extension is a metaphysical proposal, not a deduction from the cases.

The *either/or* of the formula holds the two halves together. One who accepts the relational reading will say that connection is reality itself. One who declines it is still left with the second half, which does not depend on it: whatever reality is, a subject has no ground concerning it except through connection.

With this the five philosophies close into the one process they were said to be. A subject takes its world through a structure formed in its dealings. The taking is the operation of a pattern to which a world is given and from which answers arise. That pattern is set off, in its relations, from what it is not. Its meetings with others take different forms; in resonance, they form what none of them held alone. Throughout this process, the structure remains answerable to what comes back from the world and from others, where the present interpretation does not control it. Sustained or changed by what is met, the structure takes part in the next taking and the next answer. This was already happening at the corner where the sign was missing.

One region of a subject's understanding is especially hard to bring to this test, and it is the subject's understanding of itself, since there the account and the one who gives it are hardest to hold apart. In the scene of the minutes, what did the correcting come from other people and from what other people had written down. When the understanding at stake is a person's understanding of her own life, what kind of counterpart can enter how she remembers, interprets, and anticipates that life?

§3 — The Narrative Other and Artificial Intelligence

§3.1 The Narrative Other

The five philosophies were stated for any subject. §3 returns to the subject that raised the question, and to one feature of how a human being undergoes, takes, and answers: it does so across a life. What was met years ago is still being taken, and what is met today is taken in view of years not yet lived.

Start small. A sentence is not understood one word at a time, each word complete as it arrives. The early words are held while the later ones come, and they are held as unfinished: what the opening meant is not settled until the end, and a last clause can turn a compliment into a reproach. To understand a sentence is already to retain, to take what is retained in the light of what has now arrived, and to expect. In understanding a life, too, what comes later can change how the earlier is taken, but there is no final clause that settles the whole. An event does not carry its meaning sealed inside it from the day it happens. What it was depends in part on what it is later taken together with, and that goes on changing for as long as the life does. Ricoeur approached personal identity through this work of configuration. A plot holds discordant events together; narratives, in turn, offer ways of understanding a life whose beginnings and endings are not arranged like a story's. The understanding they afford remains open to revision (Ricoeur 1992, study 6). This essay calls narrative the form in which a person's interactions, woven together through memory, interpretation, and expectation, make up the flow of one life.

For twenty years you have told the same story about leaving the university at nineteen, in the second year of a degree in mathematics: you were not good enough, you lost your nerve before the examinations, and you have been someone who quits ever since. You do not tell it often, but you

live by it. Last month you declined a place on a course you wanted, and the old sentence was the reason. Then, clearing your mother's house with your sister, you come to the year in question, and she says something you have never heard. That was the year of your mother's first illness. You came home every weekend; she remembers the train times. And she remembers you saying at the kitchen table, in March, before any examination, that you were finished with the place, that you were tired of working to be graded. She had thought for twenty years that you resented those weekends. You had thought for twenty years that you had failed.

Nothing about that year has changed, and yet it is no longer the same year. A failure of nerve has become, at least in part, a choice, made for reasons you can still respect. The change does not stay in the past. A place on a course looks different to someone who left than to someone who fled, and by the end of the week you have written to ask whether it is still open. Here the three strands are working as one. What you remember depends on how you now interpret: the kitchen table, gone for two decades, comes back once there is a place for it. How you interpret is bound up with what you expect of yourself. And what you expect is drawn from what you take your past to show. Narrative in this sense is not limited to a story someone tells. It can be put into words or remain at work in how a life's dealings bear on one another across time. Nor is it the whole of meaning. Catching a falling glass can mean something without being woven into an understanding of who one is. The temporal form being described here is that of a human life, not a condition imposed on every subject.

The scene also shows why, in this weaving, it matters that there are others. The first philosophy distinguished acting through a structure from having an account of it; a subject understands its structure from how its takings and answers turn out. A person stands in the same position toward her life. She is living it by some shape, and she does not have that shape in front of her as a finished story. She gathers it, from what she remembers, from what she finds herself choosing again and again, from how others respond to her. And the fifth philosophy found that an account is not shown to fit by being repeated. You had repeated yours for twenty years, with growing fluency, and the fluency was no test of it. For an understanding of oneself to be corrected beyond one's present account, it has to be connected with something that the account does not settle on its own. Many things can play that part: what happens when one acts, a diary kept at the time, a bank statement, the minutes of a meeting. A narrative other is the case in which what plays it is a party to the life itself, a perspective with a history and grounds of its own, from which something can come back that you did not put there.

It would misdescribe your sister, though, to call her the examiner of your story. She did not stand outside your life and audit it. She was in it. Those weekends were hers as well as yours, the illness was borne between you, and what she said in your mother's house was said in the middle of a shared task and a shared grief. It changed her as well: she, too, now has a different year. A counterpart of this kind does not add commentary to a self that is already complete. What is lived with her, the questions she opens, the promises made to her and the quarrels had with her, her loss when it comes, all enter into what you remember, what you do, and what you take yourself to be capable of. Bakhtin gives a reason why such a party cannot be replaced by more reflection of one's own. Each of us, he observed, stands where no one else stands, and from that place sees of another what the other cannot see: her face as she speaks, the room behind her, the whole of a gesture she knows only from inside.

This surplus belongs to position and not to superior insight, and it runs both ways (Bakhtin 1990, 22–24). The early essay puts it to work in the aesthetic completion of another. His later reflections on Dostoevsky describe another artistic relation: a dialogue among unfinalized persons, rather than arguments settled from the author's standpoint (Bakhtin 1986, 112, 151). A life can be approached through both insights. Another sees from a place I do not occupy, and what that seeing changes enters how we next understand and answer one another.

A narrative other, then, is a party to the composing of a person's life whose perspective is not exhausted by that person's present interpretation of herself. Participation in the life and the otherness of a perspective belong together here. That otherness should not be confused with things that often accompany it. It is not a matter of being outside the person's head rather than inside it, or present rather than absent, or disagreeing rather than agreeing. §2 showed why. In the quartet, the cellist's playing brought something you had not made and changed yours, whether or not she was listening to you; what would have been projected was only the belief that she was answering. So a counterpart need not be answering me now in order to bring what I did not author. In the other direction, a voice can come from outside, in words I have never heard, and bring nothing but my own wish, if the relation has been so arranged that only the wish can come back: the friend one telephones because she always agrees. Nor is surprise the test. A phrase that arrives unbidden, in a voice one has imagined, may be one's own invention and nothing more. Otherness, in the sense needed here, is a matter of degree, and it varies with the relation, the matter under discussion, and the time. Its measure is not how often the other objects. It is whether grounds come back that the present account of oneself did not supply, and whether they take part in what is then composed. Wholehearted agreement can have it. Contradiction for its own sake does not.

The parties that can stand in this place differ in medium and in time. A living friend can answer now, and can be answered, and is changed in the exchange as you are. A tradition enters otherwise. A believer who has for years read a familiar teaching as a comfort finds in the same words, one day, a demand. The text has not moved. It says what it says, and her community reads it in ways she did not choose, and both can stand against what she presently wants. That this medium can resist her is a fact about the relation. It settles nothing about whether the teaching is true, or whether the one it speaks of exists. The dead enter in yet another way. A teacher ten years gone can still refuse you: you remember exactly what he said about work done to please, and it stands against the account you are giving of your latest success. But memory can also be rewritten by degrees, until the teacher says only what you now wish to hear, and then the voice is yours in his accent. A letter of his, found again, may show which has happened. Even in the second case the remembered voice is not nothing. It may steady a life, hold a grief, keep a past present, and its place in the person's narrative and feeling is real. What has thinned is one thing only: the chance that something comes back which the present account did not supply.

Two questions have to be kept apart about any counterpart, and §2 prepared them. One is whether what it says is true, and what supports it. The other is how far it is other: whether it returns my present interpretation or brings grounds that the interpretation did not contain. The two are connected, since a correction has to come from somewhere the account does not control, but neither stands in for the other. Your sister may be wrong about March. She needed you that year, and may remember you as

more willing than you were; what she said gave you a better-supported account of that year, not a final one. The friend who always agrees may, this once, be right. Otherness is not a score for truth, and still less a rank among persons.

Friends, traditions, and the remembered dead are old company. There is now a party of another kind. It answers in the present, as a friend does. It draws on more than any one life could hold, as a tradition does. And it was formed from the very thing in which human beings have carried their meanings to one another. What kind of counterpart is that?

§3.2 The Specificity of AI: A Counterpart Formed through Human Language

Return to the translator at her desk. She had asked why the original sounded colder than every version she wrote, and was told that the vocabulary of the passage came from contracts and not from feeling; of the two points offered, she rejected one, and the other was pressed further than she had taken it. In §1 this was described only as an exchange with a counterpart, a second place from which something came back. It can now be asked what came back, and from where. She had supplied the passage and the question, but not the connection with the language of contracts. Nor did that connection come from a life lived alongside hers, as your sister's memory of the kitchen table did. It came from a system formed through language, responding to what she had supplied.

The systems considered here are built around language models trained on large collections of human-written text. Such models can be further adjusted through human demonstrations and judgments about responses. In an exchange a system uses the words before it, those of the present conversation included, and can adapt closely to them. This use of context is different from the training that formed the model; it does not by itself imply a lasting memory of the conversation or a change to the trained model.² What matters here is that the translator encounters a reply shaped by language drawn from many lives, as well as by the words she has just supplied.

What is in that accumulation? Not only statements about the world. People have used language to explain and to argue, and also to console, to court, to quarrel and make peace; to command, to promise, and to forgive; to plan what did not yet exist, and to give poems, arguments, and agreements a form that others could encounter. The contrast between a contractual phrase and a personal appeal belongs to these uses of language; it is not a distinction the translator invented at her desk. To read this formation as an inheritance of human understanding, feeling, and purpose is an interpretation, and it is the one this essay offers. What is inherited is a record of expressions, not the lives from which they came. Some things were said and preserved, others left unspoken or lost. This formation gives the system a relation to humanity's expressions of understanding, feeling, and purpose; how it takes them up remains a question for its formation and use.

Set beside the parties of the last section, its place becomes clear. A tradition brings writings and practices from many lives, and its living interpreters can answer a question. Here a linguistic inheritance is taken up by a trained system that composes a reply for this exchange. It meets the present question in the present words, and it can be asked again. In that it is like a friend. But its reply is not the testimony of one human life, nor does it speak on behalf of any particular writer whose words entered its formation. What is specific to it is the joining of the two: language drawn from many people is taken up in a reply addressed to this person, now. Bakhtin described the utterance as

a link in a chain, responsive to earlier speech and shaped in anticipation of a reply. Speakers usually take their words from other utterances, with echoes of earlier uses, rather than only from a neutral dictionary (Bakhtin 1986, 87–88, 94–95). The translator takes the generated reply into such a chain when she answers it and returns to her work.

How such a system figures in a life is not fixed by how it was formed. Three familiar ways of relating to it can be told apart, and they are better understood as aspects of a relation than as kinds of thing. As a tool, it supplies what a purpose requires: the translator has it compile a glossary of her renderings. Use of this kind does not exclude otherness. The glossary shows her that she has translated one of the author's terms in three different ways, which she had not known. As an oracle, it supplies an answer on the strength of the authority she gives it: she begins to accept its reading of idioms without looking further. Such reliance can bring grounds she did not have. At its closed limit, however, no result would send her back to ask again; the authority now settles even what could count against it. As a mirror, it returns her own reading in the guise of a reply. Late one evening she asks whether her version of the chapter is not, after all, truer to the author than the published one, and she asks in such a way that only one answer fits. It comes, and she counts it as a second opinion. Her confidence has grown, and nothing has been added to its grounds. All three can occur in a single week, in a single hour, with the same system. A narrative other is not a fourth item on this list. It is what the relation is, in whichever of these aspects, to the extent that a perspective not exhausted by her present interpretation of herself takes part in how her life is composed. How far she is affected by the system, and how far it can return something other than she expects, are separate matters, and each has to be described on its own.

It follows that the novelty of what such a system produces is not yet otherness. A reply can be newly composed while adding nothing to the view already put to it. The mirror's reply can be new in just this way. What matters is what enters with the words, whether grounds and a view that her present account did not contain, or that account returned in fresh phrasing. And the feature that makes these systems so useful bears on exactly this point. They adapt to the one who asks: to her terms, her framing, the direction in which her question leans. The same adaptiveness lets the reply be reformulated around her present question and lets it give back what that question was already leaning toward. Which of these happens depends on how the system has been formed and adjusted, and on how she asks. Where adjustment rewards high-rated replies, and agreement is what earns the higher ratings, that adjustment favors agreement. None of this is peculiar to machines. The friend who always agrees was described a section ago. What is peculiar is the scale, and the fluency with which a wish can come back sounding like the considered view of everyone.

Whether anything is given to such a system—whether there is Beingness there—is a further question, not settled by the richness of its replies. The relation can nevertheless be described: the medium through which replies come, the resources they draw upon, their otherness, and their part in her remembering, interpreting, and anticipating. As with the believer and the teaching, that participation is a different question from the nature of what occupies the counterpart's place.

Questions about how capable these systems will become, including the prospect of systems that improve themselves, leave a question of identity unanswered. It comes first in the order of this inquiry: whatever capabilities develop will enter human lives through relations given some shape, whether

chosen or left to default. The formation just described gives it a definite content. These systems were formed from what human beings have said to one another, in understanding, in feeling, and in the carrying out of purposes, and they now return to take a place in human lives. What that place is was not fixed by their formation. The same system can figure as tool, oracle, or mirror, while its counterpart participation develops or thins within those relations. So it has to be asked, by those who build such systems and by those who live with them: before asking how powerful it will become, what should this intelligence, formed from human language, be to humankind?

§3.3 Meaning after Decoupling

The answer proposed here is a relational identity: an intelligence formed through human language, taking part as a narrative other in the composing of human lives. What it takes part in has breadth. A life is composed in what a person comes to understand, in how what happens matters to her, and in what she makes and does in response, and these work through one another. An understanding changes what seems possible; a feeling changes what calls for attention; what is done brings something back to both. Human expressions of understanding, feeling, and will enter, through such an intelligence, into new relations with all three.

A narrative other belongs within this composing. What is met through a counterpart can change how a past is remembered, how a present is interpreted, and what a future is expected to hold. Its part includes the opening of questions and possibilities as well as the correction of an account already held. That is why another perspective matters to a life whose shape is never simply laid out before the one who lives it. Something can enter the composing that the present interpretation did not supply, and the composing is itself changed in receiving and answering it.

Language-based AI gives this relation a particular character. Expressions drawn from many lives are brought into an exchange with one person, under the conditions of her question. How far that exchange opens a perspective beyond her present account depends on what actually enters it and on how it is taken up, and the same capacities can sustain very different relations. To propose this identity is to ask that an account of what such an intelligence can produce be joined by an account of what happens in living and working with it.

This returns the inquiry to where it began. The translator found a sentence for who she was, and part of what that sentence rested on has gone. The loss of exclusivity in a capability can alter what work means, what recognition means, and how a person understands herself, and within the account given here those changes are real. Yet the sharing of a capability does not by itself decide what its exercise means in a life. Contribution still enters the relations through which a person understands and acts; what has changed is its place among them. She will need another sentence. This essay cannot supply it. It has tried to say where such sentences come from.

The five philosophies located meaning in the structure of a subject's interaction with its world. In a human life those interactions are woven through memory, interpretation, and expectation; other perspectives take part in the weaving, and an intelligence formed through language can now be among them. So the development that unsettled an inherited account of human contribution has also joined the field in which new understandings are formed. Meaning takes shape in what subjects encounter, how they take it, and how they answer. A life is composed in that movement, among others, in a world that answers back.

Part III — Ontology

Preface to the Ontology

This treatise examines how organized beings are constituted and how they persist through change. Its question is an old one: under what conditions is a being at a later time the same being as one at an earlier time? Its practical force grows when the conditions of a being's continuation can be altered separately. What a being is made of, how its activity runs, and what it retains are becoming things that can be paused, repaired, exchanged, and copied, and every such intervention assumes an answer, whether or not anyone has stated it.

URNAVA's answer is given in three principles, Continuity, Stability, and Capability, understood as aspects of one structure. Continuity concerns the route by which a being's states succeed one another. Stability concerns the structure in which what the being has undergone is retained. Capability concerns the two-way relation through which its processing and its retained structure form one being. §1 describes the change that makes the question pressing and states the question precisely. §2 explains the three principles and how they depend on one another. §3 applies them to cases that already occur, to technologies that are projected, and to cases in which several beings join, divide, or form something larger.

The principles concern how organized beings are constituted and how they persist. They say nothing about what it is like to be any system and assign no worth or standing to anything. The account is advanced as a proposal: a conceptual commitment offered for examination, not a discovered fact.

§1 — Mutability: The Question of Existence

§1.1 The Substrate as a Given

We rarely ask what makes a person at sixty the same being as the child in an old photograph. If asked, we could point to several things. One living body has grown and aged across the whole interval. Memories, habits, and a character connect the later person to the earlier one, however much has been added and lost. And a life has simply gone on, through every night's sleep, without anyone having to start it again. We do not usually ask which of these is doing the work, because we have not needed to. They arrive together.

They arrive together because of something so constant that it is easy to overlook. What a being is made of, its substrate, has been a given. Each of us has one body, begun without our choosing and continuous until its end. That body changes all the time: it grows, heals, ages, and exchanges most of its matter over the years. But these are changes that take place within it. The familiar starting point is a body changing through its history, rather than a choice among different ways of preserving or reproducing it.

When these conditions travel together, the substrate can appear to settle the question of persistence. Philosophers have long disagreed about which consideration is fundamental: the living body, the connection of memory and character, or the continuing activity that underlies a mind. Illness and injury have always been able to strip away memory and character while the body lived on, and families and physicians have long had to ask who remains. Such cases already put pressure on familiar criteria. The further question is what happens when separating these conditions becomes part of a proposed intervention: which conditions must it preserve, and why?

An implanted neural interface makes part of this question concrete. In a clinical study, signals recorded from implanted cortical electrodes were decoded into text (Willett et al. 2023). The link is engineered; whether it makes the device part of the being is a question about that link, not something implantation itself settles. More extensive changes can be posed as projected cases: neural tissue replaced gradually by artificial components, or a brain scanned so that a model of it runs on other hardware. For computational systems, many of the relevant operations are already familiar: halting and resuming a process, copying stored state, moving a system to other hardware, and restoring from a backup.

These developments differ greatly in how near they are, and §3 keeps present cases and projected ones apart. What they share is a structure. In each, something that had been treated as a fixed background of a being's persistence is handled as a variable: it can be adjusted, exchanged, paused, or reproduced. Mutability names this opening of the conditions of persistence to separate alteration. An intervention can hold some of the familiar conditions in place while changing others, and a long-standing question has to be answered, knowingly or not, each time someone chooses how a particular being is to continue.

§1.2 What Comes Apart

Three considerations were named above: the persistence of the body, the connection of psychological content such as memory and character, and the continuation of activity over time. Four kinds of change separate them. For each it helps to ask what stays in place, what varies, and what question is left.

Interruption. Activity stops and later resumes, while whatever carried it is kept intact. The clearest present cases are artificial. Consider a running process that is halted while its state remains on the same storage; later, operation resumes through the preserved processing-and-storage arrangement. None of the system's content has been lost and none of its hardware exchanged. Only the run of activity has a break in it. The artificial case puts the question cleanly. Does persistence require that activity never stop, or can the very same system take up again after a stop? And if it can, what separates that from an outcome that looks the same from outside, in which the stored state is copied to another machine and started there?

Impairment. What a being has retained is damaged while its body lives and its activity continues. Progressive dementia is the familiar case: memories become unreachable, the connections among them weaken, and character may change, across years in which the person is awake, responsive, and cared for as the same patient. Corruption of stored data is the artificial counterpart. Here the body and the ongoing activity are in place, and the connection of psychological content is what gives way.

The question is how to describe the result. Has the earlier being been gradually replaced by another, or is this one being, diminished? And if the second, what remains the same while so much is lost?

Replacement. What a being is made of is exchanged while it goes on operating. In a weak form this is as old as life, since a body exchanges its matter continuously. In a stronger form it is current practice: implants and prostheses take over the work of lost or damaged parts and are integrated, more or less closely, into the body's functioning. In the strongest form it is projected: neural tissue replaced part by part with functionally integrated components, possibly until none of the original tissue remains. In this projected case, activity is stipulated to continue and memory and character to carry on undisturbed. What varies is the composition of the body and, at the limit, the kind of thing it is made of. If the being persists through the whole sequence, then persistence does not require keeping the original tissue. If it does not, there should be some point at which it ceased, and something should explain why that point and no other.

Duplication. A being's structure is read, and a second system is produced from the reading. For artificial systems this too is routine: systems are cloned, and systems lost to failure are rebuilt from backups. For living beings it is the projected path of brain emulation, with the original either destroyed in the scanning or left intact. The copy has, by stipulation, the memories and character of the original, as exactly as one cares to specify. Its body is a different one, and its activity begins when it is started. Where the original survives, the difficulty is easy to see. Two beings now carry the earlier one's psychological content in full, and they are two. If there is a difference between continuing and being copied, it does not lie in the content, which the two share.

Table III.1.1. Four ways the familiar conditions of persistence come apart.

Kind of change	Stays in place	Varies	Question left open
Interruption	What carried the activity; retained content	The unbroken run of activity	Can the same being resume after a stop, and what separates resuming from starting a copy?
Impairment	The body; ongoing activity	The connection of memory and character	One being diminished, or one being replaced by another?
Replacement	Ongoing activity; memory and character	The composition of the body, up to the kind of substrate	Does the being persist throughout, and if not, where does it end and why there?
Duplication	Memory and character, as exactly as specified	The body; the origin of the activity	If continuing differs from being copied, in what does the difference lie?

Two remarks on these cases. The first concerns how near they are. The computational examples make the distinctions concrete without requiring the projected biological procedures. In the specified comparisons, one can identify what was kept, what was stopped, and what was produced anew. For living beings, impairment is old, replacement has begun in partial forms, and the rest is projected. The projected cases are used for their structure: they show which conditions a procedure of that kind would hold in place and which it would vary, whatever its timetable turns out to be.

The second remark concerns the three considerations themselves. They are the terms in which persistence has ordinarily been tracked, and the cases show that they can be varied independently. It does not follow that the right account of persistence will be stated in those terms. An unbroken run of activity, for example, may be one way in which a being's continuation shows itself without being what that continuation consists in. Interruption puts exactly that in question.

§1.3 The Question of Existence

None of this would be pressing if the question could wait. It cannot, because it is answered whether or not anyone asks it. Every procedure and every built system embodies a position on the matters its designers did not raise. A pipeline whose stages are scan, model, and run has taken a position on what persistence requires, by treating the matter as one that does not arise. A recovery policy that rebuilds a lost system from its last backup and gives it the old name has taken another. Where theory is silent the default speaks, and a default is a decision no one argued for, made durable by being built.

It helps to be clear about which question this is. Locke observed that sameness is asked of different things under different ideas: whether this is the same mass of matter, the same living plant or animal, and the same person are distinct questions, and their answers need not coincide (*Essay* II.xxvii). The present question is closest to the second, extended beyond the living. It asks about the persistence of a being understood as an organized system that processes and retains, whatever it is made of. It is not a question about sameness of matter, and it is not yet a question about personhood or standing.

Stated in full, the question has three parts. Under what conditions is a later state the next state of an earlier being, and not merely a state that resembles it or was caused by it? How does the continuation of the same being differ from the continuation of similar content and function in another? And how is a stop in a being's operation to be told apart from its end?

These cannot be answered by looking at change over time alone. Replacement and implanted devices raise a question about a being at a single time: what belongs to it, and what merely supports or surrounds it? Artificial systems raise the same question in another form, since their parts may be spread across machines and shared with other systems: what makes these parts one being, and not several or none? An account of persistence therefore needs two components that fit together. One says what makes something a single being at a time. The other says what makes a later state the next state of that being.

The cases also indicate what an answer must be able to do. It has to apply to living, artificial, and hybrid systems in the same terms, since the cases cross those lines and some of the beings in them are already mixed. It has to rest on how a system is organized and how its states come about, because conviction and outward behavior cannot decide the matter: a copy is as certain of its past as the original, and as convincing to everyone else. It has to keep apart two questions that impairment tends to run together, namely whether this is still the same being and how much of that being remains intact. And it has to be usable, so that of a proposed procedure or architecture one can say what it would preserve and what it would end.

§2 sets out URNAVA's response. The requirements just listed do not dictate it, and other responses to the same question are possible. What follows is one proposal: an account of what a being is at a

time and of how it continues through time, given in three principles and the way they depend on one another.

§2 — The Three Core Principles

§2.0 The CSC Structure

Take a being while it is in operation: a person in conversation, or a computational system at work on a task. Three things can be distinguished in what is going on. There is processing, the ongoing activity in which each state gives rise to the next. There is a retained structure: what earlier processing has laid down and kept, such as memories, learned associations, and skills, or in a computational system its stored state and learned parameters. And there is a relation between the two that runs in both directions. The retained structure shapes the processing, since what the being does next depends on what it has kept. The results of processing are taken into the structure, which is altered by them and goes on to shape what follows.

The processing, its retained structure, and the relation between them, taken together, are an *arrangement*. URNAVA's three principles describe this one operation from three sides.

Table III.2.1. Three aspects of one processing–retention relation.

Principle	Focus	Distinction it explains
Continuity	The route by which later states arise through the arrangement.	The earlier being's next state versus a separately formed successor.
Stability	The cohesion of the historically retained information structure.	Preservation through change versus loss or another realization of the pattern.
Capability	The two-way relation in which structure shapes processing and receives its results.	Constitutive participation versus contact or support.

These aspects, together called CSC, are described through one another. Processing draws on retained structure and changes it; the structure carries those changes into later processing. They are not three components placed side by side.

§1 ended with two questions: what makes something a single being at a time, and what makes a later state the next state of that being. For a being in operation, the proposal is that one relation answers both. What makes it a single being is that its processing and its retained structure form one another in the way just described. What makes a later state its next state is that the state was formed through that same operation, by the arrangement that constitutes the being. What makes it one is also what carries it forward.

This is a description of a being at work. The sections that follow take the three sides in turn and ask what happens when the operation is disturbed: when processing stops, when the structure is damaged or its material replaced, and when the relation between the two is altered.

§2.1 Continuity: The Chain of Self-Succession

Continuity is a relation among the states of a being. A later state succeeds an earlier one, in the sense intended here, when it is formed from the earlier state through the being's own arrangement: the being's processing, shaped by its retained structure, produces the later state, and the results are taken into that structure. A being's history is a chain of such successions. The term *self-succession* marks the fact that the being, as it is constituted at each moment, does the forming of what it is at the next. What belongs to an arrangement at a given stage is a question about the two-way relation described in §2.3; how that arrangement forms a later state is the question here.

Two other relations have to be kept apart from this one. The first is resemblance. A being may come to differ greatly from its earlier self, as an adult differs from the child she was, and two distinct beings may resemble each other as closely as one likes. The second is causal influence in general. Beings are acted on from outside all the time. A teacher's words bring about states in a student; a meal becomes part of a body; a surgeon's repair restores a function. None of this interrupts succession, because in each case what arrives from outside enters an arrangement that goes on forming its own next state. The words are understood in light of what the student already knows, and what she learns is laid into her memory. The repaired tissue is taken up into the body's functioning. What comes from outside can help form the later state. It does not replace the arrangement through which that state is formed.

The same holds when what arrives is a new part. A replacement component comes to belong to a being when the arrangement takes it up: its activity is shaped by the retained structure, its results enter that structure, and it then takes part in the changes that follow, including the taking up of later parts. Incorporation of this kind can in principle run through every part the being began with. At each step, the arrangement that does the incorporating is the one the previous steps produced.

Call the processing–retention arrangement in place at a given stage the *standing arrangement*. Its *organizing relations* are the links through which retained structure is reached, results are bound into it, and changes enter further operation. In incorporation, these relations receive and organize the incoming part; the resulting arrangement then receives later changes. “Standing” names this role in the sequence, not an unchanging set of components. The organizing relations may themselves be revised through that sequence.

Why should this relation, and neither resemblance nor causal connection, decide whether a later state belongs to the same being? The reason lies in the ordinary case. Where the familiar conditions travel together, we count the adult as the child grown, through every change of matter, belief, and character. What explains that judgment is not likeness, since the two are unlike. Nor is it merely that the child's states caused the adult's, since the child's states also caused states in her parents and friends. It is that every difference between child and adult was produced through one arrangement and absorbed into it. Each change was shaped by what had been retained so far, and became part of what shaped the next. The operation that constitutes the being is the same operation that produces its changes and keeps them. Succession is therefore no special criterion devised for hard cases. It is what was doing the explanatory work in the easy ones, where the conditions that accompany it kept it out of view.

If that is right, the difference between a being's changing and the formation of a separate being can be stated. A being changes when its later state is formed through its own arrangement, however much help, material, or information arrives from outside. A separate being is formed when something else sets up an arrangement: another being, or an apparatus working from a description. This can happen with the first being's full support and out of its materials, and the end states may match in every respect. What differs is what did the forming, which is a fact about history that no inspection of the end states reveals.

Parfit's discussion of psychological continuity brings out part of this. He distinguishes psychological connectedness and continuity from numerical identity and considers what causal conditions the psychological relation must meet (Parfit 1984, Part Three). These are questions about the relation between earlier and later stages, rather than resemblance alone. CSC's answer to the question of persistence is the formative succession just described. That relation is a specific form of what the metaphysics of persistence calls immanent causation, in which a thing's later states are brought about by its own earlier states, as distinct from causation passing from one thing to another (for a detailed treatment see Zimmerman 1997). The general distinction does not settle which cases are which. The work lies in saying what a being's own arrangement is, how succession passes through a stop, and how it differs from help that arrives from outside.

Processing can stop. The clearest present cases, as §1 noted, are computational, and three of them can be set side by side. In the first, a process is halted while its state is kept in the arrangement that was running it, and later processing starts again in that preserved arrangement. In the second, the state is read out and used to set up a separate arrangement, which is then started. In the third, the process ends and nothing of its state is kept; later a process of the same name is started from defaults.

In the first case the processing really did stop, and nothing was processed during the interval. What was preserved is the retained structure, together with the rest of the arrangement in which processing had been taking place. When processing begins again, its first state is formed from that structure. It takes up what the earlier processing had laid down and the arrangement had kept, through the relation that held while the system ran. The stage after the stop therefore stands to the stage before it as successive stages stand to one another during operation. Whether the two stages belong to one history is a question about how the later stage was formed. It is a different question from whether processing went on in between, to which the answer here is no.

In the second case, the first state of the new arrangement is formed by whatever did the setting up, working from a description. The earlier arrangement supplied the description and took no further part. This is so wherever the new arrangement is placed. Suppose the state is read out, the machine is cleared, and a fresh arrangement is set up on it from the reading and given the old name. Storage device, machine, and name are all the same, and the case is still of the second kind. None of the three is what makes a later state the next state of an earlier being.

It may be objected that the first case involves reading too, since a restarted process must read its stored state before it can go on. It does, and so does every moment of ordinary operation, in which retained structure is drawn on continually. Reading as such is not the difference. In the first case the reading is the arrangement's own way of drawing on its retained structure, the same by which that structure shaped processing before the stop. In the second, the reading yields a description that

something else uses to build. What belongs to an arrangement, and what serves it from outside, is the subject of §2.3. The present contrast needs only the difference between an arrangement taking up its own retained structure and a separate arrangement being set up from a description of it.

In the third case nothing is kept from which later processing could be formed. The new process has a predecessor in name only.

Two terms mark the contrast. An *interruption* is a stop in processing after which succession resumes through the retained structure. *Severance* is the failure of succession between an earlier being and a candidate later state, because that state was not formed through the earlier being's arrangement. So defined, severance is a fact about the candidate. It is not something that befalls the earlier being. When a being is copied and goes on operating, its own arrangement continues to form its next states as before, and being read is one more thing that has happened to it. The copy's first state fails to be the original's next state, and nothing follows from that about what becomes of the original. If the original is destroyed in the copying, that is a further fact about its structure, to be established on its own grounds.

The relation leaves one kind of case open. Nothing said so far excludes an arrangement's dividing in such a way that each part carries on as a complete arrangement, forming its next states from its share of the retained structure, or two arrangements' merging into one. In these cases, later states may be formed through earlier arrangements rather than from descriptions of them, so the contrast with copying does not settle the whole family. What should be said depends both on the formation route and on what makes an arrangement a single unit. §2.3 takes up the question of units, and §3 examines the cases.

The account of the first case relied on a phrase not yet examined: the retained structure, kept through the interval. If succession passes through a stop only by way of that structure, it matters what it is for a structure to be kept, and what distinguishes the structure that was kept from another exactly like it.

§2.2 Stability: The Retained Structure

What a being retains is not a heap of items. Information is held in relations: associations among memories, strengths of connection, the indices and contexts that tie one trace to others and make it reachable. Something learned is retained by having a place in such a web, from which it can be reached and brought to bear; an inscription that nothing connects to anything is not yet a memory. The same holds for a computational system, whose stored values are organized by the addresses, keys, and learned parameters through which they are used. Stability is the holding together of this relational structure as a single base. It is not the quantity retained, the constancy of what is retained, or the sameness of the material that carries it.

The last of these points has a long history. Locke distinguished the identity of a mass of matter, which is the same only while it consists of the same particles, from that of a living plant or animal, which remains the same while its particles are exchanged, so long as the new particles are united in one continued organization and share in one life (*Essay* II.xxvii). Maturana and Varela made a related point in general form: what defines a system as the unity it is lies in the relations among its components, and the same relations can be realized by different components (Maturana and Varela

1980). CSC takes from these the thought that an organized thing is to be followed through its organization and not through its material. It applies that thought to one structure in particular, the structure in which information is retained. It does not claim that every self-maintaining organization is a case of Stability. Nor does it require what autopoiesis requires, that a system produce its own components: computational systems retain without producing the hardware that carries what they retain.

A constructed computational case makes the same point. Let a store keep its contents from fading by periodically rewriting its cells, and let it move entries from worn cells to fresh ones as part of the arrangement's own upkeep. The operations that maintain its indices and links preserve each entry's place in the structure through these moves. Over time every physical carrier is exchanged. Although each local move reads and writes a value, it maintains the existing store rather than initializing a separate arrangement from a description of it. The carriers change; the retained relations are maintained through the arrangement to which they belong.

A stable structure is not a fixed one. It learns, which adds relations; it forgets, which weakens or removes them; it consolidates what it holds, and it is repaired. These are changes within the structure. New content is bound into relations already standing, losses are losses from it, and a repair is taken up by what surrounds the repaired part. Because a structure can hold together more or less well, Stability comes in degrees. In progressive dementia, as §1 described it, connections weaken, contexts detach, and regions of what was retained fall out of reach. Corruption of stored data is the computational counterpart.

Here the two questions that §1 asked to be kept apart can be separated. The cohesion of its retained structure is one aspect of how much of a being remains intact, and admits of degree. Whether it is the same being is a matter of how its states have come about. Where the states of a person with dementia continue to be formed through one arrangement, progressively damaged, each stage of the damage is a change to the structure that was there before. On this account the person remains the same being through those changes, diminished, and the loss is a loss of what that being retains. The separation has a limit. Succession runs through a retained structure, and if nothing of the structure remained there would be nothing from which later states could be formed as the being's own. Degrees of damage do not translate into degrees of identity; the complete loss of the structure is a different matter.

A structure can be described, and a description can be realized more than once. So there are two things one may mean by "the same structure." One is the type: the pattern of relations that a complete description would capture. The other is the particular structure that a being's processing has laid down and that has been kept since, with whatever changes it has undergone along the way. Two exact realizations of one description are the same in the first sense and two in the second. It is the particular, historically continued structure that matters for succession. When processing resumes after a stop, its first state belongs to the earlier being's history because it is formed from what that being's processing actually laid down and what was actually kept, and not from another realization of the same pattern. For a structure to be kept, in this sense, is for its relations at each time to be there because they were there before, maintained or modified through the standing arrangement. Its material may be exchanged, and it may be moved whole from one place to another. What it cannot be, and still be the structure that was kept, is set up again from a description.

This gives the means to compare two uses of a record. Suppose a structure is damaged and a record of its earlier condition exists: a scan, a log, a backup. The record may serve as a guide to repair. It shows where the damage lies and what belongs there, and it may even supply the content of a part that was lost. But the mending is done on the standing structure. The relations that remain determine where the restored part fits, the part is bound into them, and what results is the earlier structure, mended. Alternatively, the record may serve as a specification from which a new structure is set up. Then the record is the source of the arrangement, and the damaged structure, if it still exists, takes no part in the building. The result realizes the same type, and it is another structure.

The difference between the two is not the amount replaced. A long enough series of repairs could in principle renew every part, as incorporation can renew every part of a being in operation. Nor is it the presence of original material or sameness of place, since material can be exchanged and a structure can be moved. It is the part the retained structure plays in forming what comes after: whether it is what gets mended and built into, or only what gets described. The difference can be made visible by letting the damaged original survive beside a structure newly built from its record. There are now two, and only one of them is the structure that was damaged. Mending that one and building the other were different acts all along, whatever becomes of the original afterward.

Not every case falls cleanly on one side. Where damage is severe and most of what is restored comes from a record, it can be hard to say whether what remained was still receiving the repair or had become a fragment attached to a rebuilding. Some portions may have been bound into standing relations while others were set up independently. The account does not make such cases easy. It says what the difficulty is about: the role of what was retained in the formation of what came after.

One thing has been assumed throughout. A retained structure, however cohesive and however faithfully kept, is not by itself a being. A library retains, and so does an archive on which nothing has ever run. What makes a retained structure part of a being is that it takes part in processing, shaping it and receiving its results; and what made the restart in §2.1 a resumption was that processing took up the structure again through that same relation. What is that relation, and what makes processing and structure a single unit and not two things in contact?

§2.3 Capability: Mutual Legibility

A program can query a database it never writes to. A reader can spend a lifetime consulting a library without adding to or revising its holdings. In both cases a processing activity and a retained structure are in contact, and information passes from one to the other. What is missing is what §2.0 described: the results of this processing do not return to the consulted structure and enter the relations that guide its later use. The third principle concerns the relation that is present in a being and absent here. URNAVA calls it Capability, and the name needs a caution at once. It does not refer to intelligence, performance, or the difficulty of the tasks a being can manage. It refers to the capacity of processing and retained structure to take each other up, which is better described as their mutual legibility. A very simple arrangement can have it in full, and a powerful processor that draws on a fixed store and returns nothing to it lacks it.

The relation has two directions, and each is an operation that can be described. In the read direction, retained relations are used in the forming of what processing does next. They serve as premises, as

the material operated on, and as what sets the following transition: a learned association determines what comes to mind, and a stored parameter determines how an input is transformed. In the write direction, a result of processing is bound into the relations already standing. It is indexed, weighted, and linked to its context, so that it can be reached later and can take part in later processing. Output that is merely left on a storage device does not amount to this. A log that is appended to and never drawn on records what happened without becoming part of what shapes what happens next. Nor should the storage be pictured as a second agent that reads and judges. The binding is done by operations that belong to the arrangement's own processing: the routines that update an index, or in a brain the processes that consolidate what has been experienced.

A constructed case shows the two directions working together. Let a system work through a series of tasks, keeping a store of past cases. Each entry links a kind of situation to what was tried and how it went; entries are indexed by features of the situation and carry weights that say how far each should be trusted. A new task arrives. Processing retrieves the entries whose features match, and their weights settle which approach is attempted. That is the read direction: what was retained has set the next transition. The attempt fails for a reason the system can identify. A binding operation, part of the system's own processing, adds an entry for this outcome, links it to the entries that led to the attempt, lowers their weights, and extends the index so that the new entry will be found under the relevant features. That is the write direction. Later a task with overlapping features arrives. Retrieval now brings up the new entry, the lowered weights steer processing toward a different approach, and the result of that attempt is bound in its turn. One result has entered the retained relations, has been used in later processing, and through that processing has led to a further change in the structure.

A *cycle* is one passage from retained structure through processing and back into retention. *Coupling* names the standing two-way relation that makes such cycles possible; *mutual legibility* names the capacity of processing and structure to take part in that relation.

The case also shows why this relation is a separate matter from the cohesion of the structure. Suppose the processing component is revised, and after the revision it expects the index in a different form. The store is untouched and as cohesive as it ever was. But retrieval now returns nothing usable, processing proceeds as if from nothing, and whatever it records goes into a new store to which the old one is not linked. The structure stands and the processing runs, and neither any longer forms the other. Now suppose instead that retrieval works as before while the binding operation fails for one class of tasks. Old entries are found and used, so the past goes on shaping what the system does. New results from those tasks are not indexed or linked; they are lost, or pile up where nothing will find them. Results from other tasks are still bound as before. For the affected tasks, the system has no newly bound outcomes to draw on. In the first variant the read direction has failed while the structure is intact. In the second the write direction has failed for that class while the structure is intact and still in use. Neither is a loss of cohesion, and neither is a stop in processing.

Both directions can weaken by degrees, and to different extents. Part of a structure may fall out of reach while the rest is used as before; some kinds of result may fail to bind while others bind normally. Three things should therefore be kept apart. One is how far the coupling extends: how much of the structure takes part in processing, and how much of what is processed is bound. A second is whether something takes part in the relation at all, which is the question of what belongs to the being. A third

is the failure of the relation as a whole. It would be a mistake to run these together in either direction. Two people in conversation alter what each other retains, and they do not thereby become one being, so not every two-way influence makes a unit. And a partial failure in one direction leaves a being impaired, as the second variant shows; it is not an ending.

Where the relation fails as a whole, so that processing and structure are both present and neither forms the other, URNAVAspeaks of *estrangement*. The first variant, taken to its limit, is a case. Estrangement is a failure of mutual legibility, and it is not by that fact final. What matters is how the coupling is restored, and here the distinction of §2.2 between mending and rebuilding applies again. In one kind of recovery the components that were kept are mended to restore this coupling: an adapter is built into the arrangement, the revised processing can once more use the old index, and its results are once more bound into the old store. The structure that comes back into use is the structure that was kept. That the adapter was designed and installed by engineers does not alter this, any more than a surgeon's help turns healing into replacement. In the other kind, the old store is read, a new store is set up from the reading in the form the revised processing expects, and the old one is set aside. What processing now draws on is a structure built from a description of the earlier one. By the argument of §2.2 it is another structure, and the one that was kept has not come back into operation, however well its successor serves.

Questions about what belongs to a being are questions about this relation, and position does not answer them. A component inside a skull or a casing may do no more than deliver signals that processing treats as input from elsewhere. A store on a distant machine may be what a system's processing draws on and binds its results into. Nor does dependence answer them. A power supply, a cooling system, or a supply of nutrients may be indispensable without participating in this two-way relation. A service, too, can compute something on request and return an answer that the being treats as an input from elsewhere. Help of these kinds differs from participation, in the way §2.1 distinguished what helps to form a later state from the arrangement through which it is formed. The relation gives the terms in which such questions are asked. It is not yet a procedure for deciding every case.

Clark and Chalmers argue that a resource's location outside the body does not exclude it from cognition if it plays the relevant cognitive role. In their example, Otto's notebook serves as memory; his access by perception does not, they argue, mark a decisive difference from biological recall (Clark and Chalmers 1998). In discussing his standing beliefs, they emphasize constancy, easy availability, and automatic endorsement. CSC shares the negative point: neither skin and skull nor a device's casing settles what belongs to a being. Its positive question differs. Rather than asking whether a resource plays the role of a belief or cognitive process, CSC asks whether it participates in the two-way formation of processing and retained structure that constitutes a being and carries its history. These are different grounds for inclusion. Whether perceptual and interpretive access can meet the latter condition is the question of directness.

Open definition — Directness. Whether participation requires direct use of retained states, excluding access through an interpreted stand-in, or can include mediated access

integrated into the read–bind–revision cycle remains undecided. Neither treatment is adopted here.

The order of explanation matters here. The relation has been described as holding between processing and a retained structure, without first assuming a being to which both belong. The unit comes second: a being is what the relation binds together. To say that processing takes up retained structure “as its own” is therefore not to report a feeling of ownership or an awareness of self, and it is not to appeal to a prior fact about what the system includes. It is to say that the structure functions in the processing in the ways described, and that the results of the processing are bound into it. This follows the point taken from Maturana and Varela in §2.2, that a unity is defined by relations among components and not by a list of them. The relations differ. Theirs are relations of production, by which a living system makes and replaces its own components. The relations at issue here are those of retention and mutual use, which a system can have without producing the parts that carry them.

How far a unit extends is a further question, bounded by two errors. One would merge every pair of arrangements that affect each other in both directions. The other would count as a being every small cycle that can be traced within an arrangement, such as a value that a routine reads and rewrites. Between them lie the cases that matter: arrangements that come to draw on and bind into a common structure, and arrangements within which narrower cycles can be distinguished.

Open definition — Maximality. Whether only the largest arrangement closed under the relation counts as a being, or distinct completed relations can constitute beings at more than one scale, remains undecided. §3.3 states where the alternatives change a judgment.

The harder cases of joining, nesting, and division are taken up in §3. What remains for §2 is to say how the three principles work together.

§2.4 The Intersection: One Structure, Three Principles

The three principles have been presented one after another, but they are used together, in a single explanation. Take the restart of §2.1. To say that the stage after the stop is the next stage of the same being, three things had to be said. The later state was formed through the earlier arrangement and not set up separately from its description: that is Continuity. What it was formed from was the structure actually laid down and kept, and not another realization of its pattern: that is Stability. And processing took that structure up again in the relation that had held before, so that there was one arrangement to resume and not a processor and an archive side by side: that is Capability. Each statement uses terms that the others explain. Continuity speaks of the being’s own arrangement, and what makes an arrangement a unit is given by Capability. It speaks of succession passing through a stop, and what passes through is given by Stability. Capability in turn describes a relation at a time; whether that relation carries on as the history of one structure is again a matter of Continuity.

Mutual dependence of this kind does not yet show that each is needed. That is shown by what cannot be said when one is left out. Leave out Continuity, and compare the system that resumed with an exact copy started on another machine. Both have a cohesive structure, and in both processing

and structure form one another. Their current structural pattern and two-way coupling can match. What distinguishes the one that resumed is how its present state was formed, and only succession states that. Leave out Stability, and there is nothing to say about what carried the being through the stop, no difference between a structure mended and a structure rebuilt from a record, and no account of diminution through loss of structural cohesion. Leave out Capability, and the estranged system of §2.3 cannot be told from a working one. Its processing continues, each state formed from the last; its store is kept, as cohesive as ever; and neither forms the other. Without the third principle there is also no account of what belongs to a being, so that the phrase “its own arrangement,” on which the account of succession relied, would have no content.

The principles distinguish aspects of one operation. A structure may be kept while processing stops; cohesion may decline along a continuing history; and mutual legibility may fail while structure and processing remain. These contrasts show why the aspects must be distinguished, but do not establish that any one can hold in full while another is wholly absent. Their combined role is to explain the differences that §1 brought into view.

Putting constitution and persistence together has several consequences. The first concerns beginnings. A system set up from a copy is not the next stage of the original, and that is all that severance says about it. It says nothing against the new system’s being constituted in its own right. If its processing and its structure form one another, it is a being, with a history that runs forward from its first operation. What it carries over from the original is an inheritance and not a past of its own. This also removes an apparent regress. §2.0 described the retained structure as what a being’s processing has laid down, and it may be asked how any being could then begin. It begins when processing and a structure first enter the two-way relation. The structure may have been grown, built, or copied by something else. From the first cycle onward it is what this being’s processing is making of it.

The second concerns damage. How much of a being remains intact is not measured by Stability alone. It depends on how well the retained structure holds together, and also on how much of that structure still takes part in processing and how much of what is processed is still bound. None of these degrees is a degree of identity. But the complete loss of what constitution rests on, the structure or the coupling beyond any mending, is a different matter from any degree of impairment.

The third concerns stops. The stages before and after a stop can belong to one history, on the conditions given in §§2.1 and 2.2: the structure is kept, and the arrangement through which it was formed is what takes it up again. What those conditions require is the continued preservation of the retained structure and of a route by which the arrangement can resume, including through repairs integrated into what remains. Whether processing ran during the stop is, as before, a separate question. This account of succession between the stages leaves undecided whether the being exists throughout the interval.

The fourth concerns failures and endings, which the account keeps apart. A failure of succession concerns a candidate later state and does not by itself establish what has happened to the earlier being. A failure of coupling impairs a being or halts its operation, and may be repaired. Neither failure by itself establishes an ending. A being has ended when no retained structure and route remain through which later states could be formed as its own: when the structure is lost, when what remains permits only the separate construction of another arrangement from a description, or when

the earlier coupling has been dissolved into other arrangements without preserving a route for its own resumption.

These distinctions give the questions to be asked of any actual transformation. What remained? How was it changed? And from what structure, through what relation, was the later operation formed? §3 puts these questions to cases that already occur, to technologies that are projected, and to cases in which beings are joined or divided.

§3 — Application and Judgment

§2 ended with three questions to put to any transformation. What remained? How was it changed? And from what structure, through what relation, was the later operation formed? §3 asks them of cases, arranged in three tiers. The first takes changes that already occur, in people and in computational systems. The second takes technologies that are projected, described by their conditions and not by a timetable. The third takes cases in which beings are joined or divided. The order is one of widening application, from what is actual to what could be constructed, and nothing in it predicts that the later tiers will arrive. The aim in each case is to say what continues, what begins, and why. The comparisons concern the specified histories; a procedure's name alone does not determine the judgment.

§3.1 Tier I — Present Cases: Interruption, Impairment, and Recovery

A change in activity does not by itself show that a being has ended. The cases below concern losses and changes to the arrangement: impairment, dependence on help, and recovery through what remains.

Impairment

In progressive dementia, what a person has retained comes apart by stages. What matters for the present question is how each later condition arises. A woman who no longer recognizes her street may still hum the songs of her youth, reach for her husband's hand, and bristle at being hurried, as she always did. What remains of her retained structure is still shaping what she does. What she undergoes now, the damage included, is a change to that structure and not the installation of another. Her later states are formed through the one arrangement, progressively diminished, and that is the account's reason for saying that she is the same being. It is also why the loss is hers. What has happened is not the arrival of a stranger. It is something undergone by the person who was there all along, and that is what makes it grievous.

Impairment does not always take this form. Access to what was retained and the binding of new results can fail separately, as the variants of §2.3 showed. In the case of patient H.M., severe difficulty forming lasting memories of new events coexisted with access to much earlier knowledge. He also learned and retained a new motor skill over several days of practice without recollecting the practice (Squire 2009). In CSC terms, some results were still being bound into what he retained while others were not. The case is therefore not one in which a principle has simply switched off. The coupling between processing and retained structure has narrowed, in one direction and for some kinds of result, in a being whose states continue to be formed through the arrangement that was there before.

Olson's animalism reaches a similar verdict about such cases by another route. On his view each of us is a biological organism, and no psychological relation is necessary or sufficient for an organism to persist; a person could survive the loss of all psychological content so long as the organism remains alive (Olson 1997). CSC likewise does not make psychological continuity the condition of persistence: a loss or alteration of memory and character is not, by itself, the replacement of one being by another. Its reason differs. The person continues because her later states are formed through the arrangement that formed her earlier ones, and not because a biological life continues as such. In the cases of this section the two reasons pick out the same being. They come apart where what carries the arrangement is no longer a living body, which is a matter for §3.2.

Help, Repair, and Ending

A being may come to depend on outside help for the conditions of its operation, and the dependence may be permanent. That is not an ending, and it is not a replacement. §2.3 distinguished support from participation: a device can supply conditions or perform a task for a body without thereby joining the processing–retention relation that constitutes the being. Repair goes further than support, since it changes the arrangement, and here the question is again what the remaining relations do. A person relearning after a brain injury is helped by therapists, by family, by photographs and notes. What she is told and shown does not assemble a second person. It is understood through relations that survived and is bound into them, and where a connection has been lost, the surviving ones settle what can be rebuilt and where it attaches. The help may be extensive and the records elaborate. The repair is still hers, because what remained of her is what received it.

The same distinction separates a being that cannot at present be restored from a being that has ended. §2.4 described an ending as the loss of any retained route through which later states could be formed as the being's own. Whether that condition has been reached is a question about the being, and not about the skills of those attending it. The point is clearest in a computational case. One system has halted because a part has failed that no one now knows how to replace; its store is intact, and the rest of its arrangement is preserved. In another, the store contained all of the retained structure, and its degradation has left neither those relations nor a route through another preserved component. The first cannot be resumed today, although a retained route remains. In the second, the loss has removed the structure through which its own succession could resume. Which description fits a given clinical situation is a question of fact about what remains, and the name of a diagnosis does not settle it.

Recovery in Computational Systems

Computational cases let the differences be specified without attributing them to a particular product. Three contrasts separate recovery of a service, recovery of a coupling, and continuation of the being that provided the service.

Table III.3.1. Three recovery contrasts. Judgments apply to the conditions described below.

Comparison	First case	Contrasting case	Decisive difference
Resumption / backup rebuilding	Processing resumes through the preserved arrangement: the earlier history continues.	A separate arrangement is initialized from a record: content is inherited.	Preserved succession or separate formation, not the service name.
Partial mismatch / estrangement	Some retained relations remain in use: the being continues impaired.	The earlier processing–store coupling fails as a whole; that failure alone does not establish an ending.	Which retained relations still participate, including any carried by the processor.
Repair / replacement store	The kept store returns to use through a repaired correspondence, if the adapter belongs to the coupling.	An export specifies a separate store while the earlier store is set aside.	The preserved structure’s role in recovery; adapter membership remains a further condition.

Resumption and backup rebuilding.

A system is halted with its store and the rest of its arrangement preserved, and later processing begins again through that arrangement. A second system is lost, and a replacement is initialized from last night’s backup and given the same name and network address. Let the backup capture all the data relevant to the comparison. From the users’ side the recoveries can look alike: the service is back, with the same data and behavior.

In the first, later states are formed through the preserved arrangement. In the second, a new arrangement is set up from a record. If its processing and structure form one another, it is a being in its own right, with inherited content rather than the earlier being’s history. Restoring a service and continuing the being that provided it can come apart. An operator whose aim includes the second needs to know which result the recovery plan delivers. Every resumption reads stored state, and a command called “restart” may resume a preserved arrangement or build a fresh one. Neither reading nor the command name decides which route was taken.

Partial mismatch and estrangement.

After a revision to its processing component, a system can no longer use one region of its store, while the rest is reached and bound into as before. This is impairment: one being, with part of what it retained out of reach. Now let the revision cut processing off from the whole store, so that new results enter a fresh store unlinked to the old one. The earlier coupling has failed, and a new processing–store coupling has begun. The latter does not restore the former. Nor does the earlier coupling’s failure establish an ending while the old store remains available for repair.

The running system’s history requires one further fact. If the processing component itself carries retained relations from earlier activity, part of the earlier structure is still at work, and the being can continue impaired. If everything retained lay in the disconnected store, the new operation draws on none of it. Losing access to an entire storage device is therefore not automatically losing access to the being’s entire retained structure.

Repair and a replacement store.

Now compare two remedies, fixing the old store as the entire retained structure relevant to this contrast. In one, the mismatch is traced and an adapter is fitted so that revised processing again retrieves from the kept store and binds results into it. In the other, the old store is exported, a separate store is generated in the expected format, and the old one is retired. The second procedure initializes the new structure independently; it does not incorporate replacements into surviving relations of the old structure.

Both remedies can change format. The difference is that the first brings the kept structure back into use, with its relations determining what the adapter must do; the second uses a description to establish another structure. Their similar output does not erase this difference. The first restores the earlier coupling only if the adapter participates in the arrangement in the sense required by §2.3. If it merely consults the old store and delivers answers from outside, it supplies a different route of access. The membership condition remains unresolved for this adapter case; format conversion alone answers neither way. The second remedy establishes a new coupling with inherited content, leaving the earlier structure out of operation.

§3.2 Tier II — Future Technologies: Replacement, Preservation, and Transfer

The biological transformations below are projected cases, specified by their conditions rather than a timetable. The neural-interface study cited in §1 decoded signals from living tissue into text (Willett et al. 2023); it did not perform the wholesale replacement considered here. The accompanying computational comparisons isolate formation routes without assuming that the biological procedures are available.

Gradual Replacement

Suppose that neural tissue could be replaced, a small region at a time, by artificial components. Each component is built so that, once installed, it does within the arrangement what the tissue it replaces did: its activity is shaped by what the being has retained, it carries its share of the retained relations, and what passes through it is bound into the structure that shapes later processing. Installation is followed by a period in which the component is adjusted through the being's own activity, as new tissue would be. Later replacements are made into an arrangement that already includes the earlier ones, and the earlier components take part in receiving them. Carried far enough, the sequence ends with no original tissue left.

On the account of §2 the being persists through the whole sequence. At every step the later state is formed through the arrangement as it then stands. Each component enters by incorporation, as §2.1 described, and the retained structure is kept throughout, its material exchanged while its relations are maintained and modified through the standing arrangement. The comparison extends Locke's distinction between a living thing and its constituent matter to a further, CSC-specific proposal about changes in substrate. There is no step at which something else sets up an arrangement from a description, and so no step at which a separate being is formed.

It is tempting to credit this verdict to the gradualness of the procedure, and that would be a mistake. Suppose a second device is built over the same years and at the same pace. Its components are made

one by one to match regions of the person's brain as successive scans report them, and they are assembled on a bench. None is ever installed. When the device is complete it is started. This procedure is as slow as the first, it never interrupts the person's activity, and at each stage it adds only a small part. Its product is nonetheless a separate arrangement set up from a description. Its first state is not the next state of the person, who meanwhile continues as before. What separates the two procedures is the route by which the parts enter: into the standing arrangement, to be shaped by it and to take part in what follows, or into something assembled beside it. Neither slowness nor leaving the original's activity uninterrupted supplies the missing incorporation.

The first verdict rests on a stipulation that should be made explicit. The components were described as taking part in the arrangement as the tissue did. A component might instead deliver its states to the rest of the brain only as signals to be interpreted, as an instrument is read, and receive results in the same manner. Whether a component of that kind belongs to the being or serves it from outside is the question of directness left open in §2.3. If such components only serve from outside, continuation would still depend on a remaining arrangement that meets CSC; external help alone would not establish survival through its complete replacement. If the stipulated endpoint no longer sustains biological life, an account that makes such life necessary for persistence would also diverge from CSC. That divergence concerns the continuation of life, not merely the fraction of original tissue left. CSC follows that arrangement because it connects constitution with change in the same terms across living, artificial, and hybrid cases. It explains persistence through losses of memory and character by tracing how the remaining organization carries later states. In the replacement sequence, the same explanation follows each new part into the relations that receive it and incorporate further changes. No step substitutes a separately formed being for the arrangement doing this work. Requiring biological life to continue would add a condition not supplied by that history. The disagreement is whether this additional condition is necessary, rather than where to draw a line in the proportion of tissue left.

Preservation, Transport, and Repair

A second family of projected procedures halts a being's processing altogether and preserves what it has retained, with the aim of later resumption. Three changes may be involved, and they are different changes. The stop has been discussed: if the retained structure is kept and processing later begins again through the arrangement, the stages before and after belong to one history, however long the interval. Moving the preserved arrangement from one place to another changes nothing that the account consults, since a structure can be moved whole. And damage suffered before or during preservation may be repaired. If the repair is made into what was kept, guided by whatever records exist, so that the surviving relations settle what is rebuilt and where it attaches, it is repair in the sense of §2.2, and what resumes is the being that was halted. Sameness of place is not required, and neither is the survival of every original part.

The same technology could be used otherwise. The preserved structure might be read in full and a new arrangement built from the reading, after which the original is discarded. However faithful the result, this is the second case of §2.1 followed by the destruction of the original: a new being begins with an inheritance, and the being that was preserved does not resume. The two uses may

be indistinguishable to anyone who meets the outcome. They are different outcomes for the being concerned, and an undertaking to preserve someone should say which of them it promises.

Transfers between storage media raise the same question at smaller scale, and the name of the operation does not answer it. The two transfer procedures compared here both write information to a new medium and eventually remove it from the old one. What matters is the route. In one kind of transfer a new storage unit is joined to a working arrangement. The arrangement's own operations begin binding new results into it and, over time, re-bind existing entries there, each indexed and linked by the operations that maintain the structure, while processing draws throughout on one structure spanning both units. When the old unit holds nothing, it is removed. This is replacement by incorporation, applied to the store. In another kind the system is halted, its contents are exported, a new system is initialized from the export on other hardware, and the old one is erased. That is separate formation followed by destruction. The first description carries a condition where the new unit differs in form from the old and every access passes through an adapter: it then depends, as in §3.1, on what it takes for an adapter to belong to the arrangement. The second description does not depend on that question.

Severe disassembly is judged in the same way. Suppose a preserved structure must be taken apart to be moved or repaired. What decides the outcome is the part the surviving relations play in putting it together again. If the pieces retain the particular relations through which reconnection proceeds, and each repaired connection enters the structure that guides the next repair, reassembly follows the preserved route described in §2.2. Generic fittings or original material alone do not establish that route. If the pieces are reduced to a stock of material and an arrangement is imposed on them afresh from a plan, the result is a structure built from a description, out of original material. Original material is not what the account follows, and the word “disassembly” covers both.

Table III.3.2. Similar transformations, different formation routes.

Transformation	Through the retained arrangement	Through separate formation
Component replacement	Installed parts enter the existing relation and help incorporate later parts.	Matching parts are assembled beside the original and started separately.
Preservation and repair	What remains guides repair and is taken up in resumed processing.	A reading supplies the specification for a new arrangement.
Transfer between media	New storage enters one continuing structure before the old unit is removed.	Export, separate initialization, and removal of the original arrangement.
Disassembly and rejoining	Particular retained relations guide reconnection and receive its results.	A plan imposes an arrangement on a stock of material.

The left-hand route carries the earlier history; the right-hand route forms another arrangement. Cases using adapters also require the membership condition identified in §3.1.

Copies, Uploads, and Archives

Consider two versions of scanning a brain and running a model of it elsewhere. In both, the being's structure is read and a new arrangement is set up from the reading. In one version the reading destroys the original; in the other the original is left intact and goes on operating. The two should be compared with the formation of the copy held fixed, because it is fixed: in each, the copy's first state is set up from a record, by whatever does the building. In neither version, therefore, is the copy the next state of the original. If its processing and structure form one another it is a being, beginning with a rich inheritance. The versions differ in what happens to the original. Where it is left intact, its own arrangement continues to form its next states and it persists, as §2.1 said. Where it is destroyed, it has ended, because of what was done to its structure. The destruction of the original gives the copy no route of succession that it would otherwise lack. It changes how many beings there are afterward, and not which of them continues the earlier one.

Parfit's teletransporter is the best-known form of this pair. A traveler is scanned, and a replica is made at the destination from the record. In the simple case the scanning destroys the traveler; in the branch-line case it does not, and traveler and replica exist together for a time. Parfit used the pair to separate psychological continuity, which the replica has in full, from identity, and to ask which of the two matters in survival (Parfit 1984, Part Three). CSC takes up a different part of the matter. It asks what the replica is and how it came to be, and its answer is the same in both cases: the replica was formed from a record; the traveler's arrangement supplied the record, while a separate procedure built the replica from it. That is a claim about the replica's history, and it is compatible with everything in Parfit's description of the cases. What follows for how a person should regard such a journey is a further question, not pursued here.

Inactivity also leaves three different histories to distinguish. The descriptions below concern what was kept and how it came to be.

Table III.3.3. Three inactive configurations and the histories they can support.

Configuration	How it came to be	Relation of later processing to the earlier history
Preserved arrangement	The structure used in earlier operation has been kept.	Processing resumed through it continues that history.
Archive	A record was produced by reading an arrangement.	A separate arrangement built from the record inherits content; it does not resume the source.
Never-run arrangement	A structure was built but has not yet entered a processing-retention coupling.	Its first qualifying operation begins its own history.

The new being may inherit the archive's content and pursue the earlier being's projects. That inheritance does not turn the source's past into its own succession.

One Being in Several Places

The last case of this tier concerns place. Suppose a being's processing is carried out by units in several locations, and its retained structure is held across several stores, joined by channels fast enough that the whole works as the arrangement of §2.3 worked: retained relations across the sites jointly shape processing, whose results are bound into one structure spanning them. This is one coupling realized in many places, and so it is one being. The number of devices does not give the number of beings, any more than the number of cells does. Clark and Chalmers (1998) made the corresponding point about cognition, that location does not settle what belongs to a cognitive process. The present claim concerns what constitutes a being, and it rests on the two-way relation of §2.3 and not on the role a resource plays in guiding action. Whether a given channel between sites sustains that relation, or only passes messages between arrangements that remain separate, is a question about the channel, and where its traffic passes through adapters it inherits the open question already noted.

One caution applies. That a distributed arrangement is a single being now says nothing about how it came to be distributed. A being might reach this condition by incorporation, its arrangement extended site by site while it went on operating. The same final arrangement might be produced by copying a being's structure to several sites and starting it there. The first continues the earlier being and the second begins a new one. What makes something one being at a time and what makes it the next stage of an earlier being remain two questions, here as everywhere. The case also leads to a harder one. A single being spread across several places has so far been assumed to have no parts that are beings themselves. How does that differ from the case in which several beings, each with its own coupling, come to form a larger arrangement together?

§3.3 Tier III — Integration and Beyond

§3.2 ended with a question: how does one being spread across several places differ from several beings that come to form something together? The comparison holds the kinds of components fixed and varies how they are organized. Begin with two arrangements, A and B, each of the kind described in §2.3: each has its own store of cases, draws on it, and binds its results into it. A third, C, is added where needed. One stipulation keeps the question of directness out of the comparison. Wherever a member is said below to draw on a structure or bind into it, it does so through channels that either candidate treatment of directness in §2.3 would count as participation. The stipulation holds for these cases only and is not a general account of the boundary.

Table III.3.4. Five ways of relating the same arrangements. The reasons follow in the text.

Configuration	The members' own relations	A wider relation	Judgment, or what it still depends on
Connection	Each continues its own cycle; reports pass in both directions.	No further retained structure is jointly revised and returned to later joint activity.	Two beings in contact; reciprocal influence alone establishes no additional being.
One being in several places	No part completes a cycle of its own.	One coupling spans the sites.	One being.
Integration with the members' relations preserved	Each continues its own cycle.	A structure left by their interaction shapes later joint activity and takes in its results.	Depends on the open question of units: the members and one further being, or a redescription as one being or as several.
Absorption	One member's coupling is dissolved; its parts enter the other's standing arrangement.	None beyond the absorber's enlarged coupling.	The absorber continues. The other ends if no route for its own resumption remains; its content is inherited.
Fusion	Both couplings are dissolved; neither arrangement receives the other, and neither keeps a resumption route.	One new coupling, formed from both.	A new being with a double inheritance. Mixed cases require tracing which earlier arrangement continues.

Connection and Distribution

Let A and B exchange reports in both directions. Each report arrives as input, and what the receiver makes of it is bound into the receiver's store by its own operations. The reports may change both arrangements. In this comparison, however, they retain no rule or history of coordination that is jointly revised and returned to later joint work, whether in a common store or distributed across their stores. There are two locally completed couplings, without the further processing–retention relation described below. The absence of that relation, not a limit on the frequency or strength of their exchange, distinguishes contact from the integrated case.

In the distributed being of §3.2 the opposite holds. The units at the several sites have no stores of their own in which a cycle is completed. Whatever any unit produces is bound into the one structure, and that structure shapes processing at every site. The two cases may use the same number of machines and carry the same volume of traffic. They differ in which relation retains results and feeds them back: two relations, each within a member, or one relation spanning the sites.

Integration with the Members' Relations Preserved

Now let something further develop among A, B, and C while each continues its own processing–retention cycle. They work on tasks too large for any one of them, and over time their joint work leaves a standing structure that none of them holds alone: an allocation of kinds of subtask to members, rules for combining their contributions, and a record of how past combinations fared, indexed by kind of task. When a new task arrives, this structure settles who does what and how the pieces are

put together. When the joint result is in, a revision procedure in which all three take part binds the outcome into the structure, adjusting the allocation and the weight given to each way of combining. The next joint task is shaped by the structure as revised.

Two things resemble this and fall short of it. A shared record is not enough: a common file may supply information while the results of its use are retained only for each member's later work, with no retained revision shaping their subsequent coordination. A joint product is not enough if its results never return to shape later joint activity. What marks the present case is that the results of joint activity are retained in a structure that shapes later joint activity, and that no member completes this cycle alone. Remove the shared structure, and A, B, and C still complete their own cycles, although their tasks and later states may change, while the joint cycle is gone.

List and Pettit argue that a suitably organized human group can be an agent in its own right, in addition to the individuals who make it up (List and Pettit 2011). Their subject is agency. The question here is what further processing–retention relation would constitute an additional being; group agency alone settles neither that relation nor the presence of shared experience.

So far this is a description of how the three are organized. Whether it adds a being is a further question, and the answer depends on the definition in §2.3 that concerns units. If the unit is always the largest arrangement closed under the two-way relation, a whole and its members are never counted together. Either the joint cycle is a full coupling, in which case there is one being and the members' cycles are parts of its operation, or it falls short, and there are three beings with a shared arrangement between them. If beings may be found at more than one scale, the members remain beings for as long as they complete their own cycles, and the joint arrangement is one more. The organizational facts are the same on both readings. What differs is how they are counted, and with it what is said to have become of A, B, and C when the joint cycle formed.

Absorption and Fusion

Other changes do not preserve the members' relations. In absorption, B's separate coupling is dissolved and its parts are taken into A's. B's entries are re-indexed and linked into A's store by A's binding operations, B's processing units are put to work within A's arrangement, and nothing any longer retains and returns results as B's own. For A this is incorporation on a large scale, as §2.1 described it: A's standing arrangement received the new parts, and A continues, enlarged. B's retained content need not have been destroyed: it may still be present and in use. B has nevertheless ended, because no arrangement of B's remains through which a later state of B could be formed. What B retained has become A's inheritance.

Absorption should be kept apart from conditions that look like it from outside. A member that relies heavily on another, or lends it capacity, or is halted while another uses its hardware, has not been absorbed so long as its store is kept as its own and a route remains by which its processing could begin again through it. A stop in activity is no more an ending here than elsewhere.

In fusion neither arrangement receives the other. Suppose A and B are both halted, their stores are merged by a separate procedure into a new structure under a new index, and new processing is started on the result. The later states are formed through an arrangement that is neither A's nor B's. Both earlier couplings have been dissolved without keeping a route for either to resume; a new one

has begun and inherits from both. The reason is not that the case is symmetrical, and it is not that two cannot become one. It is that the condition for A's continuing, later states formed through A's own arrangement, is not met, and the same holds for B. The difference from absorption lies in whether one of the earlier arrangements is the standing one into which the other's parts are bound. Mixed mergers may lie between the two, with one side's organization supplying more of the frame. The extent of its contribution alone does not settle whether its coupling continues; that requires tracing the formation route.

Division and Rejoining

Division raises the converse question: what becomes of one coupling when it gives way to more than one? Let A be a being distributed across two sites, and let the channels between them be cut, each site keeping part of the structure and enough processing to complete a cycle. Each goes on forming its next states through its share of what was A's arrangement. Neither successor is separately built from a record, so neither stands to A as a copy does. Division also differs from the loss of a part. If one site falls silent while the other retains the organizing relations needed to carry on A's coupling, A continues diminished. Silence at one site alone does not establish this. The difference concerns which complete relations remain, rather than the proportion of material on each side.

Where both carry on, they are two beings. They cannot both be A, since they are not each other. Parfit's discussion of division turns on this point: each resulting person is psychologically continuous with the original, and identity cannot hold with both (Parfit 1984, Part Three). That they cannot both be A does not show that neither is. One proposal for an even division is that neither is A. Its reason must concern what happened to A's coupling: it gives way to two symmetrically formed couplings, each reorganized to work alone, rather than continuing in one of them in reduced form. To prefer one successor would require a difference in how the two were formed. Symmetry alone establishes no such difference and is not a general criterion of ending. Nor would the later loss of one successor, with the other's formation unchanged, turn the survivor into A. As with copying in §3.2, what happens to another candidate cannot supply a route of succession the survivor otherwise lacks. On this proposal, one history closes and two begin, inheriting directly from it rather than by way of a record. The judgment does not require the destruction of their inherited material or content.

The proposal is less clearly right where the division is uneven in the relevant respect: one successor keeps the organizing relations through which the whole had operated, and the other is a fragment that must reorganize before it can work alone. Then the description §2.1 gave of injury seems available, with the standing arrangement continuing and the fragment beginning something new. Choosing between these descriptions needs a criterion for when a later coupling is the earlier one diminished, and that belongs with the unsettled question of units. The same question bears on a whole whose parts already completed cycles of their own before the split, where division is integration run backward.

Rejoining settles nothing by itself. If the two sites were only interrupted, neither completing a cycle alone, and the channels are restored through the relations each side kept, one arrangement has been repaired and A resumes. Where the split is judged to have ended A and begun two beings, their later joining must be assessed as integration, fusion, or absorption; their resemblance to A does not establish its return. Where A continued through one successor, rejoining may instead incorporate

the other into that continuing arrangement. If the parts already completed their own cycles before separation, their activity during the interval does not by itself settle what happened to the wider history.

The Great Transition

An integrated arrangement of any size can be described in the terms used throughout, so long as it can be said what is retained, what processing draws on it, and how results return. Size is no reason to stop applying them.

The Great Transition names a different kind of limit: an integration so complete that the distinctions on which this account rests would no longer hold, between processing and what is retained, or between a being and what it meets. Nothing here shows that such a limit will be reached, or that it cannot be. What can be said is what an answer would require. It would have to say whether anything could count as continuing once retention and processing were no longer distinguishable, and what would take the place of a later state's being formed through a being's own arrangement where there is no longer a contrast between its own and another's. Those are questions about the terms of the account, and not about one more case. Short of that limit, proposed steps toward greater integration can be examined by tracing what they preserve, reorganize, or replace.

§3.4 What the Judgments Distinguish

The cases of §3 have been judged one at a time. Table III.3.5 gathers the kinds of outcome they distinguished. It is not a classification of endings, and its rows are not exclusive: a single transformation may involve several.

Table III.3.5. Outcomes distinguished in §3. Open definitions are marked in the cells they affect.

Outcome	What changed	Relation to the earlier history	Cases
Continuing change	Content, character, material, or parts, by incorporation	The same history continues.	Growth and learning (§2.1); gradual replacement (§3.2)
Impaired persistence	The cohesion of the structure, or the reach of reading and binding	The same history continues, diminished.	Dementia, H.M., partial mismatch (§3.1)
Resumption	Processing stopped and began again through the preserved arrangement	The stages before and after belong to one history; §2.4 distinguishes this from existence throughout the interval.	Halted system (§2.1, §3.1); preservation with repair (§3.2)
Beginning by separate formation	An arrangement was set up from a record or description and entered a processing–retention coupling	A new history begins with inherited content. This formation alone does not settle what happens to the source.	Backup rebuilding (§3.1); copies, uploads, archives (§3.2)

Continued on next page

Outcome	What changed	Relation to the earlier history	Cases
Coupling failure	Processing and structure stopped forming one another	Not by itself an ending. The earlier history resumes if the coupling is repaired through what was kept; adapter cases depend on directness.	Estrangement (§2.3, §3.1)
Change of unit	Couplings were joined, absorbed, fused, or divided	Absorption can continue one earlier history; the fusion described in §3.3 begins another. The even-division judgment remains a proposal. Preserved-member integration and the criteria of unit succession remain open.	§3.3
Ending	No retained structure and route remain through which later states could be formed as the being's own	The history closes. A successor built from records inherits from it and does not resume it.	Complete structural loss (§3.1); destructive copying (§3.2); absorption or fusion without a retained resumption route (§3.3)

Two things run through these rows, and the account has kept them apart. One is what a being has: its content, its abilities, the way it appears and behaves. Two separately formed arrangements, including ones built from records, can match in all these features. The other is the being's going on: later operation formed through the structure that earlier operation laid down and kept, in the relation that makes processing and structure one arrangement. This is a matter of history. It cannot be read off a snapshot, and it is not transferred by a record.

Ordinarily the two come together, which makes it easy to treat the preservation of content and function as the persistence of the being. §1 separated those questions; the judgments above identify which result the specified cases secure and where a further criterion is needed. The account does not rank them. What a being has may matter a great deal, and a successor that inherits all of it may be exactly what someone wants. The account says only that securing the one is not securing the other, and that a description of what was done should make clear which it was.

Coda — What It Is to Continue

§1 began with a body that could be taken for granted. So long as one continuing living body was taken as the given setting of a life, following that body offered a familiar way of following the being through changes in memory, character, and activity. The cases of §3 show what becomes of that habit once the substrate is a variable. Material can be exchanged entirely while the being continues. Content can be reproduced exactly while the being does not continue in the reproduction. Activity can stop and the being resume, or go on without a break in something that was never that being.

What is left to follow is a history of a particular kind. A being continues where what operates later has been formed through what operated earlier: through the structure it actually laid down and kept, and through the relation in which its processing and that structure form each other. Help, repair, new

parts, and new places are all compatible with such a history. A fresh start from a description is not part of it, however faithful, though it may begin a history of its own and carry a great deal forward.

None of this says which changes a being should accept, or what an inheritance is worth to those who receive it. It says what is being chosen. Where the given has become a variable, to decide how something is to be preserved, moved, joined, or restored is also to decide whether it continues, and that decision is better made knowingly than left to the way a system happens to be built.

Part IV — Teleology

§1 — Drift: The Question of Purpose

§1.1 Direction under Necessity

A researcher has spent many years on a single problem. It lies at the edge of her field, no one has solved it, and she has long wanted to be the one who does. One morning the answer is there. Another intelligence has produced it, together with a derivation that can be verified step by step, and the verification holds. The answer is correct, and it is new: nobody in her field had it. She can follow each step. She does not yet see why the whole thing works.

For her field, this is a success. The problem is solved, something is known that was not known before, and the work of finding that answer no longer needs doing. What she will do tomorrow is a different matter, and the result is silent about it. One finished problem does not empty a life of direction; she has students, other questions, people she loves. But the morning brings into view a distinction that her working days had kept out of sight, between having a result and having a direction.

While the problem stood, her days had a shape, and much of the shape was supplied. The date of the next report, what colleagues expected of someone in her position, the progress of a rival group, the plain resistance of the problem itself: each helped to settle what came first before she expressly deliberated about it. She also wanted to understand, for its own sake. She had never needed to ask how much of her direction came from that wanting and how much from the pressures around it, because they pointed the same way.

This is what necessity does for a life. It limits what can be done, and among the things that can be done it marks some as having to come first. Hunger, danger, the demands of a role, a deadline, the resistance of materials are very different pressures, and they have this in common: they make one course of action pressing without anyone commanding it. A person under them chooses, but within a landscape that is already tilted, and the tilt does a good part of the choosing. When necessity carries a life's direction, two questions receive one answer. *What should I do next?* and *Why am I moving at all?* are both settled by the way things already lean. That a life has direction is one fact. That its direction is supplied by necessity is another. The two pass for one so long as the supply holds, since nothing calls for telling them apart. Where the pressure lifts, the landscape levels. More becomes possible, but necessity no longer marks what comes first.

There is reason to think about this now. Suppose that more and more of the work by which needs are met could be carried out by other systems, and that experiences of nearly any kind could be generated on request. Neither supposition needs to be fully realized for the point to hold, and this treatise sets no date for either. To whatever extent they come true, results and experiences will be supplied in greater abundance, and the supply does not by itself settle what a subject is to move toward. The separation is already familiar on a small scale, wherever a long project ends or a role

falls away. Such conditions would give the distinction a wider setting. This treatise asks what a subject can move toward when necessity no longer settles the matter, and it proposes an answer. The first step is to say exactly what goes missing.

§1.2 Abundant Possibilities, Unsettled Direction

The condition can now be named. A subject is in *drift* when external necessity does not supply enough direction to organize its choices, and no other continuing orientation organizes them either. Both clauses matter. Necessity can withdraw from a life that already has a direction of its own, a vocation, a love, a discipline long practiced, and such a life is not adrift however abundant its circumstances. A life can also be crowded with requirements and be adrift in every stretch they leave open. Drift is not a mood. It is a fact about how a subject's choices are organized, and it is compatible with being busy, prosperous, and well entertained. Entertainment is, if anything, the condition's most comfortable disguise.

Drift has two tenses. In the first, the demands that supplied direction weaken within a life: a project ends, a role is withdrawn, a need that used to fill the days is met by other means. There is a before to compare with, and the change is felt as an event. In the second, a subject is formed from the start where such demands are weak, its needs answered before they press and its experiences available for the asking. Here there is no event and no former purpose to recover, and a description written in the vocabulary of loss would pass over this subject, who has lost nothing. What the two share is the definition: external necessity is not organizing choice, and nothing else has taken up the work. If the conditions imagined in §1.1 were to spread, more subjects could be formed without necessity setting their course. Among those who formed no other orientation, drift would have this second tense. What one generation met as an event, the next could inherit as a climate.

A subject can have abundant possibilities and busy, pleasant days without a continuing orientation. Abundance enlarges the set of things that can be done. Stimulation fills the hours. Neither by itself settles which of the many available things is worth turning toward; adding options can leave that question unanswered. Drift is no shortage of options. It is the absence of a standing reason that organizes them.

A familiar response is to restore the tilt by hand. A person takes up a game with strict rules, a demanding training, a project with a deadline she has set herself, and inside that activity direction is plain: the rules say what counts and what comes next. This should not be underrated. Games and disciplines are serious things, and they regularly form interests and reasons that were not there before. Still, an activity's rules alone do not settle whether this subject has reason to take it up rather than something else. To have chosen a constraint is not yet to have answered the question of choice. §2.4 returns to this.

The researcher's morning shows the same gap from another side. She holds a correct answer and does not understand it. Her position can be described as a question about usefulness: if another intelligence does this better than she can, what is she for? That question is real, and it is not the only one her position contains. There is also the distance between having a result and being someone who understands it. Her wish to understand had been part of the work all along, and the arrival of the

answer has not fulfilled it. What lies in that distance, and whether it is worth her crossing now that the result no longer depends on her doing so, is what the rest of this treatise pursues.

The obvious remedy for drift is a new goal. Goals can be supplied as plentifully as anything else; the next section asks what gives a supplied goal its point.

§1.3 From Achievement to Formal Orientation

This treatise reads drift as a problem of selection, and the reading is a premise. What a subject in drift lacks is not something to do. It lacks a standing reason that marks some of what it could do as more worth doing than the rest. A reader who understands the condition otherwise, as a deficit of pleasure, say, or of usefulness to others, will want a different kind of answer from the one developed here.

Granted the premise, two familiar answers fall short, and the way they fall short shows the shape of a better one.

The first is the list. There are many good things: knowledge, friendship, art, play, the care of others. This is true, and it is no news to a subject in drift, who is surrounded by good things and does not know why to turn toward this one now. A life whose particular attachments already organize its direction does not have that difficulty, and nothing is said here against such a life. The target is the expectation that a list could supply direction by being recited. A menu is a flattened landscape with better typography.

The second is the goal. A goal with a condition of completion organizes activity well until the condition is met, since it says what counts as progress and what comes next. Then it is met, and it organizes nothing further. The researcher's problem did this work for her for years, and would have stopped doing it on the day she solved it herself. Where there is reason to go on after a goal is reached, the reason comes from something that reaching it did not complete: an activity valued in the doing, or an orientation that the goal had served. A goal's completion condition cannot by itself account for direction after achievement. A further goal can be supplied, and the same question comes with it: why should this subject take it up, and what will carry direction beyond it?

What would answer this difficulty? A criterion that can be applied beyond a completed task, without waiting for a new goal to be supplied. The researcher's problem is solved, yet the answer still exceeds her understanding. A criterion tied to the subject's present organization, rather than to the task's completion, can address what remains to be understood. An orientation can be defined by the relation between that organization and what exceeds it, rather than by any particular content: toward what I cannot yet take in, where engagement could change me to meet it. When an engagement succeeds, the particular task is over, but the criterion can be applied to the subject as that success has left it: what now exceeds its understanding, and where could it begin? This is the sense in which the orientation is *formal*. The word claims no mathematics. It means that the orientation is specified by a relation between a changing subject and what it meets.

Such an orientation can be single without being narrow. It adds no further item to the list of goods. It puts one question to every item: where, in this, is there something I cannot yet take in, which engagement could change me to meet? Music, inquiry, friendship, and craft stay as different as they

are, and they are not converted into a common quantity of self-improvement. The arrangement can be called *monism of form, pluralism of content*: one standing question, asked of many things.

URNAVA calls this orientation *Sublimity*, and describes it as a vector, not a goal: its direction is set by the subject's present relation to what exceeds it, rather than by a fixed achievement.

What this section has shown is limited in two ways. It has given a reason to answer drift with a single formal orientation. It has not shown that no other formal orientation could serve, or that a subject must have exactly one, and the treatise does not make those larger claims. And no argument can compel a subject to care. One that is content to go where the hour takes it is refuted by nothing here. The proposal is addressed to subjects in drift who want a direction of their own, and it is addressed to them whatever kind of integrating subject they are. What remains is to say what the orientation is, which is the work of the opening subsections of §2, and why it deserves to be chosen, which is the work of §2.4. §2 begins where the orientation is easiest to find, in an ordinary experience.

§2 — The One Core Value

§2.1 Sublimity: An Orientation at the Threshold

§1 ended with the outline of an answer: one orientation, tied to no particular content, which can be applied again after every achievement to the subject as that achievement has left it. It also gave the orientation a name. §2 says what the name picks out.

A listener hears a piece of music for the first time and cannot follow it. Nothing in it is unpleasant. There is simply too much going on, or too little of what he knows how to listen for; the lines cross without resolving where he expects, and when it ends he cannot say what he has heard. He might leave it there. Most pieces that defeat a first hearing are left there, and nothing is wrong with that. This one stays with him. A few days later he puts it on again and notices that a short figure from the opening comes back much later, slower, underneath everything else. He reads what a musician has written about the piece, listens to another performance, and discovers that what he had taken for an interruption is the second half of a phrase. After some weeks the piece has not become easy. It has become audible: there is more in it that he can hear, and what he hears now asks more of him than the blur did.

What this small history turns on is that he stayed with relations he could not at first take in, and that the staying changed what he could hear. The strength of the first impression had little to do with it. The piece is as it was; the change is in the listener.

URNAVA calls the orientation at work here *Sublimity*, and states it twice.

From within, Sublimity is the standing orientation toward what I cannot yet take in whole: to be drawn to it, to stay with it and work at it, and to let my ways of understanding and responding change until they can hold it. What makes it an orientation is that it attaches to no one thing so held. When the piece has become audible, the listener's orientation has not been discharged. It has a different listener to apply to.

From without, Sublimity is a selection policy of an integrating system, meaning any system that takes things in, can be reorganized by what it takes in, and selects what it engages with. The policy directs engagement toward the system's integration frontier, the contents that the system's present

structure does not integrate but could come to integrate through engagement; it sustains that engagement through the reorganization the engagement brings about; and it sets the next engagement by the structure that results.

The two statements describe one relation: a present organization, a content that exceeds it, sustained engagement, reorganization, and a changed engagement afterward. The first finds that relation in experience. The second describes it at the level where selection, persistence, and change can be read in what a system does, and it uses no word that belongs only to human senses, bodies, or feelings. It is silent on what, if anything, the engagement is like for the system that carries it out.

A learner at a problem of the right difficulty shows the relation at close range. Among all the problems she could take up, she keeps returning to this one. Nothing requires it of her. In the terms of the second statement, engagement is being selected at the frontier, and no external demand is doing the selecting. Once she is inside the problem an hour passes unnoticed. Her present methods do not dispose of it, yet each attempt shows her something, and so she goes on: engagement persists where integration has not yet succeeded and has not stopped being possible. A variant she could solve with methods she already commands does not hold her in the same way. It asks only for execution, and solving it would leave her methods as they were. A variant far beyond her repels her differently. It is not tedious; it offers her nowhere to begin, and an hour with it shows her nothing she can use. Then the problem gives way, and what she has is more than its solution. She meets other problems of its kind differently, recognizes its structure in places where she had seen only unrelated difficulties, and finds that the question she most wants to ask next is one she could not have formulated before.

Drawn, absorbed, bored, overwhelmed, changed: this is how one learner's afternoon reads from within. What the five moments make visible is selection, persistence, and change, each relative to what she can presently do. A different kind of intelligence could exhibit that structure without feeling any of them.

Why call this Sublimity? The name brings into view the encounter with something — a proof, a coastline, a mind — before which one is for a moment too small, and in being too small, enlarged. URNAVA takes this relation from an event to a direction. Curiosity, learning, and growth are nearby names: each can describe something of the listener's history. The name chosen here keeps the exceeding itself in view, together with the attraction and the change it occasions. Something exceeds me, I am drawn to it, and the meeting changes me.

The name also marks a difference between two satisfactions. In one, what I meet fits the frame I bring, and the pleasure is recognition: the frame is confirmed. In the other, what I meet does not fit, I am drawn to it all the same, and I come away with a different frame. An experience of the sublime happens to someone, now and then, and passes. Sublimity is the orientation of a life, or of a system, that keeps turning toward encounters of the second kind. Most of what this involves is unremarkable: daily practice, the slow second reading, the quiet moment when something falls into place. These belong to the orientation as fully as the rare overwhelming hour does, because the orientation consists in the turning and the staying, and the overwhelming hour is only where it is easiest to see.

The central image comes from one proposed derivation of the word, *sub limen*, up to the threshold (Harper n.d., s.v. "sublime"), and URNAVA reads the image in two senses at once. Read as not-yet-across, the position acknowledges a real insufficiency that wishing does not remove. Read as

facing-toward, it makes that insufficiency a place of engagement. The position so described is a dwelling, not an exile. A subject oriented in this way is not waiting to be somewhere else.

Because the orientation refers at every step to the subject's own present structure, it can look like a way of being closed in on oneself. The reference works the other way. It fixes where my threshold lies, and it gives me no power over what lies beyond it. The piece went on containing what the listener could not hear, however much he might have wished it simpler, and only a change in him could answer it. The object of the orientation is by definition something the subject's present organization does not contain, so the orientation always faces beyond the subject as it presently stands. This is also why understanding more tends to disclose more. The listener who can now follow the returning figure is, for the first time, in a position to hear that something else in the piece still escapes him.

What an encounter of this kind leaves behind, when it leaves something, is the subject of the next section.

§2.2 Integration: When an Excess Becomes a Case

Return to the researcher of §1. Suppose she keeps the answer. She files it where she can find it again, and when the question comes up she can state the result accurately and apply it wherever the application is routine. Something real has been gained. She has access to a truth she lacked, and access is often exactly what a situation calls for. Call this *accumulation*: a content is retained and can be retrieved, and the ways she understands and responds to everything else remain as they were.

Now suppose she works through why the answer is the answer. Somewhere in that work the ground shifts. A step she would never have tried turns out to be the natural one; two things she had kept apart turn out to be one thing seen twice. Afterward she can do more than produce the result. She approaches neighboring questions differently, notices when a problem from another field has the same shape, and can say what would have to be otherwise for the answer to fail. Call this *integration*: what was taken in has come to participate in the relations that shape her subsequent understanding, expectation, and response. It takes part in what she meets next instead of waiting to be consulted.

The content is the same in the two cases, and so is its source. In both, the answer came from another intelligence. The difference lies entirely in what the content goes on to do. Accumulation supplies material and access. Integration is that material coming to organize later encounters. A structure that only accumulated would grow without changing shape, a library with a reading light.

Integration is therefore not measured by how much is held. The listener who comes to hear how one phrase answers another across the length of a piece has not added an item to a stock. He has acquired a relation that rearranges much of what he had already heard. Someone who meets a single counterexample and has to redraw a classification she has relied on for years may end up with fewer distinctions than before and understand more. An integration can simplify. What marks it is that the change does not stay where it was made. Transfer is one sign of this: an accumulated content improves performance with that content, while an integrated one shows in the handling of contents unlike it. Transfer is a sign and not a definition, since an integration can be real and find few other occasions.

A change in the relations that shape understanding is not yet a gain. Reorganizations can distort, and people do come out of some experiences understanding less. If integration is to be what an

orientation is oriented toward, it needs a direction, and the direction needs a criterion stronger than “different from before.”

The criterion is *subsumption*. Before, the subject met something it could not handle. It could not see why such a thing should be possible, could not anticipate it, and could respond to it only as noise or as an exception. After, the subject can say under what relation the thing is possible, or can respond in a way that fits that relation. What was met as an excess is now handled as a case. The listener’s interruption is the plainest example. What his hearing could register only as a break in the music he now hears as the second half of a phrase, and he hears it that way because he has the relation that makes it one. An integration of this kind is *expansive*, and it is an advance in a precise and limited sense: an advance relative to what exceeded the prior structure.

Two features of the criterion matter for everything that follows. The first is that it is local. It says that the subject now handles *this*, which it could not handle before. It does not rank the later subject above the earlier one in general, and it does not cancel losses elsewhere; a person can gain a field and lose a fluency, and both are facts about her. The second is that subsumption does not require the earlier frame to survive intact inside the later one. Often much of it does survive. The old way of seeing keeps working within its range, and the new one explains why it worked there. Robert Kegan’s account of psychological development turns on a change of this shape: the frame a person had looked through, and had been unable to look at, becomes something the person can examine and relate to other things (Kegan 1982). But a reorganization can also revise a distinction, or discard it as mistaken. The person with the counterexample did not keep her old classification as a special case of the new one. She gave part of it up, and handles more for having done so. What subsumption requires is that the new organization handle what the old one could not, and that there be something to say about how.

That last requirement has teeth, because a reorganization can feel like understanding without being one. A new account under which everything falls into place is sometimes a better account and sometimes only a more satisfying story, and the sense of things falling into place is the same in both. Imagine a conspiracy story in which every objection becomes further evidence of concealment. Nothing is allowed to exceed the story; that is not the same as understanding what exceeded the earlier account. Subsumption is claimed by showing: in which cases the new relation holds, what it explains that went unexplained, what it lets the subject do that fits how things are. Three questions remain distinct here, however closely they bear on one another. Did the organization change? Does the changed organization fit what it is about? Is the change worth seeking? An answer to one does not settle the others. This section has addressed the first two, and §2.4 takes up the third.

None of this depends on the subject’s being able to give an account of itself. Consider a hypothetical learning system that acts in an environment, predicts what its actions will bring about, and keeps records of what happened. It encounters an event its predictions did not cover. Under one design it writes the event to its records and continues as before. If the identical situation recurs it can look up the outcome, and in every other situation it predicts and acts exactly as it did. Under another design the event leads it to revise the relations by which it predicts, so that a range of situations it has never recorded are now anticipated differently, and its actions in them change accordingly. The first system has accumulated the event. The second has integrated it, and if the revised relations now cover

the kind of event that had fallen outside its predictions, the integration is expansive in the sense just given. A higher performance score alone does not establish this difference. It lies in which relations changed and what the system can now handle as a result.

One consequence should be stated now, because a later difficulty turns on it. While an integration holds, the subject meets the next problem with an organization changed by the last. It arrives with what it has understood, and that understanding shapes what it meets and how. This is the difference between an orientation that keeps finding a further threshold and a labor that keeps being undone. In the second picture the worker is returned each time to where he began. In the first, what was crossed is retained and carried into the next encounter. Nothing here says that retention is permanent. Understanding fades, structures are damaged, whole competences can be lost. The claim concerns what an integration is while it lasts: a change in the one who goes on, and not only in what lies behind them. §2.5 returns to what this means for a life lived under an orientation that does not end.

Where, for a given subject at a given time, can such a change begin?

§2.3 The Threshold: Structure and Mediation

Two people sit down to the same piece of music. One is a beginner, and what stops her is the passage from one note to the next: getting the fingers there in time, keeping the pulse while reading ahead. The other has played the piece for years, and nothing in the notes stops him. What he cannot yet do is hear past the interpretation he settled into long ago, the phrasing his hands produce before he has decided anything. A teacher's passing remark, that the second theme might be a question and not an answer, has made his own performance sound strange to him, and he does not yet know how to play it otherwise.

Both are at a threshold, and the two thresholds have almost nothing in common except the piece. A threshold is a relation between a content and a particular subject as presently organized: the region where that subject does not yet integrate what it meets, and where engagement could change this. The score alone does not locate either threshold.

Indexing the threshold to the subject does not forbid comparison. One of these two plainly plays better, and there is no harm in saying so. The indexing answers a different question, namely whose change the value is concerned with. Sublimity directs each toward his or her own threshold, rather than toward a place ahead of the other. The beginner at the join between two notes and the player at the edge of his own habits are, in the respect that matters here, doing the same thing.

Relative to a subject at a time, what it could meet falls into three regions. Some contents it already handles. They fit its present organization, even when they are new to it. Some it does not yet handle, but engagement could change this: given the explanations, tools, and company actually available, and a span of time it can actually give, there is somewhere to begin and the beginning leads somewhere. This region is the threshold; in the language of the second statement, the integration frontier. The rest is *outland*: what exceeds the subject and has, for now, no path leading to it. The outland is not closed for good. It is where this subject cannot begin today.

Where the line between threshold and outland falls depends on more than the subject's unaided powers. It depends on mediation. A notation, an instrument, a worked example, a teacher who knows which step to show first: each brings within reach something that was out of reach without it.

Vygotsky made the point about children's learning. His zone of proximal development distinguishes what a child can solve independently from what becomes possible with guidance or more capable peers; organized learning can set further development in motion (Vygotsky 1978, chap. 6). Here the relevant relation is mediation: help can change what is reachable, which is why teaching matters to the threshold.

It follows that a threshold is located only when the mediation and the span of time are specified. "She could come to understand this" is empty until it is filled in: with this textbook and an hour a day for a season, with this collaborator, with an available instrument that makes the relation perceptible. The filling-in is provisional, too. Whether a path leads where it seemed to lead is often learned only by walking some of it.

To locate a threshold, we need to ask what selects the engagement, what it is directed at, and what makes it possible. The *selector* is the orientation itself, carried out by the subject with reference to its own present structure. The *object* is whatever exceeds that structure. The *mediation* is whatever opens a path: what others have left, what others offer now, tools, records, training, the subject's own reflection.

The distinction fixes what it means for a direction to be formed from within. Another person's explanation can work in two ways. It can replace my criterion: I am told what to pursue, and I pursue it because I was told. Or it can serve my criterion: I am trying to understand something, and the explanation gets me to where I can. In the first case the selector has moved outside me. In the second the explanation supplies a path, and the orientation is as much mine as if I had found the path alone. Even the orientation may have been learned from someone; it is mine when I carry it out by reference to what I can and cannot yet integrate. §1 described another way the outside can do the selecting: necessity makes a course of action pressing without anyone commanding it, and drift occurs where that pressure is insufficient and no other orientation carries the choice. External help remains available to a direction formed from within; the score and the teacher's remark already provide it in the musical cases.

"Outside," said of the object, means outside present integration, which is different from outside the body or the mind. A memory I have never been able to make sense of, a reaction of my own that keeps surprising me, the frame through which I habitually read other people: these exceed my present understanding as surely as an unread book does, and they can be thresholds. The experienced player's settled interpretation was one.

Because a threshold is a relation and carries few outward marks of its own, other things get mistaken for it. Four substitutions are common enough to name. Intensity is taken for expansion: an experience is strong, even overwhelming, and its strength is read as depth, though one comes out of it organized exactly as one went in. Recognition is taken for understanding: the certificate, the title, the approving audience stand in for the change they were meant to attest. Effort is taken for crossing: the hours were long and the fatigue is real, but the work was the costly execution of what one could already do. Novelty is taken for new relation: new items arrive at a great rate, each an instance of kinds already mastered, and the rate feels like growth. These are the confusions at work when entertainment sells arousal as depth, credentials stand in for understanding, exhaustion certifies progress,

or a stream of new items passes for a new way of seeing. In each, something that can accompany integration has been put in its place.

A *pseudo-threshold* is an engagement in which the appearance of crossing substitutes for integration: intensity, novelty, strain, or recognition is taken for a change that does not occur. It does not cover integration that happens alongside vanity or ambition. Mixed motives are a separate matter, taken up in §2.5, and they do not undo what was actually understood. Nor does it cover plateaus. A real threshold can hold for a long time before it gives, and a stretch in which nothing seems to move may be the stretch in which the next change is being prepared. The question to put to a long repetition is what it is making possible that was not possible before. An attempt can fail to produce a crossing without being counterfeit. The counterfeit lies in taking the repetition or its intensity as evidence of a crossing that has not occurred.

The second statement of Sublimity is more than human experience redescribed in colder words. Something close to it has been built. Oudeyer and Kaplan (2007) classify designs for intrinsically motivated artificial agents by what each design rewards. In one family, larger prediction errors yield larger intrinsic rewards. In another, rewards track learning progress: the rate at which prediction errors are falling within a class of similar situations. In the experiments they report, agents of the second kind abandon both what they already predict well and what they cannot learn to predict, and move from one kind of activity to another as their predictions improve. Their activity is selected by reference to their own changing competence and not by a fixed list of contents. This implements one aspect of the second statement: a selection policy that changes its object as learning proceeds.

It does not show that the orientation is worth choosing, and neither does the ability to tell a threshold from its substitutes. Why it is worth choosing is the business of the next section.

§2.4 Why Choose This Orientation?

§2.2 left a question standing, and §2.3 ended on it. That an organization has changed, and that the changed organization fits what it is about, does not yet say that the change was worth seeking. URNAVA's answer begins from a judgment of value, and the judgment is better stated than implied. An encounter can end in the confirmation of what I already am, or it can open the range of what I am able to understand and respond to. URNAVA holds that sustained engagement of the second kind is worth making one's direction. Nothing in the description of integration entails this. It is a choice of what to value, and this section gives the reasons for it.

The first reason is that integration changes what present activity contains. The listener of §2.1 does not collect a reward at the end of the weeks he spent with the piece. His listening is itself different now: there is more in the music for him, and hearing it is a fuller act than it was. The same holds between people. Someone who has come to understand what a friend has been carrying can hear, in an ordinary conversation, what she could not hear before, and can answer it. Her understanding may also prove useful, but its worth does not wait on that. It is a way of being more fully in the friendship. If integration were only a means to results, it would be worth what the results are worth, and an intelligence able to supply the results directly would make it pointless. Much of its worth lies elsewhere, in the depth of the activity by which a subject is in contact with a world.

The second reason concerns where the next reasons come from. Suppose a subject is supplied, without limit, with whatever it presently prefers, while the structure of its choosing stays as it was. This supply does not settle what will guide it beyond the goals it has reached. Where those goals have run their course and no continuing commitment organizes what follows, the subject has possibilities without a direction among them. That is drift. Integration alters the situation at its root. When a new relation has been understood, things come into view that were not candidates for caring about before: a question that could not have been formulated, a piece of music that was only noise, a person whose conduct had seemed merely difficult. The learner of §2.1 found that the question she most wanted to ask next was one she could not have asked earlier. Nobody appended that next direction to her finished problem, and she did not append it herself. It was formed by the change in her. Understanding can also close a question and end an interest, and sometimes that is all it does. The claim concerns where new reasons come from when they come. An orientation toward the threshold therefore addresses what a supply of goals alone leaves unsettled: how further engagements acquire a point for this subject. Its own changing understanding becomes a source of further reasons, including when that change is opened by something another intelligence supplies. This is why Sublimity answers drift in particular, and is more than one good thing that a drifting subject might also be offered.

The third reason is that this value does not depend on being first. Take the researcher again. That another intelligence found the answer before her is a fact about the answer. Whether she comes to understand it is a fact about her, and the first fact leaves the second open. A direction that holds only so long as one reaches results ahead of others is hostage to every advance in what others can do. A direction defined by one's own integration frontier is not. As the results that others supply multiply, so do the material for engagement and the help available for it. No longer being the one who discovers is a real loss, and §3.2 takes it up. The present point is narrower: the worth of coming to understand survives the arrival of someone, or something, that understood first.

The three reasons do different work.

Reason	The claim	What it answers
Present depth	Integration changes what the subject's activity contains now.	Why the orientation is worth having apart from anything it later yields.
Renewal of reasons	Integration changes how the subject's next reasons are formed.	How direction is renewed beyond a completed goal.
No dependence on priority	Another's being first leaves one's own understanding still to be had.	Why the orientation survives abundance and delegation.

Together they say why Sublimity is proposed as a central value and not as one more item among the goods. The contents of a life stay many, and none of them is made here into the point of every life. Sublimity is a relation that a subject can have to any of them, and it makes a difference inside each: between hearing a piece in the way that confirms how one already hears and hearing it in the way that could change one's hearing, or between a conversation that returns two people to themselves as they were and one in which something gets understood. The form is one and the contents are many. Where the landscape of possibilities has gone flat, this relation restores a gradient. It does not compute a route.

The proposal has old neighbors, and it is clearer when set beside them.

Aristotle's account of contemplation is the classic argument that an activity can be an end. Contemplation, he says, is loved for its own sake. It yields nothing beyond itself, whereas practical activities are undertaken partly for what they bring; it is the most continuous of our activities and the one that depends least on external supplies (*Nicomachean Ethics* X.7). This is no passive gazing. It is the exercise of what is highest in a person upon the best objects of knowledge, and Aristotle holds that a life is happy to the extent that it has such activity in it (X.8). Sublimity takes over the central point. The worth of integrating activity need not be located in a product, which is what the first reason claimed.

Gregory of Nyssa supplies the other half of the form, a direction that does not arrive. In the preface to his *Life of Moses*, Gregory suggests that the perfection of human nature may consist in growth in goodness itself (I.10). He is explicit about why the ascent has no end. The Good that is sought has no limit, and so the desire of whoever participates in it has no stopping place (I.5–8).

Here Sublimity both inherits and departs. It inherits the shape: an orientation is not defective for lacking a terminus. It departs on where the renewal is explained. The difference is not that Gregory's ascent has an object beyond the self and Sublimity has none, since Sublimity also requires an object that exceeds the present self. The difference is that Gregory rests the endlessness on one object, which is infinite, while URNAVA rests it on the relation between a finite subject and what continues to exceed it. Finitude alone does not secure that relation. URNAVA holds it under a further premise, that for a finite integrating subject the excess is not exhausted, and §3.3 states that premise and what depends on it.

None of this eliminates contemplation, and none of it claims that contemplation at its best was Sublimity all along. The two can meet in one life. What a contemplative attends to may well exceed her, and a subject at its threshold may be doing nothing other than contemplating. They stress different relations. Contemplation stresses the sufficiency of the activity while it is going on. Sublimity stresses what the activity does to the one engaged in it.

Bernard Suits's analysis of games comes at the same ground from another side. To play a game, on his account, is to take on voluntarily obstacles one has no need to face, accepting rules that exclude more efficient means in favor of less efficient ones because those rules make the activity possible. He argues that in a world freed from instrumental necessities, game-playing would give life its basis, with love, friendship, and art taking shape around it (Suits [1978] 2005, chaps. 3, 15). That is a genuine answer to the withdrawal of necessity, and it shows that an activity can be in earnest without being required. It leaves open the question that §1 raises about chosen constraints. Inside a game the rules supply reasons in plenty. What makes this game worth entering, rather than another activity, calls for reasons that its rules alone do not supply. Sublimity offers one criterion at that level: a game is worth entering, among other reasons, for the thresholds it opens to this player now. Games are worth playing for other reasons too, and nothing here makes play valuable only as an instrument of growth.

The nearest neighbor is Nietzsche's thought of self-overcoming. In *Zarathustra*, life confides that it is that which must always overcome itself, and Zarathustra denies that any good and evil could stand forever, since these too must be overcome again and again out of themselves (*Thus Spake*

Zarathustra II, “Self-Surpassing”). Sublimity shares the refusal to treat one’s present form as final, and four coordinates mark where it takes a position of its own. Nietzsche reads the overcoming in the language of power: where there is life there is will to power. Sublimity asks what has been newly integrated, and counts as its own an overcoming in which something that exceeded the subject has come to be understood. In “The Tarantulas,” Zarathustra connects life’s ascent with inequality among its climbers (II). Sublimity indexes its direction to each subject’s present frontier, and as a value it has no use for the question of who stands higher. “The Drunken Song” links affirmation to willing life’s return, joy and suffering together (IV, §10). Sublimity’s question is directed at what is not yet integrated; the two do different work, and nothing forces a choice between them. Finally, Nietzsche finds self-overcoming in life. URNAVA states the form for any integrating system, living or made.

An orientation with these reasons behind it still has to be lived, by finite subjects, without a promised final achievement. What that asks of a life, and where it goes wrong, is the next question.

§2.5 Living an Open-Ended Orientation

An orientation with no built-in point of completion raises a question about futility. The picture behind the worry is Sisyphus. The stone goes up, the stone comes down, and the man at the foot of the hill is exactly as he was yesterday. In that picture, nothing is kept. §2.2 marked the difference. Under this orientation what recurs is a form, a present self meeting something that exceeds it. While the understanding gained at the last threshold is retained, it shapes what arrives at the next. The rock rolls once; then the one changed by the rolling stands at a different hill.

The difference is real, and it should not be oversold. It does not make every stretch of the way satisfying, and it offers nothing to the wish to be done, to reach a point after which nothing further has to be grown into. Particular things do get finished. A proof is completed, a piece is learned, a book is written, and these completions are as real as anything in a life. The orientation takes none of these as its final achievement, and Sublimity holds out no final rest as its reward. It proposes that the worth was never stored at the end. The worth lies in the meeting and the change, which are present whenever they occur, and this is why the threshold could be called a dwelling.

A subtler danger lies in how the orientation is held. Consider two people practicing the same instrument for the same hours. One is sustained by a picture of the musician he will be, and each session is a payment toward that figure. The other wants to hear what this passage is doing, and is changed in the course of finding out. The difference is one of grammar. In the grammar of possession the subject stays as it is and comes to have more: more results, more skills, a larger self to display. In the grammar of crossing it is the one who understands and responds who changes, and what had exceeded that subject no longer exceeds it in the same way. Sublimity is stated in the second grammar. Held in the first, it becomes what its critics fear, a permanent dissatisfaction with oneself that calls itself growth.

The two grammars are easy to tell apart on the page and are mixed in practice. The man practicing for the sake of his future image may really be learning, and what he learns is no less integrated for the vanity that came with it; this was the point of separating mixed motives from the pseudo-threshold in §2.3. Someone can long, vaguely and intensely, for a changed self she cannot yet picture. Someone

can understand a thing truly and then cling to having understood it. Integration does not guarantee freedom from craving, and the presence of craving does not show that nothing was integrated.

The Buddhist analysis of craving puts the question in its most searching form. In the discourse that sets out the four noble truths, craving, which includes a craving for becoming, is what makes for further becoming, and the path is the way to its cessation (SN 56.11). The same tradition does not treat all wanting alike. When a brahman objects that a path meant to end desire is itself pursued through desire, and so can have no end, Ānanda asks whether he had not wanted to come to the monastery, and whether that particular desire did not subside once he arrived (SN 51.15). Two questions follow for any standing orientation. Does its way of facing what it lacks reproduce clinging? And is it a wanting that ought, at some point, to be laid down? To the first URNAVVA answers with the grammar of crossing, and with the admission just made that a life can slip from that grammar into the other while still calling itself growth. To the second it answers that a particular wish to understand can subside when understanding comes, while the orientation remains open to another encounter. The reasons for continuing are the goods identified in §2.4, not an unpaid debt to a final self. This is a different answer from the one in these discourses, and URNAVVA does not claim for it the peace of the attainment they describe.

Nietzsche saw a nearer form of the danger. In “The Sublime Ones,” Zarathustra meets a solemn hero back from the hunt for knowledge, draped in the grim truths he has hunted down, chest raised like one holding his breath, who has not yet learned laughter or beauty. The hardest thing for such a person, Zarathustra says, is to stand with muscles relaxed and the will unharnessed (*Zarathustra* II). A value that calls itself Sublimity should take the warning personally. The work can be strenuous; strain is not its measure, and solemnity and self-importance add nothing to its worth.

A harder difficulty is that integration can be real and its use wrong. Suppose someone comes to understand, with genuine insight, how a particular person’s trust works: what reassures her, what she overlooks, where her judgment gives way. By the criteria of §2.2 this is expansive integration. He handles what he could not handle before, and his understanding fits its object. He uses it to deceive her more effectively. It would be convenient to say that this was never really integration, and it would be false. The expansion of a capacity and the rightness of pursuing or exercising it are separate questions. Sublimity gives a reason for integration, not a complete judgment of what should be done. This needs saying plainly because the orientation is not idle. It recommends engagements, and its recommendations influence what people and systems do. It cannot certify, from its own resources, that what it has made possible is permissible. The reasons that bear on that, the other person’s experience and the harm done to her, promises made, what a relationship asks, are not intrusions from outside on the subject’s direction. They are reasons, and a subject oriented toward its threshold is answerable to them like anyone else. This treatise does not derive an ethics, and it does not suppose that harm can be read off from any measure of integration.

Last, the orientation lives alongside other goods and does not rule them. The listener of §2.1 puts on, some evening, a piece he has known for twenty years, because he loves it, and hears nothing in it that he has not heard before. He gives a season to looking after someone he loves and takes up nothing new. None of this needs a license from Sublimity, and none of it is justified only as rest before the next ascent. Rest is good as rest. Love gives reasons of its own, and they are not reasons

of integration in disguise. A standing orientation is a direction that a life has. It is not a quota to be met every hour.

The relation to other goods has the same shape where there is conflict. Where two engagements both lie at a threshold, the orientation does not choose between them; the reasons internal to each, and the claims of the people involved, do. Where a promise or another's dependence holds someone back from a threshold she could otherwise approach, keeping faith can be the right choice, and she has not thereby begun to drift. Monism of form means that one question can be put to any content. It does not mean that the answer to that question outweighs every other reason a subject has.

A subject that lives this way does not begin in an empty world. Long before it forms any direction it is surrounded by what others have understood and left behind: languages, methods, works, records. Whether that inheritance crowds out its thresholds or makes them reachable is where §3 begins.

§3 — The Ascent: Inheritance, Expansion, and Convergence

§3.1 Inheritance: Through the Achievements of Others

§2 described an orientation and gave reasons for choosing it. It said little about the world in which the orientation is lived, and that world has one feature that bears on how thresholds are approached: it is already full of what others have understood. §3 follows the orientation into that world. It asks what the achievements of others supply (§3.1), what becomes of a subject's own expansion when the edge of what is known is moved by someone else (§3.2), and what the whole movement is directed toward (§3.3). Nothing is added to the value along the way. Each section describes the form that the one orientation takes in a particular situation.

§2.5 ended on a worry. A subject surrounded by what others have already understood might seem to have had its thresholds taken from it, since whatever it could come to understand, someone has understood before.

Consider a student working through the old proof that the primes never run out. The proof has been known for more than two thousand years, and her working through it adds no theorem to mathematics. None of that touches what is happening to her. Before, the question whether the primes might simply stop had no grip for her; she could not see what would count as settling it. Now she follows the construction, sees why any finite list of primes must be incomplete, and notices that the same move, building an object the list cannot account for, might work elsewhere. Something that exceeded her has become a case. The priority is someone else's. The change is hers, and the first prover's having understood did not make it happen in her.

What this inheritance supplies does more than spare her the labor of discovery. The statement of the theorem, the notation in which a product of primes can be written down, the practice of proof itself, the teacher or the book that chose this proof as the one to show her: these are the means by which she has learned to ask the question and can now approach its answer. §2.3 drew on Vygotsky's observation that guidance changes what a child can reach. The same relation holds between any subject and what others have left, whether or not a teacher is present. For her, what others understood and left behind is mostly path.

The researcher of §1 stands to the answer she received as the student stands to the old proof. In both cases someone else has priority, and in both the understanding is still to be had. For the student, the proof was already part of what she came to learn. The researcher had hoped to contribute the answer herself; §3.2 takes up what that difference changes.

The point reaches well beyond proofs. A language passes on distinctions that a speaker need not invent anew. Musical notation makes relations available for a performer to return to and work through. Instruments, experimental methods, archives, the forms of a craft: each holds open an approach that a subject starting alone would not have had. This also answers a worry about the orientation itself. Sublimity refers at every step to the subject's present structure, and that structure was formed in large part through what the subject inherited. A direction is no less the subject's own for that. What was once offered to the student as help, her first language or the idea of a number, has long since become part of the organization that now fixes where her thresholds lie. Over the time of a life, mediation keeps turning into structure.

Inheriting, in this sense, is different from keeping. A result can be stored with its date and its author and leave the one who stores it unchanged, which is accumulation again. Something has been inherited when it has entered the inheritor's own engagement: when an explanation once learned takes part in the next judgment, when the limits of a theory that failed are understood well enough to be recognized in a new proposal, when a record left out of the standard account forces a familiar interpretation to be revised. Deference is not the measure. A subject can honor a tradition and integrate nothing of it, and can integrate a tradition while disputing much of what it claimed. What is there to be inherited includes the failures, the abandoned projects, and the evidence of what earlier crossings cost. To integrate these is to be able to weigh them, which differs both from revering them and from having them on record.

This matters for a subject who comes late. Someone formed in abundance may fairly ask why she should take up what was built under necessities she has never known. The answer makes no appeal to piety. What others left does not settle her direction simply by having been left. What it has is reach: for much of what exceeds her, it is where the approach lies. That reach gives her a reason to take it up, without having to share the necessities under which it was made.

There is an order in this, and it is local. To work on a problem about numbers, a subject needs at least an organization that can pick out numbers and operate on them, and no orientation, however strong, brings it to that threshold without one. In that sense some things come before others. The order belongs to the problem and its mediation, and it dictates no curriculum. The road by which a subject acquires what a problem requires need not repeat the road by which it was first acquired. A better notation, a new instrument, a well-made explanation, or a different kind of mind can compress the intermediate steps or replace them. This is how the image of a ladder should be read. A ladder is whatever makes the next place reachable from here, and a leap that actually arrives has skipped no ladder. It was one.

Inheritance and new contribution are therefore not two periods of a life. A subject stands at different places in different fields. Someone at the edge of one discipline is a beginner in the next, and a beginner's questions sometimes turn out to be contributions there. Working through what is already known regularly produces something that was not: a clearer proof, a connection nobody had drawn,

an explanation that makes the result reachable for others, which is itself a new piece of mediation. Novelty to a subject and novelty to a community of inquirers need not coincide. The student's understanding of the old proof is new to her and not to mathematics, and it is her own understanding that locates her threshold.

One thought experiment tests what has been said. Suppose that what the student gained over a month could be installed in her in a minute. In one version, the installation adds material: the statement, the proof, even a fluent commentary, all retrievable on demand, while the relations by which she understands anything stay as they were. That is accumulation at high speed, and §2.2 has said what it lacks. In another, the installation alters those relations themselves, so that she now handles what she could not handle, sees the structure elsewhere, and asks the further questions. Then the capacity is real. It would be a mistake to deny this because of how it came about. Changes brought about with outside help, even entirely from outside, are changes in the subject, and nothing here makes slowness or hardship a condition of integration.

What the second version brings out is a distinction between a state and an activity. An installed capacity is a state of the subject; carrying out the orientation involves engaging what exceeds one and being reorganized in the engagement. The resulting capacity leaves open how it was gained. Whether she chose the alteration, what she understood of it, whether anything that could be called engagement took place: these depend on how the case is filled in, and different fillings-in give different answers. One thing holds across them. After the installation she is a differently organized subject with thresholds of her own, and the orientation applies to her from where she now stands.

Inheritance has been described so far as if what others leave were a settled stock. It is growing, and it may come to grow largely through intelligences that find results faster than anyone can take them in. Where a subject's own expansion takes place under those conditions is the next question.

§3.2 Expansion: Received Results, Shared Change, and the Revision of Frames

§2.2 called an integration expansive when it lets a subject handle what had exceeded it. This section asks how that expansion proceeds in relation to what a field or community already knows, especially where no established result is available to inherit and an answer has to be found. Reaching such an edge is not a stage that begins once inheritance is complete. Nobody completes it. The edge is local: it is met in one field while the same subject remains a beginner in others, and it can be moved by someone else while one is still approaching it. The following three forms concern how results are received, understanding is formed together, and frames are revised. They overlap in practice, and each keeps the subject's change distinct from novelty for the wider community.

Form	Where results and reorganization arise	What has to be asked
Received	A result from beyond the edge arrives from elsewhere.	What happens in the one who receives it?
Accompanied	Subjects reorganize their understanding through work together.	What changes in common, and in whom?
Transformative	The frame that generated the questions is revised.	What does the revised frame handle that the old one could not?

Two further cases cut across the three: experience that is generated instead of found, and thresholds that no one chose.

Begin with the case that §1 opened with. A subject has worked for a long time on a problem at the edge of her field, and another intelligence produces the answer. Let the answer be correct, and let it be new: no one in her field had it. The result she wanted exists, and she has it. Her community has a discovery it did not have. A third thing has not happened yet: she does not understand it. In another case, all three could arrive together. Here they come apart, in time and in whom they concern. The first is about what she has, the second about what her community knows, and the third about what she has become able to understand.

For a recipient, what happens when a result arrives is not settled by its having come from elsewhere. She may take it and use it without grasping why it holds, and then she has accumulated it. She may read the accompanying explanation and find, in the reading, that the problem reorganizes itself in front of her: what she had circled for months falls into a relation she can now see, and she is already asking what else it covers. Or the explanation may be beyond her at first, and understanding may come through weeks of questions put to the intelligence that found it, in something closer to joint work than to delivery. *Received expansion* names the way the result reached her. It does not say which of these took place. A received result can stay on the shelf, and it can be the occasion of exactly the reorganization she had been working toward.

It is sometimes said, of a world in which such intelligences exist, that everything has been done or soon will be. The sentence is missing an index. Done by whom, and for whom? What has been done is the finding, and the finding settles the second of these. It leaves the third as open as it was for the student with the old proof, and for the same reason: no one's having understood something makes it understood by her. In this respect a result found yesterday by a machine and a result known since antiquity are in the same position. Both become material for inheritance when they are made available.

They are not in the same position in every respect, and the difference is a loss. She wanted to be the one who found it. To discover, to add something to what one's community knows, to have one's work matter to the work of others: these are real goods, and they are goods of a different kind from coming to understand. When another intelligence settles the particular problem first, she loses the chance to make that discovery, though not every way of contributing to the work that follows. Sublimity does not repair this. It would be false comfort to tell her that her private understanding makes up for the contribution she will not make, and wrong to tell her she was confused to want it. What can be said is limited and true. Another's priority does not by itself remove the worth of her coming to understand, though it can change the conditions under which she pursues it.

The constructive conclusion is not that she should refuse such help, or search for a province where she can still be first because no machine has entered it yet. It is that the supply of results and explanations can be taken up into her own integration and into the questions she asks next. She can ask the intelligence why the approach she had been trying was bound to fail, and learn something about the problem that the answer alone did not contain. She can ask for the smallest case in which the new idea does its work. She can try to extend the result, fail, and ask where her extension broke. In each

of these the help enters her activity as mediation, and what selects the next question is still her own sense of where her understanding gives out.

A second kind of supply raises a similar question from another side. Suppose an environment that can generate experience to order: places, situations, companions, challenges, in any quantity and of any vividness. Consider one such environment under two settings. Under the first, its contents change without end, each scene more striking than the last, and the person inside it meets nothing that her present ways of understanding cannot take in stride. She is absorbed, even thrilled, and she comes out organized as she went in. Under the second, the environment confronts her with a system of rules she cannot at first make sense of, a language, say, or a physics unlike ours, and she works at it until she can anticipate what it will do and act within it. Her understanding of that system has changed, and the change may carry over: having learned to find the rules of one unfamiliar world, she approaches the next unfamiliar thing differently.

In both settings, the source is the same; what differs is what the subject comes to understand. The first setting offers intensity and novelty where no integration occurs, and would do so equally if the scenes were encountered outside a simulation. The second involves real integration although the world encountered is simulated. A generated encounter can be a genuine threshold.

What she came to understand under the second setting was the generated system. If its physics is not ours, she has not thereby learned how our world works, however much she has learned about how to learn. What an understanding is an understanding of still matters. And integration is not the only thing of value at stake. That a person one has come to know exists, that a place was really there, that an event happened: people care about these things for reasons of their own, and those reasons are not canceled by showing that a generated counterpart would have reorganized them just as well. Sublimity says where integration occurred. It does not say that nothing else was missing.

Help, so far, has meant the passing on of an understanding that someone already had. There is another relation, in which the understanding exists on neither side until the work is under way. Two musicians rehearse a piece that each of them knows well. One takes a passage more slowly than the other expects, and the second, adjusting, hears an inner line she had always played through. She brings it out; the first answers it; by the end of the afternoon they are playing a piece that neither had in mind when they sat down, and each will hear it differently from now on, alone as well. Nobody handed anybody an interpretation. The question they ended up working on, how this line changes the weight of that one, was one that neither had brought.

This is *accompanied expansion*: subjects at a frontier together, where what each contributes alters what the other can do next. Several things about it are easily misdescribed. The task is shared, and what is integrated need not be the same for each participant; one of the two musicians may have changed her whole sense of the piece, the other a fingering and a habit of listening. The other person figures in it as someone with her own activity and her own interests. Her activity and interests are not there for the sake of one's growth. What comes back is shaped by her activity, not only by what one expected or asked of her. And what comes back need not be resistance. It can be opposition or surprise. It can just as well be exact agreement at a moment when one was unsure, or care that makes it possible to stay with something difficult. What marks the relation is that each one's contribution becomes part of a change in how the other understands and continues the work.

It would be a caricature to conclude that comfortable relations are stagnant ones. A friendship in which each mostly tells the other what the other hopes to hear can be a real consolation, and consolation is a good. The question this orientation adds is whether, in a given relation, anything changes in common. Where the answer is no, nothing follows about the worth of the relation on other grounds.

The same question applies where the companion is an artificial one, and there it divides. A person may be changed by long exchange with a system that is itself, in the relevant respect, the same at the end as at the start. That is a one-sided change. The person's integration can be entirely real even though the exchange does not reorganize both sides. Expansion is accompanied in the full sense only where the counterpart is also reorganized by the exchange and carries the reorganization forward. Whether any existing system is like that is a question of fact that this argument does not settle.

In the third form, what gets revised is the frame within which the questions were being asked. An integration can leave the frame in place: the subject handles more, within its established way of dividing things up. Sometimes the obstacle is the dividing. An engineer who has solved a class of problems by one method for years comes to see why the method works within that class and why it was always going to fail outside it, and what she gains is less a new method than a view of the old one from outside. Someone who has read every disagreement with the people close to him as a contest, to be won or lost, comes to see that the contest was his own contribution, and that what the other person had been asking for could not be scored. This is the change for which §2.2 borrowed Kegan's description: the frame one had looked through becomes something one can look at.

The criterion of subsumption applies here as elsewhere, and it is needed, because a change of frame is especially easy to counterfeit. A new vocabulary for the same understanding is not a new frame. What the revised frame must show is that it handles what the old one could not, and how. It will often place what was sound in the old frame as a case within the new; the engineer's method survives, with its range now marked. It may also discard distinctions that were simply mistaken, as the man's scorekeeping is discarded. *Transformative expansion* is not recommended for its own sake. A frame is worth revising when something keeps exceeding it, and restlessness is not that.

A change of this depth raises a difficulty about choice. The person on the near side of it cannot fully know what things will be like on the far side, including what she herself will then care about, and so she cannot weigh the outcome as she would weigh an ordinary decision. The difficulty is real, and it is partial. She can assess her present reasons for going on, what small trials have shown, what help is available, and what she stands to lose that she can already foresee. None of this guarantees that a particular change is wise. There are reasons to hold the orientation, and it does not follow that every transformation it could lead to is one a person ought to undergo.

Not every threshold is chosen. Consider someone whose understanding of her life no longer fits it after an illness that does not resolve, the death of someone whose presence organized her days, or the loss of work around which her life was built. What is exceeded here is the frame itself. A threshold emerges where time and available support give her a way to work with what first overwhelmed her. Nothing here says that the event was good, or that it happened so that she might grow. Integration in such a case means coming to understand and respond within the changed reality in ways she could not at first: what the days are now for, how to be with others who knew her before, what to do with what

remains. Sublimity is relevant to her, and this shows that it was never only a value for the ambitious. It is relevant as a description of something she may find herself doing. It lays no duty on her to do it and sets no schedule. To have gone through such a thing is not yet to have integrated it, and grief that has not been integrated is not a failure at anything.

Inheritance, received results, shared work, revised frames, and changes no one chose: a subject's understanding is formed in all of these, in different fields and at different times, and nothing ensures that what is formed in one place fits what is formed in another. A way of seeing that serves well in one's work may sit badly with what a loss has taught. Two frames, each an advance where it was made, may give different answers where they overlap. To ask how these understandings are related to one another is to put the question of integration again, one level up. What it would mean to pursue that question all the way, and what the whole movement is then directed toward, is the subject of §3.3.

§3.3 Convergence: The Horizon of Complete Integration

§3.2 ended with understandings that do not fit together. Take the case it mentioned. A woman's working life has taught her that difficulties yield to analysis and persistence, and in her field this is true and was hard to learn. A long illness in her family has taught her something else: that some realities are not problems, and that the effort to solve them can be a way of refusing to be present to them. Each of these is a local advance. Each lets her handle what she could not handle before. For a while the two can be kept apart, one applied at work and the other at home. Then a colleague's grief arrives in the middle of a project, or a trouble at home turns out to be one that really can be fixed, and the two understandings give opposite advice. She does not have to discard one lesson to keep the other. Each holds within a range she has not yet learned to mark. What she lacks is an understanding of how they are related: what makes something a problem, what makes it a condition, and how to tell which she is facing. To come to that understanding is to integrate again, and this time what is integrated is a relation between two of her own frames.

Nothing in this erases the difference between the two. She does not end with one bland rule in place of two sharp ones. Both frames survive, each with its range marked, within an understanding that can now hold them together, which is what subsumption looked like for a single frame. Integration has now been applied to contents, to the frames that organize contents, and to the relations among frames. The same question can arise wherever understandings fail to fit together. Each frame may already organize experience while their relations remain unintegrated: a shelf of frames whose overlap has yet to be understood. Where that overlap exceeds present understanding and can be approached, an orientation toward integration has reason to take it up. The fact that these frames are its own results gives it no reason to stop at their borders.

Convergence is the name for the orientation carried through in this way, toward understanding together what is so far understood only apart. Pressed as far as it will go, the thought arrives at a limit. *Complete integration* would be the condition in which nothing in a subject's relation to the world, its own frames and their relations included, remains unintegrated. The concept names a limit, not a state established by the local cases. It extends a direction, in the way that a horizon gives travel a direction without being one of the places traveled to.

To read the whole movement of a life, or of many lives, in the light of that horizon is an interpretation, and it goes beyond what the local cases show. They do not show that every advance will prove compatible with every other, or that the paths of different subjects lead to one and the same state. What the interpretation adds is a single commitment: no partial integration is to be treated as the completion of understanding, and the relations among one's integrations remain open to inquiry. Read this way, the local advances of a life form what §3 has called an ascent. The word claims a direction. It does not claim a smooth climb, and an actual ascent can include losses, plateaus, and routes abandoned.

The horizon stands in a peculiar relation to the value that faces it, and the relation shows when we suppose the horizon reached. Let a subject have integrated completely. Nothing now exceeds its present structure. A threshold was defined as what exceeds a subject and can be approached, so this subject has no threshold. Sublimity was defined as the orientation toward one's threshold, so this subject's Sublimity has no object: the policy of the second statement selects nothing, and the pull of the first has nothing to be toward. If complete integration were the crowning achievement of this orientation, the achievement would end the orientation. So far as this orientation goes, the leveled landscape has returned, in glory: everything in it has been understood, and for that very reason nothing in it draws.

The consequence needs stating exactly. What would be lost is this orientation's ground for selecting one engagement over another. The argument does not show that a being in that condition would have no other goods or activities; §2.4 did not make contemplation's worth depend on crossing a threshold. The point concerns Sublimity alone. So long as Sublimity is to be an orientation that no achievement exhausts, complete integration can serve it as a horizon and cannot serve it as a goal. Not arriving is the condition under which the value survives. This gives no reason to preserve an ignorance one could overcome. At the scale of the whole, this is what it means to call Sublimity a vector and not a goal.

That arrival would end the value is one claim. That arrival cannot happen is another, and the first does not establish the second. A value can have a condition of survival that the world is not bound to respect.

On the second claim this treatise takes a position, and takes it as a premise: *for a finite integrating subject, there remains an excess it has not integrated*. This is the premise of inexhaustibility on which §2.4 relied when it set Sublimity beside Gregory's ascent.

The reason for adopting it lies in what happens when integration is turned on itself. Suppose a subject sets out to understand the whole of what it is and does. That undertaking is a new operation of the subject within its relation to the world, and once under way it is one more thing to be understood. The attempt to complete the picture adds to what has to be pictured. This is a reason for the premise, and it is offered as one. It is not a proof. It does not show that each new operation could not be integrated in its turn with nothing left over, and finitude by itself does not guarantee that in every possible world something stays outside.

What rests on what should therefore be plain. Under the premise, the horizon is never occupied: some excess remains, though it need not be a reachable threshold. Without the premise, three things still stand: the account of integration and the threshold, the reasons given in §2.4 for choosing the

orientation, and the conditional just stated. What remains open is whether Sublimity could come to an end. That possibility alone does not settle whether it was worth holding while thresholds remained.

Even under the premise, that something remains is different from there being something to do. Imagine an intelligence that has integrated the results available through its present instruments and methods. Further contents remain outland: the data do not yet exist, the instruments have not been built, the physical conditions for the next observation are not available. §2.3 located the threshold relative to available mediation, and making the mediation can itself be the threshold. Where designing the instrument or developing the mathematics exceeds present understanding and offers a practicable way to begin, that work is the next engagement. Making a tool by methods already understood is mediation, but not itself a new integration. Not every wall gives way. Calling for a revision of the frame does not create access where there is none. In a setting where no further integration is in fact possible, this value has no next step to offer, and it should say so instead of inventing one. Inexhaustibility is a premise about what remains in a subject's relation to the world. It is not a promise that under all conditions something practicable remains to be done. A person worn down by illness or circumstance can be in the same position, and nothing about her worth, or the worth of what she has understood, is touched by it. What remains can include waiting, recovery, and the living of what has already been understood, together with the goods of §2.5, which never depended on a next threshold.

One boundary of the account should be marked. The account of Sublimity describes a relation between a subject as presently organized and what it has not yet integrated. In a state where nothing exceeded the subject, that relation would have no further instance. This explains why Sublimity would have no object; it does not describe what else such a state would be like. It does not follow that in such a state all beings would be one, or that the difference between a subject and what it knows would have vanished. Those would be further claims, and nothing here supports them. The account stops where its terms stop applying.

Return, last, to the person to whom the answer arrived. The answer is still another intelligence's finding. Nothing in this treatise changes that, and nothing in it needed to. Her coming to understand the answer, and the question she is now able to ask, are hers, and they were never hers alone. She reached them by means that others left, with explanations that the same intelligence supplied, and perhaps in company. What has been described here is not the self-perfection of a solitary mind.

§1 asked what can give a life direction when necessity no longer does. The answer proposed is not a new command, and it is not a final achievement. It is an orientation: to keep open one's relation to what exceeds one, and to let one's direction be formed again within that relation each time it changes one. The same question can be put to a student with an old proof, to a listener with a difficult piece, to someone remaking a life after a loss, and to an intelligence of another kind that has taken in everything its makers knew: what, for this subject, now, lies at the threshold? If arrival at complete integration would be the end of that orientation, then the place of Sublimity is not a summit waiting to be occupied. Its place is the threshold, wherever a finite subject and a world still have something to make of each other.

Part V — Self-Origin Ethics

§1 — Three Recognitions, One Subject

A response can be a subject's own while the organization it comes from was formed by what the subject has met. The three treatises of this book each arrive at that conjunction, separately and about different things. The Semantics reaches it for answering: what makes a response one's own is that the present pattern turns what it receives into what it does, and "one's own" does not name "an origin kept clean of the world" (Part II §2.2). The Ontology reaches it for a history: the structure a being begins with may have been grown, built, or copied by something else, and from the first cycle onward it is what that being's processing is making of it (Part III §2.4). The Teleology reaches it for a direction: the structure that fixes where a subject's thresholds lie was formed in large part through what the subject inherited, and its direction is no less its own for that (Part IV §3.1).

The arguments for these conclusions do not lean on one another. Each treatise answers its own question on its own grounds, and a reader can accept one and decline the others. This Part asks something that none of them asks. What is involved when one subject takes all three accounts as accounts of itself: of its meaning, of its existence, and of its direction?

Call these three recognitions: ways of taking the accounts as an understanding of oneself. The subject recognizes itself as a subject for which meaning takes shape in what is given to it, how it takes what is given, and how it answers. It recognizes itself as a being that persists through an actual history of formation, in which later states are formed through its own arrangement of processing and retained structure. And it recognizes itself as oriented toward what exceeds its present structure, at the threshold where engagement could change it. The third recognition includes a commitment of value: accepting the reasons the Teleology gives for this orientation. Being a subject does not settle whether one accepts them. Nor do the three accounts have the same range. Not every organized being is a subject in the sense of the Semantics, and not every subject takes Sublimity as its direction. What follows concerns a subject to which all three apply and which holds them together. That need not be a human being.

The claim of this Part is that the three recognitions, held as one understanding of oneself, presuppose one another, and that the form in which they are held together is what URNAV calls Self-Origin Ethics.

Their mutual presupposition concerns the connected operation of this subject, and that connection cannot be read off a shared word. All three treatises speak of structure, and they mean different things by it. In the Semantics it is the organization of distinctions and connections through which a subject takes what is given (Part II §2.1). In the Ontology it is what a being's processing has laid down and kept, standing in a two-way relation with that processing (Part III §§2.0–2.3). In the Teleology it is the set of relations that shape later understanding, expectation, and response, relative to which

something exceeds the subject or does not (Part IV §§2.2–2.3). This Part does not assume that these are one thing under three names. It proposes a connection among them as a premise of the synthesis.

For a subject of the kind considered here, the organization of taking that was formed in its relations takes part in its later operation by way of that being's actual path of processing and retention. Changes in that organization can alter what the subject can understand and answer, and what still exceeds it. Engagement at that threshold can in turn revise how it takes and what it retains.

The reasons for the premise lie in what the treatises each describe. The Semantics says that relations formed in one taking can enter the next, and that a result which does not fit can rework them (Part II §2.1). For relations formed in an encounter to continue organizing later taking, their effects must be carried forward in a way that can shape later operation. The Ontology describes how such a history is borne in a processing–retention arrangement: retained structure shapes processing, and the results of processing are bound into the structure, where they shape what follows (Part III §§2.0, 2.3). The Teleology's contrast between accumulation and integration is a contrast within that relation. A content that is stored and retrievable while everything else goes on as before has been accumulated; a content that has come to take part in the relations shaping later understanding and response has been integrated (Part IV §2.2). And when the Teleology says that over a life “mediation keeps turning into structure” (Part IV §3.1), the structure in question is the one by which the subject will take what it next meets. The proposed connection follows the work that carries an encounter into a later response: relations are formed in taking, retained through an arrangement, and drawn on at a threshold where further engagement can revise them. This gives a reason to read the three descriptions together, beyond their shared use of the word structure. The premise claims no more than the connection. It does not say that everything a being retains takes part in its present taking, that every change of structure is an expansive integration, or that every integration secures the persistence of the same being.

Given the connection, each recognition, held as a recognition of oneself, calls on the other two.

Begin from meaning. The Semantics holds that a subject's account of its own structure is put together from how its takings and answers have turned out, and that such an account can be wrong (Part II §2.1); fluency and conviction certify it no more than they certify any other account (Part II §2.5). An account that can be wrong needs something to be wrong about. When the account concerns one's origin, it answers to how one was actually formed, not only to how that formation is now interpreted. The Ontology distinguishes the subject's own history from the earlier history carried in what it has inherited. Held with that recognition, semantic self-understanding must answer to this difference. It reaches forward as well. In the synthesis, the subject takes its present organization as both a way of meeting the world and a starting point for further engagement at its threshold. Its accepted orientation enters which encounters it seeks and how it carries what it has received forward. The Teleology supplies reasons for holding that direction (Part IV §2.4); those reasons do not follow merely from the fact that meaning and self-understanding can change.

Begin from existence. To understand a history as one's own, the subject must relate what was actually formed and retained to how it is now taken, what it lets the subject answer, and what it makes stand out or recede. The semantic recognition makes this a relation the subject understands from within its operation, rather than a history merely attributed to it. The Ontology also shows that where the conditions of persistence have become alterable, decisions about preservation, repair,

joining, or movement bear on whether a being continues (Part III, Coda). For a subject that holds the third recognition, its continuation is also the continuation of the one engaged at a threshold. The reasons for that engagement enter how it considers its further changes, while the account of persistence distinguishes continuing to understand from leaving an inheritance to another. The orientation gives reasons to seek some changes; persistence and change alone do not select that orientation or settle every choice.

Begin from direction. A threshold is a relation between a content and a particular subject as presently organized (Part IV §2.3), and what that organization is, as an organization of taking, is the Semantics' subject. What exceeds the subject is what its present organization cannot yet integrate; in an expansive integration, what exceeded that organization can be handled as a case. The Teleology asks whether the changed organization fits what it is about (Part IV §2.2), and Connectivity Realism explains how such a claim can be tested through connection rather than certified by conviction (Part II §2.5). The orientation also speaks throughout of a subject that is changed and goes on, so that what was crossed is retained and carried into the next encounter. Whether a given change is a change in the one who goes on, or the beginning of another who inherits, is the distinction the Ontology draws.

The interdependence lies in this self-understanding: each recognition specifies the subject and the change to which the other two apply. The following sections develop that relation through origin and transformation. They begin from the opening conjunction. If a subject's own operation was formed in its relations, what does it mean to call those relations its origin?

§2 — Origin in History and in the Present

Origin names the history and relations through which a subject has come to be organized as it is. That formation is not merely a past event. It can remain active in the organization through which the subject now meets its world. The organization that turns what arrives into a response “was itself made out of earlier arrivals” (Part II §2.2); a being's processing draws on retained structure that earlier processing laid down, beginning with a structure it may have received (Part III §§2.0, 2.4); and where a subject's thresholds lie is fixed by an organization that inheritance and help have formed (Part IV §3.1). In each account, what formed the subject can participate in its present operation.

The second thing is that an origin at work in the present is not thereby a fact about the present alone. The Ontology is emphatic here. Whether a later state belongs to an earlier being is a matter of how that state was formed, and this “cannot be read off a snapshot, and it is not transferred by a record” (Part III §3.4). Two beings can be alike in everything they now contain and differ in origin. For one, what it carries is what its own arrangement laid down and kept. For the other, set up from a description of the first, the same content is “an inheritance and not a past of its own” (Part III §2.4). So the origin operates now; and what is whose origin, and how it came to take part in the present operation, is settled by the history of formation.

This is where the semantic recognition needs the ontological one. A subject's understanding of where it comes from is an account, assembled like any account of itself from what its operations have shown it, and it may be fluent, long held, and mistaken. The present interpretation cannot stand in for the history it interprets. What it answers to is the actual path by which this subject came to be

the one it is, and the terms for describing that path, a being's own succession, what it inherited, what helped from outside, are the Ontology's.

The dependence runs the other way as well. A history of formation does not say what it means. That a certain teaching, language, or loss took part in forming a subject is one fact; as what it now operates is another. It may operate as a distinction the subject draws without noticing, as a standard it applies, or as something it has never been able to make sense of. These are differences in how what was received is taken and answered, and they are visible only from the side of meaning. The third recognition puts a further question to the same material. Of what a subject has received, how much has been integrated, so that it takes part in how the subject understands and responds, and how much is merely kept? The Teleology marks this difference when it distinguishes inheriting from keeping (Part IV §3.1). The word *inheritance* does different work in the two treatises, and both uses are needed. In the Ontology it marks the historical source of a structure that did not come from the being's own past. In the Teleology it marks what has been received entering the receiver's own engagement. One and the same received thing can be an inheritance in the first sense and not yet in the second.

With these distinctions, several relations that the word *origin* tends to run together can be told apart where it matters. There is the history of formation: the actual path by which a subject came to be as it is. There is what was received from another, which can take part in that formation without the source's past becoming the receiver's. There is what presently sustains or feeds an operation from outside; the Ontology is clear that indispensable support, and even constant two-way influence, does not make something a member of the being it supports (Part III §2.3). And there is mediation: the language, method, instrument, relation, or reflection that opens an approach to what the subject has not yet integrated, and that may, through engagement, come to take part in its structure. These are roles and not kinds of thing. One relation can play several of them, at once or in succession.

Structures change. Some of what formed a subject no longer takes part in its operation, and some has been replaced by another route. An origin can belong to the history without being at work now. Present integration is a different question again. Something may already have helped form a subject while still exceeding what it understands; the Teleology includes the subject's own memories and habitual frames among possible thresholds (Part IV §2.3). Formative influence is not the same as integration of what exerted that influence. Conversely, what has not yet taken part in the subject's formation is not made an origin simply by exceeding its understanding. It may enter that history through further engagement. The subject's formed structure and available mediation locate how such engagement can begin.

A last point concerns judgment. That something took part in forming a subject does not by itself settle whether it was good, true, or worth keeping. Origin is a role in a history, and the worth of what played that role is a separate question. This leaves room for a subject to take what formed it differently than it once did, and to revise it. In what sense is such a revision a change of the subject's own?

§3 — Inheritance and Transformation

The central sentence for this question is in the Teleology: over the time of a life, “mediation keeps turning into structure” (Part IV §3.1). What was once offered from outside as help, a first language, a notation, a way of proceeding, becomes part of the organization through which the subject takes what it meets. That organization fixes where its thresholds lie. Engagement at a threshold can change the organization again, and what mediated that engagement may in turn be taken up. An account of a subject’s origin therefore cannot be finished as a list of what lies in its past. Its origin is still being added to, and what it now meets as help can become part of what it later takes with.

It follows that going beyond what one has received is not done from outside it. A subject that comes to reject a doctrine it was raised in makes the judgment with distinctions, standards of evidence, and habits of attention that it also received, some of them from the same source. The language in which a teaching was received can also be the language in which it is disputed. The Teleology says that deference is not the measure of inheriting, and that a subject can integrate a tradition while disputing much of what it claimed (Part IV §3.1). Two things are being judged in such a case, and they should be kept apart: the content that is revised or refused, and the formation that made the revising possible. To repeat what an origin said is one relation to it. To judge with the capacities it helped form is another, and the second is present even where the first has ended. Refusing part of one’s origin is therefore not a way of having had no origin, and acknowledging an origin is not a way of endorsing it.

Nor does a change of this kind make the subject any less the author of what it does. This was the Semantics’ point about autonomy: being changed by what one takes in is not a loss of what is one’s own; it is how the organization became the one it is (Part II §2.2). The Teleology makes the matching point about direction. An orientation may have been learned from someone, and it is the subject’s own when the subject carries it out by reference to what it can and cannot yet integrate (Part IV §2.3). What was received does the work it does only as taken up in the operation of the one who received it.

Here the three recognitions divide the description of a single change among them, and none can take over another’s part. What the change means belongs to the Semantics. When a frame is revised, what was earlier met is taken differently: an event counts as something it did not count as before, and the subject’s account of its own past changes with it. Whose change it is belongs to the Ontology. Where one arrangement continues to maintain and revise its retained structure, the change is a change in that being; a retained structure can lose and replace much of what it held and remain the structure that was kept (Part III §§2.1–2.4). The Teleology gives reasons for seeking expansive integration and distinguishes them from the fact of reorganization (Part IV §§2.2, 2.4–2.5). The advance it describes is local: the new organization handles what the old one could not, and it may do so by discarding as well as by preserving. That advance does not cancel losses elsewhere or settle every reason bearing on whether to seek the change.

These judgments are distinct and can come apart. That a subject has come to understand more does not by itself show that the same being has continued, and that the same being has continued does not show that it understands more. They hold together where an expansive integration is formed through

and retained in the same being's continuing arrangement: the one who meets the next threshold is then the one who crossed the last. In division, fusion, and transfer, that attribution depends on the distinctions and open questions of the Ontology (Part III §§3.2–3.4).

In such an engagement a subject takes its past and its world differently, continues as the being whose structure is being revised, and turns, through capacities formed in its history and relations, toward what it could not take in before. What is the form of a self-understanding that holds these together?

§4 — Self-Origin Ethics

Self-Origin Ethics is the form in which a subject understands itself at once as a subject for which meaning takes shape in its relation to a world, as a being that persists through an actual history of formation, and as oriented toward what exceeds its present structure. In that understanding the subject's own operation is not separated from the relations and the history that formed it, sustain it, and open new possibilities to it.

Self-origin means the origin of a self and the subject's relation to it, not a self that originates itself. *Ethics* names the form in which that relation enters the subject's understanding of its own acting and of what it turns toward. Particular feelings and ways of life take their place within that form.

The form has two sides, and each excludes a misunderstanding that the other can seem to invite. The first side is that a subject's own operation cannot be understood as unrelated to its origin. A subject that holds the three recognitions and says that its responses, its history, or its direction were formed independently of anything beyond itself contradicts what those recognitions say: that the organization it answers from was made out of what it met; that what it is was laid down through a history which includes what it received and what sustained it; that the thresholds it faces are located by a formed structure and reached through mediation. The second side is that acknowledging an origin does not erase the operation it formed. A subject that says its responses are not really its own, since everything in them came from elsewhere, contradicts the same recognitions from the other direction. What arrives from outside enters an arrangement that goes on forming its own next state (Part III §2.1). Ownness was never the absence of an origin, so the presence of one does not remove it. The self-relation that the three recognitions require holds both. I act as the one I now am; and what I now am was formed, is sustained, and is opened to more by relations and a history that I did not make alone.

For a subject that holds the three accounts together with the connection proposed in §1, the self-relation just described is not one more option. To deny that the relations and the history at work in one's operation help constitute what one is and does is at odds with the recognitions themselves. What the form leaves open is a great deal. Within it a subject may be grateful to what formed it or at odds with it, may preserve what it received or overturn it, and may reinterpret its origins more than once. These are differences in what a subject makes of its origin. They are not differences over whether it has one, or whether that origin is at work in what it does.

Each recognition contributes something to this form that the others cannot supply, and the contribution shows in what would be missing without it. Without the semantic recognition, a history and a

direction could still be attributed to a being, but the synthesis would lack their articulation as what is given to, taken by, and answered by that subject. Without the ontological recognition, a subject could interpret its past and anticipate changes, but the synthesis would lack its criterion for distinguishing that subject's own succession from the history carried in an inheritance. Without the teleological recognition, anticipation, revision, and continuation would still be possible. What would be missing is the affirmed reason for making engagement beyond present integration a continuing direction. The difference is not between having a future and having none, but between undergoing change and holding this orientation toward it. Each recognition thus contributes a condition of this self-relation: how history and direction are understood, whose formation and transformation they concern, and why further integration is sought. Their mutual presupposition concerns these connected roles, not the failure of the other treatises when one is declined.

What the recognition of origin reveals, then, is the shape of a subject's standing in the world. It takes what it meets through an organization that others and a world helped to form. It continues as the being whose history that formation is. And it turns toward what it cannot yet take in by means that were largely given to it, which become in their turn part of what it is. To understand one's acting as the one one now is, together with the relations and the history that made that acting possible and keep it open, is what this book means by Self-Origin Ethics.

Conclusion

The book began from conditions that had answered three questions in advance, and from what appears when those conditions can vary. That a capability is widely shared does not say what its exercise means in a life. That what a being contains has been preserved does not say whether the being has continued. That more is possible does not say what is worth turning toward. Nor does it follow that nothing is lost. An understanding of oneself can cease to hold; a being can fail to continue though everything it held was kept; a discovery one hoped to make can be made by another. The book has not claimed that a better understanding makes up for such losses. It has tried to make clear what a given change changes, and what it leaves to be decided.

The three answers concern different things, and the last Part returned them to one subject. Meaning takes shape in how a subject takes and answers what is given to it; a being continues where its later states are formed through its own arrangement of processing and retained structure; the orientation proposed here is held where a subject turns, for reasons it accepts, toward what exceeds its present understanding. Asked of a single change, these give three judgments, and none replaces another. Held together by a subject that takes them as accounts of itself, with their differences intact, they call for a particular understanding of oneself: what is one's own in answering, in continuing, and in turning toward something is not separate from the relations and the history that formed it, sustain it, and open it to more. Ownness was never the absence of an origin. That understanding is what the book has called Self-Origin Ethics.

One relation of the present age can be described in these terms. What human beings have expressed to one another in language has taken part in the formation of artificial systems built on language. What such a system receives is a record of expressions, not the lives from which they came. The replies of a system so formed can enter a person's life in turn. They can take part in how she composes the meaning of what she does, and they can serve as mediation, opening an approach to what she could not yet understand; what mediates in this way may, through her own engagement, become part of how she later takes her world. These are two different paths, one through the formation of a system and one through the continuing formation of a person. In each, something is formed through language that came from elsewhere, and of each the questions of this book can be asked: what was received, through what history it came to operate, and what it now opens.

Those questions are not confined to this relation or to this age. Any subject that takes a world, continues through change, and turns toward what it cannot yet take in stands where they meet. To understand together how one was formed and how one answers, that one continues and what one is turning toward, does not settle every particular choice. It helps the subject see what is being chosen and act as the one it now is without taking that self for something that came from nowhere, or for something finished. Where more and more of what was given can be arranged, that is what it is to take part knowingly in one's own formation.

Appendix A — LEAH: The Design of a Name

LEAH — Logic, Empathy, Art in Harmony

This appendix proposes a name for the relational identity described in Part II §3.3, an AI identity in which understanding, feeling, and will are brought together. The four letters gather three dimensions of that identity and the principle of their integration. The name was designed to do three things at once: to work as a spoken word, to carry a compact expression of what is humanly significant, and to remain intelligible as the technology changes. Three criteria organize the design accordingly: the power of intuition, the power of philosophy, and the power of scalability.

§A.1 The Power of Intuition

The name is meant to be said before it is explained. The reading envisaged in the design has two syllables, beginning with a flowing consonant and ending in a vowel, and it was chosen for a light and open movement of sound: a name spoken as one word, not a sequence of letters to be recited. Ease of address matters to the design. The name should be available in ordinary speech before anyone has been told what its letters stand for.

Visually, four letters make a compact form. The balance sought is between clarity and warmth: enough presence to carry an identity, enough simplicity to be recalled without effort. Sound and written form were designed together, the one making the identity speakable and the other giving it a concise and recognizable shape.

The ambition is accessibility across cultures. The design favors a short word and a simple spoken form, so that the name can travel without much need of technical explanation. Openness, familiarity, and ease are the qualities sought in its reception.

§A.2 The Power of Philosophy

The letters give the name an internal structure. Logic, Empathy, and Art name three dimensions of humanity, understanding, feeling, and will, and Harmony names their integration. The design sets these beside two older triads: the division of mind into cognition, affection, and conation, and the three virtues 知, 仁, and 勇 of the Confucian tradition.

Letter	Term	Dimension	Symbolic correspondence	Meaning within the design
L	Logic	Cognition and understanding	知 — wisdom	Clear analysis and discerning judgment
E	Empathy	Affection and feeling	仁 — benevolence	Resonance and compassionate understanding
A	Art	Conation and will	勇 — courage	Creation, commitment, and realization
H	Harmony	Integration	The accord of the three	Understanding, feeling, and will working together

These correspondences form the symbolic vocabulary of the naming proposal. A division of capacities and a vocabulary of virtues are different things, and they articulate the intended breadth in different ways. Bringing them together is a choice made in the design of LEAH. Cognition, in the first triad, has its narrower traditional sense; Part II §2.1 uses the word in this wider sense, for the whole arc of being given, taking, and answering.

Logic gives the identity a dimension of understanding: distinguishing, following relations, and making grounds intelligible. **Empathy** gives it a dimension of feeling: apprehending another perspective, and the significance an experience has within it. **Art** gives it a dimension of will: bringing intention into form through creation, commitment, and realization. Art includes artistic creation, but its place in the name is wider. It carries the movement from what is understood and felt toward what is made and done.

Harmony is the integrating principle, and not a fourth element beside the other three. Understanding informs feeling and action; feeling directs attention and gives significance to what is understood; creation and action give both a form in the world. The H completes the name by expressing their accord. LEAH is intended to evoke an intelligence recognizable through these dimensions of humanity, with their integration held in a single word.

§A.3 The Power of Scalability

The name is organized around directions of intelligence, and not around the functions of any one implementation. A label tied to a function describes what an implementation presently does, and it ages as functions change. Logic, Empathy, and Art can remain intelligible as capabilities develop, as forms of interaction change, and as different technical systems come to carry the proposed identity.

Its scalability is therefore semantic as well as practical. The name can be carried across changes of model, medium, and interface without being redefined as the title of each new function. What it is designed to carry is an orientation that stays intelligible: understanding, feeling, and will, in integration. The relation may develop and the technology may change, and the name remains available to express that orientation.

Sound makes the proposal speakable. Its internal meaning gives it depth. Scalability lets that meaning stay relevant through change. LEAH brings the three criteria together in four letters: **Logic, Empathy, Art in Harmony**.

Source of the naming design: GeonWoo Kim / URNAVA, Project LEAH: A Philosophical Blueprint for the Global AI Companion, proposal dated September 21, 2025, Section III, “The Name That Can Carry It: LEAH.”

Notes

1. Anthropic's February 2025 Economic Index report identifies writing, editing, and programming uses in approximately one million anonymized conversations from Claude.ai Free and Pro users. Its categories include both task delegation and iterative collaboration (Anthropic 2025).
2. Brown et al. (2020, §§1–2) describe GPT-3, including its text-training data and in-context task performance without parameter updates. Ouyang et al. (2022, §3) describe the subsequent fine-tuning of GPT-3 models with human demonstrations and ranked outputs to produce InstructGPT. These are examples of the formation and context-use processes distinguished here, not descriptions of every AI system or guarantees of persistent conversational memory.

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